Vektor Engineering & Consulting Services, Inc.

2603 Camino Ramon, Suite 417, San Ramon, CA 94583

January 26th, 2026

SLAC National Accelerator Laboratory
2575 Sand Hill Road
Menlo Park, CA 94025-7015

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization ☑ No ☐ Yes	
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

PROJECT MANUAL COVER

B950 MECH ROOM CO2 CHILLERS

PROJECT: LCL2024-034

(REV E, BIO REVIEW)

Electrical Contractor shall use these specifications in conjunction with those listed on Electrical Sheets. If conflict between these Specifications, those listed on actual Electrical Construction Drawings, or general Project Specifications arises, then the stricter requirement(s) shall govern and shall be provided by Electrical Contractor.

Approvals:

ELECTRICAL ENGINEER:	Vektor Engineering and Consulting Services, Inc. 2603 Camino Ramon, Suite 417 San Ramon, CA 94583 Sergey Korolev, P.E. (E20491)
SLAC APPROVALS:	

DIVISION 1 – GENERAL REQUIREMENTS

DIVISION 7 – ARCHITECTURAL SPECIFICATIONS

DIVISION 22 AND 23 – MECHANICAL SPECIFICATIONS

DIVISION 26 – ELECTRICAL SPECIFICATIONS

26 08 05	ELECTRICAL ACCEPTANCE TESTING
26 24 00	PANELBOARDS
26 27 26	WIRING DEVICES
26 28 00	LOW VOLTAGE OVERCURRENT PROTECTIVE DEVICES
26 28 16	SWITCHES AND CIRCUIT BREAKERS
26 28 19	CIRCUIT AND MOTOR DISCONNECTS
26 29 13	ENCLOSED CONTROLLERS

DRAWING LIST

SLAC Drawing Number	Sheet Description
GENERAL	
B0950-G00-0415	G0.01 COVER SHEET
MECHANICAL	
B0950-M00-0416	M0.01 COVER SHEET
B0950-M01-0417	M2.00 3RD FL CHILLED WATER PIPING PLAN
B0950-M01-0418	M2.01 3RD FL CO2 PIPING PLAN
B0950-M01-0419	M2.02 BASEMENT CO2 PIPING PLAN
B0950-M01-0420	M2.03 SUB-BASEMENT CO2 PIPING PLAN
ELECTRICAL	
B0950-E00-0421	E0.01 COVER SHEET
B0950-E06-0422	E0.02 EQUIPMENT SCHEDULES
B0950-ED1-0423	E3.01D LCLS B950 R302 DEMO POWER PLAN
B0950-EP1-0424	E3.01 LCLS B950 R302 POWER PLAN
B0950-ED7-0425	E5.01D SINGLE LINE DIAGRAM DEMO
B0950-E07-0426	E5.02 SINGLE LINE DIAGRAM
B0950-E07-0427	E5.03 SINGLE LINE DIAGRAM NEW
B0950-E06-0428	E5.04 PANEL SCHEDULES
B0950-E06-0429	E5.05 PANEL SCHEDULES
B0950-E05-0438	E6.01 DETAILS
STRUCTURAL	
B0950-S00-0439	S0.01 DETAILS

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SECTION 01 00 00 – GENERAL REQUIREMENTS

PART 1 - GENERAL

The following specification outlines the requirements for the SLAC National Accelerator Laboratory.

1.1 LOCATION

A. Perform the work outlined in this specification at the SLAC National Accelerator Laboratory, 2575 Sand Hill Road, Menlo Park, San Mateo County, California.

1.2 GENERAL

A. The purpose of this specification is to describe the intended results of the work, identify certain reference standards, and provide certain technical details. Where standards are not specified, the Subcontractor shall perform the work in accordance with the best general construction practices and industry standards, codes, and provide all new materials and workmanship of the first quality. Where the specifications do not provide complete details, the Subcontractor shall provide a submittal to SLAC for prior approval, showing the details of the planned installation for meeting the Subcontract requirements. The term “contractor” as used in the Specifications and Drawings shall mean “Subcontractor”. The term “should" as used in the Drawings shall mean “shall”. The term “Provide” as used in the specifications and drawings shall mean “Furnish and Install”.
B. The Subcontractor shall verify all dimensions, locations and quantities in the field prior to construction, and shall furnish all construction surveys and tests necessary for the correct location and installation of all work.
C. The terms of the Subcontract, General Provisions, and these General Requirements apply to each section of these specifications as fully as if repeated within that division.
D. The Subcontractor is required to comply with all articles, attachments, exhibits, and provisions of the solicitation and resultant contract.
E. Items listed under each division or section of the specifications are not necessarily all inclusive. Subcontractor shall be responsible for performance and completion of the work in accordance with the scope of work and the Subcontract Documents.
F. Portions of these specifications are of the abbreviated, simplified type and may include phrases instead of complete sentences.
G. Omissions of words or phrases such as "Subcontractor shall," "in conformity with," "shall be," "as noted on the drawings," "in accordance with details," "a," "the" and "all" are intentional. Omitted words or phrases shall be supplied by inference in the same manner as they are when a "note" occurs on the drawings.
H. Such terms as "approved," "approved equal," "as directed," "as required," "as permitted," "acceptable," and "satisfactory" are defined as by or to SLAC.
I. Order of Contract Document Precedence: Refer to the Subcontract General Terms and Conditions for Fixed Price Constructions Subcontracts, Article 2 Order of Precedence.

1.3 DESCRIPTION OF WORK – GENERAL

A. Refer to subcontract documents, drawings and specifications for details, and refer to Statement of Work.

B. The Subcontractor shall coordinate with SLAC all utility outages, new connections, traffic disruptions, access restrictions and other activities which may impact the operations of existing building and experimental facilities. Refer to the subcontract documents for additional information.

1. The following specific potential constraints have been identified for this project. This list is not inclusive of all constraints for this project, but an overall guide for the Subcontractor's work:
    a. Utility Outages: Electrical
    b. Site Access: Stairs, Elevators, Freight Elevators (in vicinity area, heavy traffic), utility chases, lab spaces
    c. Traffic.
    d. Hours of work.
    e. Working in or near an occupied building.
    f. Experimental Integrity: Acoustic noise, vibration/shock, particles.

1.4 DESCRIPTION OF WORK

A. Electrical Energization Plan. The Subcontractor shall prepare an Electrical Energization Plan that complies with the following SLAC requirements:

1. Prior to connecting the new installation to electrical power, work on electrical circuits and equipment does not require lockout/tagout. After the new installation is tied into the electrical supply, work on electrical circuits and equipment must be performed under the SLAC Control of Hazardous Energy Policy.
2. Attendance and participation in Electrical Energization Readiness Review meetings.
3. Installation of new electrical circuits and equipment shall be substantially complete prior to first energization, including all of the following: Installation of new electrical circuits and equipment shall be substantially complete prior to first energization including the following:
    a. All main switchgear or main switchboard or motor control center incoming and outgoing feeders must be terminated at both ends
    b. All National Electrical Code (NEC) required working space shall be clear of tools, materials, and other objects
    c. NEC required dedicated space shall have leak protection installed if required
    d. All electrical enclosures shall be intact with covers installed and doors closed and locked
    e. Electrical room doors shall be locked with entry restricted to authorized personnel only
    f. SLAC review of electrical analyses must be complete with comments incorporated
    g. All required arc flash hazard labels must be affixed
    h. All required testing/commissioning activities for equipment to be energized must be complete and associated test reports reviewed and accepted by SLAC

i. SLAC BIO inspection of equipment to be energized must be complete with no significant open items. BIO will install BIO Energization tag on panel or equipment indicating BIO acceptance for code compliance.
j. SLAC F&O inspections of electrical equipment to be energized must be complete with no significant open items. Inspection shall include internal and external installation of equipment including but limited to switchboards, motor control centers, panelboards, disconnect switches, transformers, variable frequency drives, motor connections, wiring termination inside j-boxes, wiring terminations inside junction boxes, cabling installation inside manholes, cable installation inside vaults, motor starters, lighting control panels, and control panels with 120V connections. Spot checking of wiring termination of branch wiring devices such as light switches and 120V receptacles.
k. All equipment hazard labels shall be affixed
l. All equipment ID labels shall be affixed
m. Subcontractor shall maintain up to date marked up top-level electrical drawings at the jobsite. These marked up drawings shall be made available to all SLAC and subcontractor workers for their use in planning lockouts in support of construction activities.
n. All subcontractor LOTO training must be complete
o. LOTO training records for all electrical workers must be verified by the SLAC FCM
p. Subcontractor shall certify to the SLAC PM that all pre-energization requirements have been satisfied.

1. The Subcontractor will incorporate the details of this plan into the CPM schedule. The Subcontractor will update this plan at each additional design submittal as necessary. This plan will be provided to the pertinent sub-tiers as part of the bid package.
2. The Subcontractor will work with the A/E to coordinate the design with the energization plan (formal comments to be submitted during PD/FPD review). Any modifications to the electrical design shall also consider the maintainability of the electrical system and shall minimize the required shutdown of equipment critical to the science mission during routine maintenance.

1.5 HAZARDS

A. Specific hazards have been identified, including, but not limited to:
   1. Low voltage work,
   2. Confined Spaces,
   3. Working on elevated surfaces,
   4. Airborne fiber and materials,
   5. Radiation,
   6. Laser Exposure,
   7. Vibration,
   8. Noise.

1.6 SPECIAL REQUIREMENTS AND CONDITIONS: The items below are unique expectations that are applicable to this project.

A. Safety and environmental protection are of the utmost importance.
   1. Refer to SECTION 01 35 23 – OWNER SAFETY REQUIREMENTS for a list of documentation and training requirements.

B. Waste Management:
   1. Refer to SECTION 01 74 19 – CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

C. Sustainable Acquisition:
   1. To the greatest extent possible, construction products and materials including indoor furnishing must be environmentally preferable. Environmentally preferable products include but are not limited to those products designated by the EPA's. Comprehensive Procurement Program, Biobased projects as defined by the USDA Bio-preferred program, Energy Star and FEMP designated projects, EPEAT certified electronics, Non-Ozone Depleting Alternatives, and Water Sense products. Environmental Preferable Purchasing data will need to be provided to SLAC to support end-of-Fiscal Year reporting requirements. Water Sense products. Environmental Preferable Purchasing data will need to be provided to SLAC Project Management to support end-of-Fiscal Year reporting requirements. Allow a space to add additional items, or the ability to remove some of the above (e.g. RP requirements).

1.7 SLAC FURNISHED EQUIPMENT AND SERVICES:

A. SLAC Furnished Services: See SOW on Construction Drawings.

1.8 RESPONSIBILITIES

A. University Subcontract Administrator: The Subcontract Administrator has sole authority with respect to the administration of the Subcontract and any and all changes involving price or schedule.
   1. Refer to General Terms and Conditions and Section H Special Terms and Conditions of the Subcontract.

B. SLAC Management Responsibilities: The SLAC Project Management team will monitor the Subcontractor's work for conformance with the Subcontract requirements, monitor Quality Assurance, review and analyze the Subcontractor's schedule, manage change control, oversee document flow, approve percent complete estimation/ pay recommendation, provide daily oversight and inspection of the Work for conformance with the Contract Documents and:
   1. Coordinate the construction decisions required of SLAC relating to the drawings, specifications, and other construction information pursuant to the Subcontract or necessary for successful performance of the work
   2. Monitor Subcontractor's safety program for compliance and general surveillance over the implementation of security procedures

3. Direct the initiation and preparation of technical changes in the applicable drawings, specifications, and other construction data
4. Review and approval of Subcontractor submitted CPM schedules
5. Quality Assurance oversight and monitor Subcontractor's quality program for compliance and general surveillance for work in progress or completed
6. Provide technical direction
7. Identify and oversee the required work by Subcontractor to correct defects discovered in partially or fully completed construction work
8. Review for SLAC approval, Subcontractor 's invoices for payment
9. Review for SLAC acceptance, Subcontractor's safety plans such as Elevated Surface Work Plans and Hoisting and Rigging Plans.
10. Coordinate for SLAC in the areas designated herein. Subcontractor shall refer all technical questions, submittals, and like items, in these designated areas to the SLAC Project Manager.
11. Provide coordination of multiple Prime Subcontracts, including scheduling and interface between Prime Subcontractors.
12. Conduct coordination meetings; Subcontractor shall conduct weekly construction meetings and Subcontractor QC Manager shall initiate and conduct Pre-Installation meetings with SLAC participation.

1.9 PARTNERING: N/A

1.10 SLAC CONSTRUCTION MANAGEMENT RESPONSIBILITIES:

A. Provide oversight to ensure the safety, quality, environmental, health, safeguard, and security requirements specified in the Subcontract are met by construction Subcontractors.
B. Coordinate with Subcontractor to schedule special inspections required for construction work in the scope of the Subcontract
C. Assist Subcontractor with permits and plans required by SLAC for construction work in the scope of the Subcontract
D. On behalf of the project, coordinate with SLAC Facilities Operations and Maintenance, ES&H Subject Matter Experts, Waste Management, Site Security and other SLAC departments.
E. Provides review and acceptance, and oversight of the Subcontractor's Work Planning and Control, (WPC), documentation.
F. Request for SLAC Operations services for all systems LOTO; All systems LOTO shall be executed by SLAC F&O personnel. G. Review RFIs and Submittals.
G. Review of monthly billings and as-builts.
H. Provide technical guidance based on contract documents.

1.11 SLAC ESH FIELD SAFETY REPRESENTATIVE RESPONSIBILITIES:

A. Review construction documents as needed.
B. Provide approval signatures on forms and permits as needed.
C. Participate in project meetings as needed.
D. Perform job walks with emphasis on worker safety and report any findings to the FCM.

1.12 SLAC PROJECT MANAGER RESPONSIBILITIES:

A. Act as primary and technical point of contact for the sub-contractor.
B. Provide oversight to ensure the safety, quality, environmental, health, safeguard, and security requirements specified in the Subcontract are met by construction Subcontractors.
C. On behalf of the project, coordinate with SLAC Facilities Operations and Maintenance, ES&H Subject Matter Experts, Waste Management, Site Security and other SLAC departments.
D. Review RFIs and Submittals.
E. Review of monthly billings and as-builts as applicable.
F. Provide technical guidance based on contract documents.

1.13 SLAC PROJECT MANAGEMENT SPECIALIST RESPONSIBILITIES:

A. Provide oversight and guidance for technical matters during design, construction, testing, and startup of the facilities.
B. Provide Subcontractor interpretations and clarifications to the applicable drawings and specifications and other construction data and reconcile discrepancies in the aforementioned documents and data as may be required.
C. Issue written technical instructions within the scope of work stated in the Subcontract that do not provide price or schedule impacts.
D. Review for SLAC approval, vendor submittals, and shop drawings.

1.14 SLAC ELECTRICIANS RESPONSIBILITIES:

A. Provide inspection of electrical installation prior to energization. Inspection shall include internal and external installation of equipment including but limited to switchboards, motor control centers, panelboards, disconnect switches, transformers, variable frequency drives, motor connections, wiring terminations inside junction boxes, cabling installation inside manholes, cable installation inside vaults, motor starters, lighting control panels, and control panels with 120V connections. Spot checking of wiring termination of branch wiring devices such as light switches and 120V receptacles.
B. Provide switching of energized breakers and disconnect switches. For all work requiring LOTO, SLAC-authorized workers shall lock first, then Subcontractors shall lock over the SLAC locks.
C. Inventory and asset tagging of electrical equipment.

1.15 SLAC INSPECTOR'S RESPONSIBILITIES:

A. As delegated by the Construction Manager, SLAC inspectors will be responsible for field observations of the work.
B. As required by the Subcontract, SLAC will provide special inspections, and acceptance testing by an Independent Testing Laboratory for Quality Assurance purposes.
C. Subcontractor and Sub-Subcontractors shall provide their own qualified and SLAC approved inspectors for Quality Control purposes at no additional cost to SLAC.

1.16 PROJECT SCHEDULER RESPONSIBILITIES:

A. Subcontractor shall provide their own qualified/certified and SLAC-approved scheduler(s) for planning and updating the baseline and forecast schedules.

1.17 Subcontractor Work, Planning, and Control Documentation - Including applicable ESH Manual section references:

A. Job Safety Analyses
B. Pressure Test Plans
C. Silica Exposure Plan
D. Lock Out Tag Out (LOTO)
E. Control of Hazardous Energy (CoHE)
F. Hoisting and Rigging Plan
G. Method of Procedures
H. Excavation/Shoring Plans
I. Mechanical and Electrical Energization Plans
J. Daily planning and release forms

1.18 INFORMATION TRANSMITTAL

A. Submittals to SLAC: Submit the information and documents described in the following subparagraphs:

1. Progress Schedule as required by Owner Representative.
2. Project Record Documents: Two complete marked-up sets of drawings and specifications fully illustrating all revisions made by all the crafts in the course of the work, due prior to final acceptance inspection. Also, make available to the SLAC PM updated (as-built) drawings and specifications to be reviewed with the monthly progress payment. Refer to Terms and Conditions for Payment requirements.
3. Construction Document Submittals: See Appendix 16.6.2.
4. Quality Program as required by Owner Representative.
5. Hazardous Material Inventory Sheets: See SECTION 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.
6. Cost Breakdown/Schedule of Values as required by Owner Representative.
7. Submit within 5 calendar days after the notice of award, a list of Subcontractor's Sub-Subcontractors. Include Sub-Subcontractors' telephone numbers and addresses.
8. Stormwater Pollution Prevention Plan (SWPPP): Projects disturbing >1 acre must submit a project-specific SWPPP and construction monitoring plan, developed by a Qualified SWPPP Developer (QSD) and implemented by a Qualified SWPPP Practitioner (QSP). The SWPPP and a Notice of Intent (NOI) must be submitted to the SLAC project manager at least 10 days before work begins.

1.19 ACCURACY OF DATA

A. The data in these specifications and on the drawings are as exact as could be secured, but their absolute accuracy cannot be guaranteed. The drawings and specifications are for the assistance and guidance of Subcontractor and exact locations, distances, levels, and like items will be governed by the work.

1.20 GEOTECHNICAL DATA REPORTS

A. Record of geotechnical investigation is available upon request in electronic format.

B. Records of investigations of soil or subsurface conditions and logs of test borings that SLAC makes available are not part of the Subcontract and are solely for the convenience of the Subcontractor.

1.21 SITE FIRE SAFETY

A. Subcontractor shall maintain the area within the construction fence free of grasses and other combustible native species as directed by SLAC.
B. Fire suppression equipment (such as fire extinguishers) shall be provided to the work crews in accordance with NFPA requirements.
C. A nonsmoking policy shall be instituted in accordance with SLAC's nonsmoking policy.
D. Before furniture is delivered to the building, the building fire sprinkler system shall be operational and accepted by SLAC.
E. Subcontractor shall comply with the SLAC Hot Work Program for all hot work. Hot work is defined as any flame or spark-producing construction activity.

1.22 CONSTRUCTION LIMITS

A. The limits of work under this Subcontract are shown on the drawings. Confine construction operations to the area shown in Subcontract drawings.

1.23 MODIFICATIONS OR CONNECTIONS TO EXISTING UTILITIES

A. A written SLAC-approved work plan or Method of Procedure (MOP) shall be provided to SLAC for approval if modifications or connections to the existing utilities (e.g., electric power, water, communications, and air) require an interruption of services. This written notice is required 14 calendar days prior to the desired modification or connection or as defined in the specifications.

1.24 SITE STAFFING

A. As a minimum, Subcontractor shall provide staff positions for the following:

1. Construction Superintendent (full-time on-site)
2. Project Manager (full-time on-site)
3. Project Engineer (full-time on-site)
4. Safety Professional (full-time on-site)
5. Quality Control Manager (full-time on-site, dedicated only to QC)
6. Scheduler/Planning Staff
7. BIM Manager/Coordinator

B. Staff the project with a construction superintendent, project manager and a project engineer with adequate experience working on projects similar to that described in these Specifications. Submit resumes and experience for proposed staffing of these key positions to SLAC for approval. SLAC will have the final determination of the adequacy of the proposed personnel's experience.
C. Sub-tier Subcontractors must also provide a superintendent or foreman with adequate experience working on projects like that described in this Specification. Exception: If

performing "limited scope" work, then no foreman is required (Refer to Chapter 42 of the SLAC ES&H Manual for more information: https://esh.slac.stanford.edu/eshmanual).

D. The qualified Superintendent must have completed OSHA 30 construction safety training and be dedicated entirely to the safety oversight of the project, including weekends and after hours as required.

E. Sub-tier foremen must have completed OSHA 30 construction safety training unless exempted due to performing "limited risk"/"low risk" work (Refer to Chapter 42 of the SLAC ES&H Manual for more information: https://esh.slac.stanford.edu/eshmanual).

F. Provide a QC manager with adequate experience working on projects similar to that described in these Specifications.

G. Safety Officer qualifications and duties may be found in SECTION 01 35 23 OWNER SAFETY REQUIREMENTS.

H. Scheduler/Planning Staff Qualifications: (As applicable to specific projects)

1. Bachelor of Science degree in Construction Management, Engineering, Project Management, or related field.
2. Planning and Scheduling Professional (PSP) or Earned Value Professional (EVP) from the Association for the Advancement of Cost Engineering (AACE). In lieu of certification, 15 years of hands-on planning and scheduling experience with demonstrated competence in earned value concepts as they pertain to construction progress payments using resource and cost–loaded CPM schedules.
3. Demonstrated prior experience with the construction work process to the extent that a logical CPM schedule can be developed, maintained, and progressed which accurately represents the scope of work to be performed.
4. Knowledge of CPM scheduling for medium and large buildings, and the ability to analyze schedules to determine duration and logic issues. analyze schedules to determine duration and logic issues.
5. Ability to demonstrate, in an interview, CPM, proper CPM scheduling protocols such as sequencing, redundant activity relationships, types of relationships and their proper uses, EV principles, and how these will be applied to the project schedule.
6. Due to the use of the Collaborative Construction Planning Process, experience with implementing and maintaining schedule consistency between P6 and Last Planner® System is highly preferred.

1.25 SUBCONTRACTOR SUPERVISION, WORK HOURS, AND OVERTIME

A. Subcontractor shall always provide adequate supervision of the project, including overtime hours and shift work hours, when work is being performed by Subcontractor or sub-tiers. It is expected that the Subcontractor's superintendent or other designated company agent with authority to make on-site decisions will provide this supervision. With the approval of the SLAC PM or FCM, on a case-by-case basis, the Subcontractor can delegate this responsibility to another qualified person. If the Subcontractor fails to provide adequate supervision, work will be stopped in accordance with the Terms and Conditions.

B. Work hours are limited to 60 hours per week,10 hours per day. Workers are not allowed to work 7 days in a row. Exceptions require advanced SLAC management written approval. Saturday, Sunday, and holiday work require SLAC Approval.

1.26 SITE PREPARATION, DEMOLITION AND EXCAVATION

A. During the construction, excavated material shall be reused, recycled, or disposed of in accordance with the SLAC Excavation Permit for the project.

1.27 EXCAVATED MATERIAL HANDLING, DISPOSAL AND WASTE MANAGEMENT

A. SLAC Excavation Permit shall be obtained prior to starting any activity that involves excavation of one foot or more, or relocation of soil at SLAC. Permit is required for activities such as trenching, drilling and removing soil that meet any of the following: depths of 1 foot or more; power tools will be used; utilities are identified; any hazardous condition is likely to be encountered. Allow 14 business days for issuance of excavation permit.

B. Excavated materials shall be subject to inspection and testing by SLAC.

C. Implement the following regulatory requirements:

1. Title 8 California Code of Regulations, Health and Safety Requirements of the California Occupational Safety and Health Administration (Cal/OSHA).
2. Department of Transportation requirements, as applicable.
3. San Mateo County Health Services Agency, Environmental Health Division requirements for subsurface drilling permit. Applies to soil borings that are anticipated to contact groundwater, soil borings deeper than 10 feet, and groundwater monitoring wells. SLAC will obtain permit. Subcontractor must conduct work in accordance with permit conditions.

D. Excavated soils with chemical contaminants shall be legally disposed of offsite at either a Class I or class II facility as determined by SLAC. Transportation and disposal of all hazardous, Class II, Universal wastes must comply with all federal, state and local laws and regulations and be coordinated with SLAC Waste Management (WM). Wastes must be disposed only at SLAC approved disposal facilities. All waste documents [waste profiles, hazardous waste manifests, Land Disposal Restriction (LDR) forms, etc. must be signed by SLAC WM.

E. Excavated materials that have been identified as contaminated shall be segregated and stockpiled in designated areas to prevent release of contaminants, fugitive dust and runoff. Refer to SECTION 01 74 19 WASTE MANAGEMENT.

F. Existing asphalt and base rock shall be recycled off-site

1.28 ENVIRONMENTAL MANAGEMENT

A. Prepare excavation, construction, and staging areas to minimize impacts to wildlife, as practical, including avoiding the bird and bat breeding seasons as follows:
B. If initial site work begins during seasonal periods of bat activity, 2/15 to 4/15 and 8/15 to 10/30, a biologist must conduct a site reconnaissance to determine presence of roosting bats and develop a buffer zone and work with the Subcontractor to minimize construction noise.
C. If initial site work begins during the bird breeding season, 2/1 to 8/15, a biologist must conduct site reconnaissance to determine presence of nesting birds and develop a buffer zone.
D. SLAC will approve selection of biologist and resulting report from site reconnaissance.
E. Can add more items or remove others

1.29 SITE RESTORATION

A. All staging areas and areas disturbed by the construction shall be restored to their preconstruction condition including regrading, repaving and reseeding as applicable.

B. All synthetic material used for erosion prevention and other stormwater BMPs, such as wattles encased in plastic netting and synthetic silt fences shall be removed at the end of construction and shall not be part of the final site restoration.

1.30 AIR QUALITY

A. As required by SLAC PM, or FCM, fugitive dust control measures shall be implemented in compliance with the approved Stormwater Pollution Prevention Plan (SWPPP) and Bay Area Air Quality Management District requirements.

1.31 CONSTRUCTION EQUIPMENT

A. Construction equipment shall be cleaned prior to mobilization to the site to prevent the introduction of non-native species.
B. All equipment shall be in proper operating condition prior to mobilization. Documentation of inspection shall be provided to SLAC upon request.
C. Subcontractor's equipment, including trucks, shall be cleaned of loose debris, dirt and mud prior to exiting the SLAC site.
D. Subcontractor shall follow SWPPP Best Management Practices for equipment, fueling,

1.32 HYDROLOGY AND WATER QUALITY

A. Construction projects that exceed 5,000 square feet must comply with the stormwater runoff requirements of Section 438 of the Energy Independence and Security Act (EISA), which calls for federal developments to maintain or restore pre-development hydrology. Project must develop and submit to SLAC a report demonstrating EISA 438 compliance.
B. Construction projects that disturb >1 acre must comply with the State Water Resources Control Board (SWRCB)'s Construction General Permit (CGP). Project must develop a project-specific SWPPP and submit to SLAC. Project must develop a Notice of Intent (NOI) and submit to SLAC at least 10 working day prior to commencing construction.
C. Construction projects that disturb <1 acre must submit an Erosion and Sedimentation Control Plan to SLAC.

1.33 WASTEWATER

A. Wastewater may not be discharged to SLAC's sanitary sewer without prior approval from the SLAC Environmental Protection (EP) Department. Any projects proposing to generate wastewater, either during construction or operation, must quantify and describe the wastewater streams and provide this information to EP for approval and incorporation into SLAC's Industrial Wastewater permit. Wastewater with elevated dissolved solids may be required to be on a time-delayed discharge schedule.

1.34 TRAFFIC AND CIRCULATION

A. Subcontractor shall inform their personnel to arrive/depart from SLAC during noncommute hours, where possible. When staff and material must arrive at SLAC outside work hours, advance permission from PM and SLAC Security is required.

B. Subcontractor shall prepare and submit for approval a Traffic Control Plan to identify transportation routes including access from local roadways. The plan shall accommodate safe pedestrian traffic adjacent to the construction area.

C. Construction traffic shall be coordinated with construction from other SLAC projects per SLAC FCM direction.

1.35 EMERGENCY REPAIRS

A. SLAC reserves the right to make emergency repairs to Subcontractor-installed equipment as required to keep equipment in operation without voiding Subcontractor's guarantee or relieving Subcontractor of its responsibilities.

1.36 SLAC'S PARTIAL OCCUPANCY OR USE

A. As determined by SLAC Owner Representative and BIO.

1.37 FINAL ACCEPTANCE

A. As determined by SLAC Owner Representative and BIO.

1.38 ACRONYMS

A/E	Architectural and/or Engineering firm
AACE	Association for the Advancement of Cost Engineering
BIO	Building Inspection Office (SLAC)
BMP	Best Management Practices
Cal/OSHA	California Occupational Safety and Health Administration
CGP	Construction General Permit
CPM	Critical Path Method
CoHE	Control of Hazardous Energy
DOR	Designer of Record
EISA	Energy Independence and Security Act
EP	Environmental Protection (SLAC)
EPA	U.S. Environmental Protection Agency
EPEAT	Electronic Product Environmental Assessment Tool
ES&H	Environment, Safety and Health
EVP	Earned Value Professional
F&O	Facilities & Operations
FCM	Field Construction Manager
FEMP	Federal Energy Management Program
FPD	Federal Project Director
LDR	Land Disposal Restriction
LOTO	Lock Out Tag Out
MOP	Method of Procedure
MUA	Make Up Air

NEC	National Electrical Code
NFPA	National Fire Protection Association
NOI	Notice of Intent
OSHA	Occupational Safety and Health Administration
PD	Project Director
PM	Project Manager
PSP	Planning & Scheduling Professional
QC	Quality Control
QSD	Qualified SWPPP Designer
QSP	Qualified SWPPP Practitioner
RFI	Request For Information
RP	Radiation Protection
SLAC	SLAC National Accelerator Laboratory
SOW	Statement of Work
SWPPP	Storm Water Pollution Prevention Plan
SWRCB	State Water Resources Control Board
USDA	US Department of Agriculture
WM	Waste Management (SLAC)
WPC	Work Planning & Control

PART 2 - PRODUCTS NOT USED

PART 3 - EXECUTION NOT USED

END OF SECTION 01 00 00

SECTION 01 11 11 – HOLD POINTS

PART 1 - GENERAL

1.1 HOLD POINTS:

Prior to commencement of the following activities, action is required as indicated:

Prior to:	Approval Required:
Provide baseline schedule, Pre-Work Hazard Analysis, SSSP, SSQF, IIPP	CA, PM/FCM, ESH
Commencement of on-site work, provide written Job Safety Analysis (JSA)	PM, FCM, ESH
De-energization and energization of affected circuits, provide MOP for SLAC Provided LOTO	PM, FCM, Operations Elec., Building Manager
Selective demolition plan. Confirm what is to be demolished and to remain. Coordinate with SLAC I&C.	PM, FCM, Operations Elec., Building Manager
General Contractor Rough In	PM, FCM, Operations Elec., Building Manager
Electrical Contractor rough in	PM, FCM, Operations Elec., Building Manager
Mechanical Contractor rough in	PM, FCM, Operations Elec., Building Manager
Inspect receptacles, plugstrip, conduit & wiring, and j-box installation (after initial construction phase/return to site)	PM, FCM, Operations Elec., Building Manager
Complete SOW Installation	PM, FCM

Final BIO Inspection	PM, FCM
Final tests, energization, acceptance and handover of system	PM, FCM
Red Lines	PM, FCM
Project Close-out	PM, FCM

END OF SECTION 01 11 11

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization	☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 01 35 23 – OWNER SAFETY REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. General:

1. The following applies to this section:

a. The terms of the Subcontract, including General Provisions, and General Requirements.

b. Related requirements in other Subcontract Documents including, but not necessarily limited to, the Drawings and the Specification Sections in the Project Manual.

c. Review these documents for coordination with additional requirements and information that apply to work under this Section.

B. Related Sections:

1. SECTION 01 00 00 – GENERAL REQUIREMENTS
2. SECTION 01 11 11 – HOLD POINTS
3. SECTION 01 45 23 – OWNER SAFETY REQUIREMENTS
4. SECTION 01 74 19 – CONSTRUCTION WASTE MANAGEMENT
5. SECTION 01 78 36 – WARRANTIES
6. SECTION 01 91 13 – GENERAL COMMISSIONING REQUIREMENTS

1.2 SUMMARY

A. This section supplements the requirements in the Terms and Conditions of the Subcontract, for the Subcontractor safety program.
B. The SLAC construction subcontractor safety and health requirements are found in the SLAC Construction Safety Requirements Manual (Document Number CP-SPECESH16325)

1.3 CONSTRUCTION SAFETY REQUIREMENTS MANUAL (CSRM)

A. The Construction Safety Requirements Manual has been developed to define the minimum safety and health requirements for Prime Subcontractors and their sub-tier subcontractors performing construction activities at the SLAC National Accelerator Laboratory (SLAC).
B. It is SLAC policy that all subcontractors shall provide a safe and healthful workplace for their personnel as well as SLAC and other affected personnel. The purpose of this document is to assist subcontractors in understanding and implementing the applicable regulatory requirements and SLAC-specific safety requirements. The requirements found in this document will assist

subcontractors meeting the requirements of SLAC's Worker SECTION 01 35 23– OWNER SAFETY REQUIREMENTS Safety and Health Program (WSHP). Notwithstanding the foregoing and notwithstanding any oversight or direction SLAC may provide, subcontractors, their sub-tier subcontractors, and their agents remain responsible for safely accomplishing the statement of work, and for complying with all applicable laws and regulations.

C. SLAC's goal is ZERO INCIDENTS. The Prime Subcontractor and each sub-tier subcontractor's line management are expected to promote and model this concept and develop, implement, and enforce a safety and health program that will result in a safe work environment. Safety is not to be compromised for production and shall be considered an integral part of the work planning process.

D. The CSRM requirements apply to all Prime Subcontractors and sub-tier sub-contractors (collectively referred to as "subcontractor" throughout this document) performing construction activities on the SLAC Site. Specific responsibilities of Prime Subcontractors are delineated as such.

E. Subcontractors are obligated to flow down the requirements to all sub-tier subcontractors.

1.4 QUALIFIED ELECTRICAL WORKER

A. Only qualified workers are permitted to work in electrical systems at SLAC. A "Qualified Electrical Worker" is a person who has skills and knowledge related to construction and operation of electrical equipment and installations and has received safety training on the hazards involved. Such a person shall be familiar with proper use of precautionary techniques, personal protective equipment, insulating and shielding materials, insulated tools, and test equipment in addition to SLAC specific procedural requirements. All Qualified Electrical Workers are required to present a letter from their employer asserting the employee's knowledge/training per NFPA 70E Chapter 1 "Electrical Safety Program" and Cal/OSHA "Qualified Electrical Workers".

B. Subcontractor journeyman electricians shall be certified electricians in the State of California.

C. Apprentice electricians shall be registered with the State as "electrical trainees" or enrolled in a state-approved apprenticeship program. Apprentices must work under the supervision of Journeymen Electricians to assign work that is appropriated for the apprentice's experience, skill level, and training.

PART 2 – PRODUCTS NOT USED

PART 3 – EXECUTION NOT USED

END OF SECTION 01 35 23

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization	☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 01 45 23 – TESTING AND INSPECTING SERVICES

PART 1 - GENERAL

The following specification outlines the requirements for the LCLS-II-HE EI B950 Mech Room 310 CO2 Chillers, SLAC National Accelerator Laboratory, Menlo Park, California.

1.1 RELATED DOCUMENTS

A. General:

1. The following applies to this section:
    a. General provisions of the Contract, including General and Supplementary Conditions.
    b. The Contract Drawings.
    c. Division 01 Specification Sections and related requirements in other sections of the Project Manual.
    d. Related requirements in other Contract Documents listed in the Agreement.
2. Review these documents for coordination with additional requirements and information that apply to work under this Section.
3. ESH Chapter 14 – Pressure Systems for pressure test requirements.

1.2 QUALITY ASSURANCE

A. General:

1. Work shall be subject to inspection, testing and approval by testing agency, inspector, and Authorities Having Jurisdiction.
2. Approval as result of inspection or testing shall not be construed to be an approval of a violation of provisions of Contract Documents, or by governing codes, laws, ordinances, rules or regulations.
3. Testing, inspections, and approvals presuming to give authority to violate or cancel provisions of Contract Documents, or by governing codes, laws, ordinances, rules or regulations shall not be valid.
4. It shall be the duty of Subcontractor to cause Work to remain accessible and exposed for testing and inspection purposes.
5. It shall be duty of Subcontractor to give timely notification to the FCM who will notify the testing agency, inspector and Authorities Having Jurisdiction when Work is in conformance with Contract Documents and is ready for testing and inspection.
6. Any portion that does not comply shall be corrected and shall not be covered or concealed until authorized by testing agency, inspector and Authorities Having Jurisdiction.

7. Tests, inspections and approvals of portions of Work required by Contract Documents or by codes, laws, ordinances, rules, regulations or orders of Authorities Having Jurisdiction shall be made at an appropriate time.
8. Subcontractor shall give SLAC FCM, testing agency, inspector, Authorities Having Jurisdiction, Commissioning Agent, and Architect, if requested, timely notice of when and where tests and inspections are to be made so that they may be present for such procedures.
9. In event such procedures for testing, inspection and approval reveal portions of Work fail to comply with requirements established by Contract Documents, or by governing codes, laws, ordinances, rules or regulations, all costs made necessary by such failure, including those of repeated procedures and compensation for designer's services and expenses, shall be at Subcontractor's expense.
10. Required certificates of testing, inspection and approval shall, unless otherwise required by Contract Documents, be secured by Subcontractor and promptly delivered to SLAC, SLAC FCM, inspector and Authorities Having Jurisdiction.
11. If SLAC, testing agency inspector, or other party is to observe tests, inspections and approvals required by Contract Documents, or by governing codes, laws, ordinances, rules or regulations or orders of Authorities Having Jurisdiction, they will do so promptly, and where practicable, at normal place of testing.

B. Special Inspectors:
   1. SLAC will employ one or more special inspectors to provide inspections during construction on types of Work required by governing codes.
   2. Special inspections are in addition to inspections by Authorities Having Jurisdiction and the Subcontractor.

C. Test and inspection method standards: See technical sections and governing codes, laws, ordinances, rules, and regulations.

D. Qualifications of Subcontractor's testing agencies:
   1. Meet American Council of Independent Laboratories, "Recommended Requirements of Independent Laboratory Qualification", latest edition.
   2. Meet requirements of ASTM-E329, "Standards of Recommended Practice for Inspection and
   3. Testing Agencies for Concrete, Steel and Bituminous Materials as used in Construction", latest edition.
   4. Satisfy inspection criteria of Construction Materials Reference Laboratory.
   5. See technical sections for additional requirements.

E. Testing equipment calibration shall be by accredited calibration agency, at maximum 12 month intervals, by devices of accuracy traceable to either:
   1. National Institute of Standards and Technology (NIST).

F. Nuclear Moisture/Density Gauge: Shall be authorized for use by SLAC Radiation Protection Department using Radiation Generating Device Authorization Sheet (RGDAS) before any device is allowed on site.

1.3 DESCRIPTION

A. SLAC will arrange and pay for the following QA testing and inspections performed by testing agency or special inspector:
   1. Division 31 - Earthwork: for tests required for earthwork:
      a. Compaction testing, at rate of one test per 2,000 sq. ft. or fraction thereof, for each lift of compacted fill material, but not less than three tests per maximum designed lift.
      b. Certification of asphaltic concrete pavement thickness and in-place density against laboratory standard mix design. Minimum density is 95% of the recorded laboratory mix design density.
      c. Site excavation and rough grading inspection and testing
      d. Soil compaction inspection and testing
      e. Excavation inspection and testing
   2. Division 03 Concrete: for tests required for concrete: Field testing in accordance with ACI 301 and CBC Section 1704. Provide test reports to SLAC.
   3. Division 21 – Fire Suppression: Fireproofing testing and inspection
   4. Division 07 – Thermal and Moisture Protection: Fire barriers and fire barrier penetration sealing systems testing and inspection
   5. All special inspections as required per CBC chapter 17 and all applicable referenced standards.
   6. Specific inspections as directed by SLAC.
   7. Pressure Test requirements in accordance with ESH Chapter 14- Pressure Systems.

B. SLAC will provide copies of the inspections and test results to the Subcontractor.

C. Subcontractor shall arrange and bear all related costs for following tests, inspections and approvals with an independent testing agency or entity acceptable to SLAC:
   1. Rebar locating for core drilling or cutting of concrete.
   2. Testing of manufacturers' products for compliance with specifications.
   3. All other testing and inspections specified.
   4. Testing and inspections of Subcontractor provided shoring or forming.
   5. Any additional inspection and testing required by Authorities Having Jurisdiction. The AHJ in this case is SLAC Building Inspection Office.
   6. Subcontractor's duties for SLAC provided tests, as specified.
   7. Underground utility location survey including ground penetrating radar (GPR) and potholing as needed.

D. Subcontractor shall arrange for, and bear all related costs for following with SLAC provided independent testing agency, or entity acceptable to SLAC:
   1. Re-testing due to failure of initial test or due to nonconformance with Contract Documents.
   2. Re-inspections of Work due to failure of Work to pass initial inspection or due to nonconformance with Contract Documents.

1.4 JOB CONDITIONS

A. Employment of independent testing agency does not relieve obligation of Subcontractor to comply with Contract Documents.

PART 2 – PRODUCTS NOT USED

PART 3 – EXECUTION

3.1 PERFORMANCE

A. Perform indicated inspections, sampling and testing of materials and methods of construction.
B. Use test and inspection or sampling methods or both conforming to methods indicated.
C. Report each test and inspection or sampling or both as indicated.
D. Report results called for by test method, in form specified.
E. Retest failed products and systems.

3.2 REPORTS

A. Submit reports and logs promptly to SLAC, SLAC FCM, Architect, Structural Engineer, Subcontractor, inspector, and Authorities Having Jurisdiction.
B. Subcontractor's Inspectors shall submit a copy of each daily report to the ESH Building Inspection Office, prior to leaving the site, at the time of each site visit.
C. Submit Pressure Test Plans to the Pressure Systems Safety Manager for approval at least 2 weeks prior to performing any pressure tests.
D. Submit Pressure Test Records to the Pressure Systems Safety Manager after performing any pressure tests.
E. Include the following for test or inspection reports or both:
    1. Project name and number.
    2. Project location.
    3. Product and specification section applicable.
    4. Type of test or inspection or both.
    5. Name of testing agency, if used.
    6. Name of testing or inspecting personnel, or both.
    7. Date of test or inspection or both.
    8. Record of field conditions encountered; i.e., temperature, weather.
    9. Test location.
    10. Observations regarding compliance.
    11. Test method used.
    12. Results of test.
    13. Date of report.

14. Signature of testing or inspecting personnel or both.

F. Maintain log of tests which have failed:

1. Type of test or inspection or both.
2. Date of test or inspection or both.
3. Test or inspection number or both.
4. Reason failed.
5. Date of retest or inspection or both.
6. Results of retest. 7. Method of retest.

3.3 INDEPENDENT TESTING AGENCY DUTIES AND LIMITATIONS OF AUTHORITY

A. Cooperate with Architect, SLAC Commissioning Agent, SLAC, and SLAC FCM.
B. Provide qualified personnel promptly on notice.
C. Promptly notify Architect, Commissioning Agent, SLAC, SLAC FCM, and Building Inspection Office and Subcontractor of irregularities, or deficiencies of work which are observed during performance of services.
D. Testing agency is not authorized to:
   1. Release, revoke, alter, or enlarge on requirements of Contract Documents.
   2. Approve or accept any portion of Work.
   3. Perform any duties of Subcontractor.

3.4 SUBCONTRACTOR'S DUTIES

A. Cooperate with testing agency personnel, inspector and Authorities Having Jurisdiction and provide access to work.
B. Provide preliminary representative samples of materials to be tested, in required quantities.
C. Furnish copies of mill test reports.
D. Furnish labor and facilities:
   1. To provide access to work to be tested.
   2. To obtain and handle samples at site.
   3. To facilitate inspections and tests.
   4. Storage and curing facilities for testing agency's exclusive use.
E. Notify appropriate testing agency, inspector or Authorities Having Jurisdiction sufficiently in advance of testing and inspection

END OF SECTION 01 45 23

SLAC BUILDING INSPECTION OFFICE
Construction Authorization (IFC)
BY: cnadler
DATE: 2/18/2026 PRS#: 26-009
Revised Authorization ☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.
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SECTION 01 74 19 – CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 - GENERAL

1.1 RELATED DOCUMENT

A. General:

1. The following applies to this section:

   The terms of the Subcontract, including General provisions and General and Supplementary Conditions.

   Related requirements in other Subcontract Documents including, but not necessarily limited to, the Contract Drawings and the Specification Sections and related requirements in other sections of the Project Manual.

B. Related Sections:

1. Section 01 35 23 – OWNER SAFETY REQUIREMENTS
2. ES&H web link: https://esh.slac.stanford.edu/eshmanual (SLAC ES&H Manual)
3. Section 01 78 36 – WARRANTIES
4. Section 01 91 13 – GENERAL COMMISSIONING REQUIREMENTS

1.2 SUMMARY

A. All waste, including but not limited to hazardous and non-hazardous construction and demolition debris and land-clearing debris, shall be minimized and managed in accordance with applicable federal, state, and local laws and regulations.
B. Subcontractor shall implement procedures to minimize the creation of waste on the job site.
C. This section includes, but is not limited to, the requirements for achieving a minimum salvage/recycling rate of 75% for non-hazardous wastes, development and implementation of a Waste Management Plan for non-hazardous wastes, and transportation and disposal requirements for hazardous, Class II universal, and nonhazardous wastes.

1.3 DEFINITIONS

A. Class II: Also called “designated waste”, is waste that exhibits a low level of contamination not classified as hazardous but is required by the State to be managed more strictly than other kinds of non-hazardous wastes. Examples are soil/sediment with contaminated petroleum hydrocarbons, metals, other organics/inorganics, drilling sludge, other cementitious materials).
B. Clean: not contaminated with oils, solvents, lead, PCBs, asbestos or other hazardous materials.
C. Construction and Demolition Debris: non-hazardous wastes including, but not limited to, building materials, packaging materials, debris, and trash resulting from construction and demolition operations.
D. Disposal: Removal off-site of demolition and construction waste and subsequent sale, recycling, reuse, or deposit in landfill acceptable to authorities having jurisdiction.

E. Hazardous waste: Also referred to as “Class I”, waste that is defined as hazardous by either the Federal (EPA) or State of California regulations. Hazardous waste is highly regulated and requires special handling, transportation, and disposal.
F. Land-clearing Debris: includes clean soil, vegetation, and rocks.
G. Landfill: Land used as a site for the disposal of trash by burial.
H. Non-hazardous waste: Waste that is not classified as hazardous, Class II, or universal waste. The term waste includes salvageable, returnable, recyclable and reusable construction materials that would otherwise be discarded or destroyed.
I. Recycle: Recovery of demolition or construction debris and other wastes with subsequent processing in preparation for reuse. Recycling does not include burning, incinerating, or thermally destroying waste.
J. Salvage and Reuse: Recovery of demolition or construction waste and subsequent incorporation into the Work.
K. Salvage: Recovery of demolition or construction debris and other wastes with subsequent sale or reuse in another facility.
L. Source Separated: Materials that are separated on-site by category.
M. Trash: Non-hazardous waste or “garbage”, acceptable for disposal at a municipal solid waste landfill.
N. Universal waste: Hazardous waste that can be collected under streamlined collection standards for universal wastes as defined by the U.S. Environmental Protection Agency EPA or the State. Examples include spent batteries, mercury containing equipment (vacuum gauges, flow regulators, thermostats), electric lamps (fluorescent bulbs, HIDs, sodium vapor lamps), aerosol cans (empty and non-empty) and electronic waste, commonly abbreviated as “e-waste”. E-waste includes electronic discards that are regulated by either federal or state hazardous waste laws, or both. Examples of e-wastes include but are not limited to televisions/monitors, computers, keyboards, cell phones, circuit boards, and microwave ovens.
O. Waste: Any material that has reached the end of its intended use. Waste includes salvageable, returnable, recyclable, and reusable construction materials that would otherwise be discarded or destroyed.

1.4 HAZARDOUS WASTE, CLASS II, AND UNIVERSAL WASTE

A. General

1. Wastes falling in this category may include, but not be limited to the following:

a. Asbestos abatement waste (tiles, mastic, cables, wallboard, wastewater)
b. Materials Contaminated with polychlorinated biphenyls (PCBs)Soil, asphalt, or baserock classified Class I or Class II by SLAC
Oily or solvent absorbent/debris from spills
Spent aerosol cans
Adhesives, epoxy, paint, grease, oil, lubricants, cutting fluid, cleaning solvents, or petroleum-based fuels, and any rag with these hazardous materials on them
Light ballasts, (PCBs or non-PCBs, small or large capacitors (PCBs or non-PCBs)
Wastewater from flushing operations
Lead paint chips (flaking)
Beryllium-contaminated debris or equipment
Empty containers last containing certain hazardous materials

Mercury switches, batteries, and bulbs
Treated wood
Electronic waste or “e-waste”
Sludge removed from sanitary sewer lines
Sediment removed from storm drain lines
Other hazardous, Class II, or universal wastes

2. Hazardous waste, Class II, and universal waste must be handled, separated, stored, and packaged for shipment in accordance with federal, state, and local laws and regulations.
3. Transportation and disposal (T&D) for hazardous waste, Class II, and universal waste are detailed in Part 3 –EXECUTION of this section.
4. As part of Decommissioning and Demolition work, equipment that has or may have stored, used, dispensed or conveyed hazardous materials may require decontamination prior to recycling or disposal.
5. The SLAC FCM shall identify any items including equipment removed during Decommissioning and Demolition work that the subcontractor shall turnover to SLAC Facilities HVAC or SLAC Salvage for reuse or recycling. Equipment and materials sent to SLAC Salvage must not contain hazardous waste or hazardous materials.

1.5 NON-HAZARDOUS RECYCLABLES AND WASTE

A. General and Performance Goals

1. End-of-project salvage/recycling rates for the non-hazardous waste generated by the Work shall be a minimum of 75 percent by weight of total, with a goal of 95 percent or more. Although excavated soil and land-clearing debris shall be minimized, it may not contribute to the final 75% salvage/recycling rate calculation.
2. Wastes falling in this category may include, but not be limited to the following material categories:
   a. Concrete, Masonry, and Other Inert Fill Material: concrete, brick, rock, broken-up asphalt pavement, clay, and other inert (nonorganic) materials. Materials resulting from excavation and land-clearing require clearance by SLAC, per requirements of Section 01 35 43 – ENVIRONMENTAL PROTECTION PROCEDURES
   b. Metals: metal scrap including iron, steel, copper, brass, and aluminum; includes beverage containers, packaging materials (such as metal banding), fencing, reinforcing bar, wiring, plumbing, etc. Metals may require clearance by the SLAC Radiation Protection Department prior to removal, depending on the location of the project; coordinate with SLAC FCM.
   c. Untreated Scrap Wood: unpainted, untreated dimensional lumber, wood edging, wood shipping pallets, etc. [Note: treated wood waste is categorized as hazardous].
   d. Engineered Wood Products: plywood, oriented strand board, Masonite, particleboard, manufactured trusses and beams, and glue-laminated timbers.
   e. Gypsum Wallboard: excess drywall construction materials including cuttings, other scrap, and excess materials.
   f. Cardboard: clean, corrugated cardboard such as used for packaging, etc.
   g. Paper Goods:

h. Office paper: includes any paper, such as manufacturer instructions, specification sheets, files, correspondence, packaging, stiffeners, etc.
i. Newsprint: shredded or whole newspaper goods.
j. Plastic: beverage containers, recyclable plastic packaging materials, containers (other than those used for hazardous materials), vinyl products, etc.
k. Glass: includes glass beverage containers, and recyclable glass building materials.
l. Insulation: rigid foam, batt, and loose-fill insulation materials.
m. Carpet: face fiber, backing, padding, and carpet cushion scrap.
n. Fabric: uncontaminated fabric scraps.
o. Rubber: uncontaminated rubber scraps, including but not limited to recycled-content rubber flooring, rubber edging, tires that are no longer serviceable, etc.
p. Greenwaste: organic landscape materials such as tree branches and vegetation.
q. Other: any additional non-hazardous construction and demolition debris identified that may be salvageable, reusable, or recyclable.

3. Non-hazardous waste must be handled, separated, stored, and packaged for shipment in accordance with federal, state, and local laws and regulations.
4. Use only Disposal Sites, Recyclers, and Waste Materials Processors properly permitted by state and local authorities.
5. The SLAC FCM shall identify any items including equipment removed during Decommissioning and Demolition work or certain metals that the Subcontractor shall turnover to SLAC \ SLAC Salvage for reuse or recycling. Equipment and materials sent to SLAC Salvage must not contain hazardous waste or hazardous materials.
6. Materials sent to SLAC Salvage must be identified and tracked as part of Waste Management Plan, Waste Reduction Reports, and performance requirements.

B. Waste Management Plan
1. Waste Management Plan for Non-Hazardous Waste
2. Develop Waste Management Plan for non-hazardous waste that results in end-of-Project rates for salvage/recycling of a minimum of 75 percent by weight of total non-hazardous waste generated by the Work, with a goal of 95 percent or more. Although excavated soil and land clearing debris shall be minimized and reported, it may not contribute to the final 75% salvage/recycling rate calculation.
3. The Waste Management Plan shall be delivered as a project submittal for SLAC approval and shall include at a minimum the following information:
a. Estimated quantities of non-hazardous construction and demolition debris to be diverted from landfills (i.e., salvaged or recycled) and disposed, in tons by material category, excluding excavated soil and land-clearing debris.
b. Estimated quantity of excavated non-hazardous soil and land-clearing debris salvaged, recycled, disposed, in tons for each material category.
c. Total estimated quantity of waste to be diverted from landfills (salvaged plus recycled) as a percentage of total waste, excluding excavated soil and land-clearing debris.
d. Describe waste diversion strategy, including how materials will be separated, the sizes of containers, container labeling, location

e. on Project site where materials separation and staging will occur, and employee training on the Waste Management Plan.
f. List local receivers and processors of salvaged/recycled/disposed materials. Include names, addresses, and telephone numbers.

4. Distribute Waste Management Plan to all necessary parties within seven days of submittal approval.
5. Distribute Waste Management Plan to entities when they first begin work on site. Review plan procedures and locations established for salvage, recycling, and disposal.

PART 2 - PRODUCTS NOT USED

PART 3 - EXECUTION

3.1 GENERAL

A. Waste Management Conference: The Construction Manager and/or Subcontractor shall conduct a conference at Project site with applicable SLAC department representatives to review methods and procedures related to waste management.
B. Training: The Construction Manager, with resources and support from Waste Management, trains workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work occurring at Project site.
    1. Provide on-site instruction on appropriate separation, handling, recycling, and salvaging methods to be used by all parties at the appropriate stages of the work at the site.
C. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.
D. Storage and Handling
    1. SLAC FCM shall: Designate and label specific areas on Project site necessary for storing and separating materials. Do not store within drip line of remaining trees.
    2. Keep hazardous waste, Class II, universal, and non-hazardous wastes and recyclables segregated and stored in accordance with local, state, and federal regulations. SLAC will provide separate collection containers to prevent contamination of materials, including protection from rain as applicable. Define specific areas to facilitate separation of materials.
    3. Transport recyclable waste materials from the Work Area to the recycle containers daily and carefully deposit them in the containers to minimize noise and dust. Close the container covers immediately after materials are deposited. Do not place recyclables and waste materials on the ground adjacent to containers. Stockpiled soils must be covered.
    4. Request replacement containers as demand requires.
    5. Deposit all indicated recyclable materials in the containers in a clean (no mud, adhesives, solvents, petroleum, PCB, lead contamination), debris-free condition. Do not deposit contaminated materials into the containers until such time as such materials have been decontaminated.

6. If contamination chemically combines with the material so that it cannot be cleaned, do not deposit into the recycle containers.
7. Comply with hauling and disposal regulations of authorities having jurisdiction.
8. Unused fertilizers shall not be co-mingled with construction waste.
9. Comply with environmental controls.

3.2 TRANSPORTATION AND DISPOSAL OF HAZARDOUS WASTE, CLASS II, AND UNIVERSAL WASTE

A. SLAC FCM is responsible for requesting hazardous waste and Class II waste containers from SLAC WM at least 7 days in advance using the Hazardous Waste Pick-Up and Empty Container Request form. Form should include a clear description of wastes and hazards to assist SLAC Waste Management with waste characterization. Hazards can be confirmed with coordination with SLAC Industrial Health (IH), SLAC Environmental Protection (EP), and/or SLAC Radiation Protection (RP) groups. SLAC FCM is responsible for inspecting containers weekly using SLAC WM-provided template and managing the project site to ensure containers are compliant and accessible when service is scheduled. When service is needed, SLAC FCM is required to provide ample time (minimum 7 days) for SLAC WM to schedule equipment and pick-up, and coordinate with SLAC RP as needed to confirm pick-up.
B. SLAC FCM is responsible for requesting containers for metals recycling and e-waste recycling through SLAC Salvage.
C. At the discretion of SLAC, the Waste Subcontractor shall perform the transportation and disposal (T&D) for hazardous waste, Class II, and universal waste (not including electronic waste, see below) for the Project. The waste shall be transported by a licensed, registered, and certified hazardous waste transporter. (Certified hazardous waste transporters will not be required for transport of Class II waste). SLAC's Waste Management (WM) Group must review, approve and sign all associated shipping documentation (waste profile, land disposal restrictions (LDR), hazardous waste manifests, Class II non-hazardous manifests).
 1. At the discretion of SLAC, Subcontractor shall supply appropriately labeled hazardous waste containers (drums, roll-off bins, end dumps, etc.). SLAC WM may inspect trucks prior to loading. Trucks not passing the SLAC Vehicle Inspection will be rejected at the Subcontractor's expense. The Subcontractor shall ensure that the handling on-site and transportation of all wastes comply with all federal, state, and local laws and regulations. The Subcontractor must deliver the entire quantity of waste that the Subcontractor accepted from SLAC to the designated facility listed on the manifest. All hazardous wastes and Class II wastes must be disposed at SLAC-approved Treatment, Storage, and Disposal Facilities (TSDFs). Ensure loads are secure and securely covered prior to departing site.
 2. Electronic Waste: The Subcontractor is responsible for the T&D of electronic wastes, including providing appropriate containers and packaging and all regulatory required paperwork, with the exception of items designated by the SLAC FCM to be transferred to SLAC Salvage. Electronic wastes must be recycled at a legally permitted facility that is certified to manage electronic waste through either the E-Stewards or R2 standards. Documentation of shipments shall be retained by the Subcontractor and made available to SLAC upon request. Ensure loads are secure prior to departing site.

3.3 TRANSPORTATION AND DISPOSAL OF NON-HAZARDOUS RECYCLABLES AND WASTE

A. General: Implement Waste Management Plan as approved by SLAC. Provide handling, containers, storage, signage, transportation, and other items as required to implement the Waste Management Plan for non-hazardous recyclables and waste during the entire duration of the Contract. Comply with the following procedures:

1. Recycle and waste bin areas are to be maintained in an orderly manner and clearly marked to avoid contamination of materials. Inspect containers and bins weekly for contamination and remove contaminated materials if found.
2. Do not mix recyclable materials unless co-mingled recycling is planned.
3. Store components off the ground and protect them from weather.
4. Arrange for timely pickups from the Site or deliveries to approved recycling facilities of designated waste materials to keep construction site clear and prevent contamination of recyclable materials.
5. Remove trash off Owner’s property and transport in a manner that will prevent spillage on adjacent surfaces and areas and legally dispose of them at a waste facility acceptable to authorities having jurisdiction. Do not allow trash materials to accumulate on-site or be disposed of in SLAC trash dumpsters.
6. Refer to other related sections of the specifications for management of excavated materials and soil.
7. Keep and maintain records of all material loads delivered or picked up to send to salvage (including equipment or materials to SLAC Salvage), recycling, and waste facilities. Include at a minimum, time/date, tonnage of material by material category, and receipts/tags by the receiving entity.
8. Ensure loads are covered and secured prior to departing Site.

3.4 CONSTRUCTION WASTE MANAGEMENT SUBMITTALS

A. HAZARDOUS WASTE, CLASS II, AND UNIVERSAL WASTE

1. Provide T&D paperwork as specified in Section 3.2 of this section.

B. NON-HAZARDOUS RECYCLABLES AND WASTE

1. Waste Management Plan: The Waste Management Plan shall be delivered as a project submittal for SLAC approval within 14 days of the Notice to Proceed and shall include information specified in Part 1 of this section.
2. Waste Reduction Report:

   a. Prepare a Waste Reduction Report, demonstrating conformance to the End-of-Project salvage/recycling rates specified in this section. The report shall include but not be limited to:

      i. Breakdown of quantities of non-hazardous construction and demolition debris diverted from landfills (salvaged, recycled) and disposed, in tons by material category, excluding excavated soil and land clearing debris, with corresponding shipping date and load number if applicable. If materials were salvaged through SLAC Salvage, the quantities should still be accounted for in this report.

ii. Quantity of excavated non-hazardous soil and land clearing debris salvaged, recycled, disposed, in tons for each material category, with corresponding shipping date and load number if applicable.

iii. Total estimated quantity of waste diverted from landfills (salvaged plus recycled) as a percentage of total waste, excluding excavated soil and land-clearing debris.

iv. Indicate receivers and processors of salvaged/recycled/disposed materials and include tags/receipts for the materials as supporting documentation.

a) Provide a monthly Waste Reduction Progress Report as part of Application of Payment. Include the data as specified above.

b) Provide by Oct. 15 of each year of Work, a Waste Reduction Progress Report covering the project period for the fiscal year, (i.e., Oct. 1 through Sept. 30) and submit to the SLAC Environmental Protection Department for use in annual reporting to DOE. Include the data as specified above.

c) Submit the final Waste Reduction Report at completion of waste generating activities. Include the data as specified above. Before request for Substantial Completion, submit calculated end-of-Project rates for salvage, recycling, and disposal as a percentage of total non-hazardous waste generated by the Work.

END OF SECTION 01 74 19

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization	☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 01 78 36 – WARRANTIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. General

1. The following applies to this section:

a. The Subcontract General Terms and Conditions for Fixed Price Construction Subcontracts.

b. Related requirements in other Subcontract Documents including, but not necessarily limited to, the Drawings and the Specification Sections in the Project Manual.

1.2 FORM OF SUBMITTALS

A. Bind in commercial quality 8-1/2 x 11 inches, three side-ring type "D" binders with durable plastic cover.

B. Cover: Identify each binder with typed or printed title WARRANTIES. Identify with title of project; name, address, and telephone number of Subcontractor and equipment supplier; and name of responsible company principal.

C. Table of Contents: Neatly typed, in the sequence of the table of contents of the project manual with each item identified with the number and title of the specification section in which specified and the name of product or work item. The table of contents or index should also include the expiration date and POC for quick reference

D. Separate each warranty with index tab sheets keyed to the table of contents listing. Provide full information using separate typed sheets as necessary. List Subcontractor, supplier, and manufacturer with name, address, and telephone number of responsible company principal.

1.3 PREPARATION OF SUBMITTALS

A. Obtain warranties, guarantees, bonds, and service and maintenance contracts executed in quadruplicate by responsible Subcontractors, suppliers, and manufacturers, within 14 calendar days after acceptance of the applicable item of work. Except for items put into use with the University's permission, leave date of beginning of time of warranty until the date of substantial completion is determined.

B. Provide a copy of each warranty/guarantee and service contract issued. Include an information sheet for SLAC personnel giving:

1. Proper procedures in the event of failure
2. Required maintenance to maintain contracts
3. Instances which might affect the validity of contracts

C. Verify that documents are in proper form, comply with subcontract documents, contain full information, and are notarized.
D. Co-execute submittals when required.
E. Retain warranties until time specified for submittal.
F. Submit two (2) original, signed copies, each, and an electronic copy. Electronic copy shall contain the project name, the building or structure name, the subject of the warranty, and the date of warranty expiration in the file title.

1.4 TIME OF SUBMITTALS

A. For equipment or component parts of equipment put into service during construction with the University's permission, submit documents within 14 calendar days after acceptance of said equipment by the University.
B. Make other submittals within 14 calendar days after date of substantial completion, prior to final application for payment.
C. For items of work for which acceptance is delayed beyond date of substantial completion, submit within 14 calendar days after acceptance, listing the date of acceptance as the beginning of the warranty period.

1.5 SPECIFIC WARRANTIES

A. Specific warranties of construction items, called out in individual Sections of Divisions 2 through 33 shall take precedence over the warranties stated above.

END OF SECTION 01 78 36

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization	☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 01 91 13 - GENERAL COMMISSIONING REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

B. Basis of Design Document (BOD) is included by reference for information only.

1.2 SUMMARY

A. Section Includes:

1. General requirements for coordinating and scheduling commissioning activities.
2. Commissioning meetings.
3. Commissioning reports.
4. Use of commissioning process test equipment, instrumentation, and tools.
5. Construction checklists, including, but not limited to, installation checks, startup, performance tests, and performance test demonstration.
6. Commissioning tests and commissioning test demonstration.
7. Adjusting, verifying, and documenting identified systems and assemblies.

As applied to the commissioning requirements for mechanical (including building automation system), electrical, plumbing (MEPCx), low voltage systems (LVCx), security systems, renewable energy, and irrigation controls.

B. Commissioning is a systematic process of verifying that the building systems perform interactively according to the construction documents and the Owner’s operational needs. The commissioning process shall encompass and coordinate the system documentation, equipment startup, control system calibration, testing and balancing, performance testing and training. Commissioning during the construction and post-occupancy phases is intended to achieve the following specific objectives according to the contract documents:

1. Verify that the applicable equipment and systems are installed in accordance with the contact documents and according to the manufacturer's recommendations.
2. Verify and document proper integrated performance of equipment and systems.
3. Verify that Operations & Maintenance documentation is complete.
4. Verify that all components requiring servicing can be accessed, serviced and removed without disturbing nearby components including ducts, piping, cabling or wiring.
5. Verify that the Owner’s operating personnel are adequately trained to enable them to operate, monitor, adjust, maintain, and repair building systems in an effective and energy efficient manner.
6. Document the successful achievement of the commissioning objectives listed above.

C. Various sections of the project specifications require equipment startup, testing, and adjusting services. Requirements for startup, testing, and adjusting services specified in the technical sections of these specifications are intended to be provided in coordination with the commissioning services and are not intended to duplicate services. The Contractor shall coordinate the work required by individual specification sections with the commissioning services requirements specified herein.

D. The commissioning process does not take away from or reduce the responsibility of the DB Contractor to provide a finished and fully functioning product.

E. Related Requirements:

1. SECTION 01 00 00 – GENERAL REQUIREMENTS
2. SECTION 01 35 23 – OWNER SAFETY REQUIREMENTS
3. SECTION 01 74 19 – CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
4. SECTION 01 78 36 – WARRANTIES
5. SECTION 01 91 13 – GENERAL COMMISSIONING REQUIREMENTS
6. DIVISION 26: Electrical – Commissioning Requirements Apply to the System(s) Shown in Section 3.8 COMMISSIONED SYSTEMS, of this document.

*Specifications not provided, to be included within Design-Builder’s specifications.

F. References

1. ASHRAE Standard 202-2018 Commissioning process for Building and Systems.
2. ASHRAE Guideline 0-2019 The Commissioning Process.
3. ASHRAE Guideline 1.1-2007, HVAC Commissioning Guidelines.
4. ASHRAE Guideline 1.4-2014, Procedures for Preparing FSM.
5. 2022 California Green building Standards Code.
6. LEED™, New Construction Reference Guide.

1.3 DEFINITIONS

A. Acceptance Criteria: Threshold of acceptable work quality or performance specified for a commissioning activity, including, but not limited to, construction checklists, performance tests, performance test demonstrations, commissioning tests, and commissioning test demonstrations.

B. Acceptance Phase: Phase of construction after startup and initial checkout when functional performance tests, O&M documentation review and training occurs.

C. Approval: Acceptance that a piece of equipment or system has been properly installed and is functioning in the tested modes according to the Contract Documents.

D. Basis of Design (BoD) document: A document that records concepts, calculations, decisions, and product selections used to meet the OPR and to satisfy applicable regulatory requirements, standards, and guidelines. The document includes both narrative descriptions and lists of individual items that support the design process.

E. Building Automation System (BAS): The central building control system and energy management system. Also referred to as a Controls System.

F. Building Management System (BMS): In general practice, this is just a different name for the BAS.

G. Commissioning (Cx): - A quality-focused process for enhancing the delivery of a project. The process focuses upon verifying and documenting that the facility and all of its systems and assemblies are planned, designed, installed, tested, operated and maintained to meet the Owner's Project Requirements.

H. Commissioning Activity Schedule: A commissioning schedule designed to provide team members with a descriptive overview of commissioning activities as they relate to parallel construction activities regardless of changes to the construction schedule. (See Exhibit B this specification and Cx Plan)

I. Commissioning Authority (CxA): An entity identified by the Owner who leads, plans, schedules, and coordinates the commissioning team to implement the Commissioning Process. This role is also responsible for commissioning building enclosure systems. The CxA reports directly to the Owner without assuming oversight responsibilities.

J. Commissioning Coordinator (CxC): A person or entity employed by the Contractor to manage, schedule, and coordinate the commissioning process.

K. Commissioning Plan (Cx Plan): A document that outlines the organization, schedule, allocation of resources, and documentation requirements of the Commissioning Process.

L. Field Construction Manager (FCM): Owner's on-site representative in day-to-day activities, including supervision and management of construction activities.

M. Contract Documents: They include a wide range of documents that will vary from project to project and with the Owner's needs and regulations, laws, and countries, construction management process, subcontractor agreements or requirements, requirements and procedures for submittals, changes, and other construction requirements, timeline for completion, and other Construction Documents.

N. Control System: A component of environmental, HVAC, security and fire systems for reporting, monitoring, and issuing of commands.

O. Data Logging: Monitoring flows, currents, status, pressures, etc. of equipment using standalone data loggers separate from the control system.

P. DDC: Acronym for Direct Digital Controls. This is the most typical controls platform for BAS' / BMS'.

Q. Deferred Functional Tests: Functional Tests that are performed later, after substantial completion, equipment, seasonal requirements, design, or other site conditions that disallow tests from being performed.

R. Deficiency or Commissioning Issue: A condition identified by the Commissioning Agent or other member of the Commissioning Team that adversely affects the commission ability, operability, maintainability, or functionality of a system, equipment, or component. A condition that is in conflict with the Contract Documents and/or performance requirements of the installed systems and components.

S. Design Builder (DB): Team under contract with the Owner which serves as the Subcontractor and Architect/Engineer (A/E) for the design and construction of the project. As it pertains to commissioning, the A/E within the DB team is the design professional responsible for the design of the portion of the project being commissioned.

T. Factory Testing: Testing of equipment on-site or at the factory, by factory personnel with District's representative present.

U. Facilities Manager (FM): The Owner representative responsible for the operation and maintenance of the physical facilities and grounds.

V. Functional Performance Test (FPT): Test of the dynamic function and operation of equipment and systems using, simulations, manual (direct observation) and/or monitoring methods. Systems are tested under various modes. The systems are run through all the control system's sequences of operation and components are verified to be responding as intended per the specified sequence of operations. FPTs are performed after pre-functional checklists and startup is complete and are performed by the contractor and witnessed by the CxA.

W. Installation Verification: Observations or inspections that confirm the system or component has been installed in accordance with the contract documents and to industry accepted best practices.

X. Integrated System Testing: Integrated Systems Testing procedures entail testing of multiple integrated system's performance to verify proper functional interface between systems. Typical Integrated Systems Testing includes verifying that building systems respond properly to loss of utility, transfer to emergency power sources, re-transfer from emergency power source to normal utility source; interface between HVAC controls and Fire Alarm systems for equipment shutdown, interface between Fire Alarm system and elevator control systems for elevator recall and shutdown; interface between Fire Alarm System and Security Access Control Systems to control access to spaces during fire alarm conditions; and other similar tests as determined for each specific project.

Y. Issues Log: A formal and ongoing record of problems or concerns and their resolution that have been raised by members of the commissioning team during the course of the commissioning process. Maintained by the CxA.

Z. Monitoring: The recording of equipment operation parameters (flow, current, status, pressure, etc.) using data loggers or the trending capabilities of control systems.

AA. Owner's Project Requirements (OPR): A collection of documents that details the functional requirements of Project and expectations of how it will be used and operated. This document includes Project and design goals, measurable performance criteria, budgets, schedules, success criteria, and supporting information. Reference the SOW and Design Narrative Guidelines.

BB. Owner: Project Owner or designated representative.

CC. Owner-Contracted Tests: Tests paid for by the Owner outside the Contract and for which the CxA does not oversee. These tests will not be repeated during functional tests if properly documented.

DD. Overwritten Value: Writing over a sensor value in the control system to see the response of a system (e.g., changing the outside air temperature value from 50ºF to 75ºF to verify economizer operation). See also “Simulated Signal.”

EE. Phased Commissioning: Commissioning that is completed in phases (by floors, for example) due to the size of the structure or other scheduling issues, in order minimize the total construction time.

FF. Pre-functional Checklists (PFC): Checklists prepared by the CxA, in conjunction with the Subs, and provided to the contractor to document the complete installation of equipment or systems. These checklists are essentially elementary component tests to verify proper installation of equipment and are primarily static inspections and procedures to prepare the equipment or system for initial operation (e.g., belt tension, oil levels OK, labels affixed, gages in place, sensors calibrated, amp readings, etc.) Pre-functional checklists are completed by the contractors prior to or along with the manufacturer’s start-up procedure and checklist.

GG. Pre-Functional Test (PFT): An inspection or test that is done before functional testing. PFT’s include installation verification and system and component start up tests.

HH. Sampling: Functionally testing only a fraction of the total number of identical or near identical pieces of equipment.

II. Seasonal Performance Tests: Functional Tests that are deferred until the system(s) will experience conditions closer to their design conditions.

JJ. Site Observation Visit: On-site inspections and observations made by the Commissioning Agent for the purpose of verifying component, equipment, and system installation, to observe contractor testing, equipment start-up procedures, or other purposes.

KK. Start-up: The initial starting or activating of dynamic equipment or the initial energization and programming of control systems.

LL. Systems, Subsystems, and Equipment: Where these terms are used together or separately, they shall mean "as-built" systems, subsystems, and equipment.

MM. Test, Adjust, Balance (TAB): A systematic process or service applied to heating, ventilating and air-conditioning (HVAC) systems and other environmental systems to achieve and document air and hydronic flow rates. The standards and procedures for providing these services are referred to as “Testing, Adjusting, and Balancing” and are described in the Procedural Standards for the Testing, Adjusting and Balancing of Environmental Systems, published by NEBB or AABC.

NN. Test Procedures: Written details developed by the CxA, and included in the FPTs, that details the expectations for tests conducted on components, equipment, assemblies, systems, and interfaces among systems.

OO. Thermal Scans: Thermographic pictures taken with an Infrared Thermographic Camera. Thermographic pictures show the relative temperatures of objects and surfaces and are used to identify leaks, thermal bridging, thermal intrusion, electrical overload conditions, moisture containment, and insulation failure.

PP. Trends / Trending: Recordings of control point value history or monitoring value history over time. Trends are automatically recorded by the building automation system or other electronic data gathering equipment. Trend data gathered over a period of time is often used to analyze and verify proper performance of equipment, systems, or sequence of operations.

QQ. Training Plan: A written document that details the expectations, schedule and deliverables of commissioning process activities related to training of project operating and maintenance personnel, users and occupants.

RR. Trending: The monitoring by a building management system or other electronic data gathering equipment and analyzing of the data gathered over a period of time to verify proper equipment or systems sequence of operations.

SS. Warranty Phase Commissioning: Commissioning efforts executed after a project has been completed and accepted by the Owner. Warranty Phase Commissioning includes follow-up on verification of system performance, measurement and verification tasks and assistance in identifying warranty issues and enforcing warranty provisions of the construction contract.

TT. Warranty Visit: A commissioning meeting and site review where all outstanding warranty issues and deferred testing is reviewed and discussed.

1.4 COMPENSATION

A. If the Commissioning Authority is required to perform additional services or incur additional expenses due to actions of the DB Contractor as listed below the DB Contractor agrees to compensate the Owner for such additional services and expenses.

1. Failure to provide timely notice of commissioning activities schedule changes.
2. Failure to meet acceptance criteria for test demonstrations.
3. Additional work required by the CxA to coordinate and correct commissioning issues.
4. Additional work required to coordinate and repeat tests when equipment and systems fail initial acceptance criteria testing.
5. See 3.11 Exhibit A: Commissioning Readiness Letter (CxRL).

B. Contractor shall compensate Owner for such additional services and expenses at the rate of $250.00 per labor hour, plus travel and per diem allowances for meals and lodging according to current U.S. General Services Administration (GSA) Per Diem Rates.

1.5 COMMISSIONING TEAM INTERACTION & PROCESS

A. A project team will be created to coordinate the commissioning effort. This team will coordinate and communicate with the rest of the project team, attend meetings, and solve problems. This team includes representatives from the DB contractor, subcontractors, and owner.

B. The Design Builder shall in addition to their representative also appoint a representative from each subcontractor involved in commissioned systems including mechanical, electrical, controls, test adjust balance, plumbing, and low voltage systems.

C. With these fundamental practices in mind, the commissioning process described herein has been developed to recognize that, in the execution of the Commissioning Process, the Commissioning Agent must develop effective methods to communicate with every member of the construction team involved in delivering commissioned systems while simultaneously respecting the exclusive contract authority of the Project Manager (PM). Thus, the procedures outlined in this specification must be executed within the following limitations:

1. No communications (verbal or written) from the Commissioning Agent shall be deemed to constitute direction that modifies the terms of any contract between the Owner and the Contractor.
2. Commissioning Issues identified by the Commissioning Agent will be delivered to the Construction Manager and copied to the designated Commissioning Representatives for the Contractor and subcontractors on the Commissioning Team for information only in order to expedite the communication process. These issues must be understood as the professional opinion of the Commissioning Agent and as suggestions for resolution.
3. In the event that any Commissioning Issues and suggested resolutions are deemed by the Construction Manager to require either an official interpretation of the construction documents or require a modification of the contract documents, the Construction Manager will issue an official directive to this effect.
4. All parties to the Commissioning Process shall be individually responsible for alerting the Construction Manager of any issues that they deem to constitute a potential contract change prior to acting on these issues.
5. Authority for resolution or modification of design and construction issues rests solely with the Construction Manager, with appropriate technical guidance from the Architect/Engineer and/or Commissioning Agent.

1.6 OWNER'S RESPONSIBILITIES

A. Participate in resolution of issues that may occur as a result of the commissioning process.

B. Assign operation and maintenance personnel and schedule them to participate in commissioning team activities including, but not limited to, the following:

1. Coordination meetings.
2. Training in operation and maintenance of systems, subsystems, and equipment.
3. Testing meetings.
4. Demonstration of operation of systems, subsystems, and equipment.

5. Provide feedback to Cx for the project warranty phase testing and review, as well as access to the site and controls system during the warranty test.

1.7 COMMISSIONING COORDINATION RESPONSIBILITIES

A. Design Builder's and Subcontractor’s Responsibilities
   1. Provide utility services required for the commissioning process.
   2. DB is responsible for construction means, methods, job safety, or management function related to commissioning on the job site.
   3. DB shall assign representatives with expertise and authority to act on behalf of the DB and schedule them to participate in and perform commissioning team activities including, but not limited to, the following:
      a. Participate in construction-phase commissioning meetings including controls coordination meeting to review and resolve any issues with the sequence of operations.
      b. Participate in maintenance orientation and inspection.
      c. Participate in operation and maintenance training sessions.
      d. Certify that Work is complete and systems are operational according to the Contract Documents, including calibration of instrumentation and controls.
      e. Perform quality control of all work and certify it is complete prior to request for inspection.
      f. Evaluate performance deficiencies identified in test reports and, in collaboration with entity responsible for system and equipment installation, recommend corrective action.
   4. Contractor shall integrate all commissioning activities into Contractor’s master construction schedule.
   5. Contractor shall provide a means to effectively commission the BMS system including the following at minimum:
      a. Schedule the controls contractor that was an integral part of programming the building BMS to run the tests
      b. Provide a table with chairs
      c. Provide a 17” 1080p monitor with 10' cables for connection to the controls contractor’s laptop.
   6. Subcontractors shall assign representatives with expertise and authority to act on behalf of subcontractors and schedule them to participate in and perform commissioning team activities including, but not limited to, the following:
      a. Participate in construction-phase coordination meetings.
      b. Participate in maintenance orientation and inspection.
      c. Complete pre-functional checklists for all equipment. Submit completed forms with start-up reports immediately after start up.
      d. Schedule and perform duct air leakage testing as specified in the technical specification sections. Make testing available for CxA to witness, and provide testing reports to CxA.
      e. Provide flushing plans, disinfection reports and water treatment reports to the CxA for review.
      f. Participate in pre-TAB meeting and jobsite inspections to verify TAB readiness.

g. Provide draft completed TAB report to CxA for review. CxA will identify up to 20% of TAB report for TAB contractor to demonstrate compliance to the completed TAB report.
h. Participate in procedures meeting for testing.
i. Perform point-to-point, calibration and checkout of the building automation system and provide completed report to the CxA for review.
j. Participate in final review at acceptance meeting.
k. Provide schedule for operation and maintenance data submittals, equipment startup, and testing to CxA for incorporation into the commissioning plan. Update schedule on a weekly basis throughout the construction period.
l. Provide information to the CxA for developing construction-phase commissioning plan.
m. Participate in training sessions for operation and maintenance personnel.
n. Verify that all systems function correctly by testing each mode of operation, alarm and system function.
o. Gather and submit operation and maintenance data for systems, subsystems, and equipment to the CxA, as specified.
p. Perform quality control of all work and certify it is complete prior to request for inspection.
q. Complete and sign Commissioning Readiness Letter (CxRL) and provide to CxA (See EXHIBIT A of this specification section).
r. Provide technicians who are familiar with the construction and operation of installed systems and who shall develop specific test procedures and participate in testing of installed systems, subsystems, and equipment.
s. Perform seasonal testing, at the direction of the CxA, to prove functional performance of the HVAC and controls in the opposite season.

B. Design-Build Architect and Design Engineer Responsibilities

1. Responsible for developing the construction contract documents and clarifying the design intent during the construction phase of the project.
2. Performs construction observation.

C. CxA's Responsibilities
1. Organize and lead the commissioning team.
2. Prepare a Commissioning Plan. Collaborate with design team, owner, contractor and subcontractors to develop test and inspection procedures. Identify commissioning team member responsibilities, by name, firm, and trade specialty, for performance of each commissioning task.
3. Work with the Contractor to schedule commissioning activities. The Contractor shall integrate all commissioning activities into the master construction schedule. All parties will address scheduling issues in a timely manner in order to expedite the commissioning process.
4. Review and comment on submittals for compliance with the approved project documents and identify any potential conflicts.
5. Conduct commissioning team meetings for the purpose of coordination, communication, and conflict resolution; discuss progress of the commissioning processes. The CxA shall prepare and distribute minutes to commissioning team members and attendees within five (5) workdays of the commissioning meeting.

6. At the beginning of the construction phase, conduct an initial construction-phase coordination meeting for the purpose of reviewing the commissioning activities and establishing tentative schedules for permanent power; operation and maintenance data submittals; operation and maintenance training sessions; TAB Work; and Project completion.
7. Periodically observe and inspect construction and report progress and deficiencies. In addition to compliance with the Contract Documents, inspect systems and equipment installation for adequate accessibility for maintenance and component replacement or repair.
8. Prepare Project-specific pre-functional checklists and functional test procedures checklists.
9. Compile test data, inspection reports, and certificates and include them in the systems manual and commissioning report.
10. Review and comment on operation and maintenance documentation for compliance with the Contract Documents.
11. Review Contractor's operation and maintenance training program.
12. Prepare commissioning status reports.
13. Assemble the final commissioning documentation, including the Commissioning Report including applicable Project Record Documents.

1.8 COMMISSIONING DOCUMENTATION

A. Commissioning Plan: A document, prepared by CxA, that outlines the process, schedule, allocation of resources, and documentation requirements of the commissioning effort, and shall include, but is not limited to the following:

1. Description of the organization, layout, and content of commissioning documentation to be provided along with identification of responsible parties.
2. Identification of systems and equipment to be commissioned.
3. Description of the level of commissioning for each system
4. Description of schedules for testing procedures along with identification of parties involved in performing and verifying tests.
5. Identification of items that must be completed before the next operation can proceed.
6. Description of responsibilities of commissioning team members.
7. Description of observations to be made.
8. Description of requirements for operation and maintenance training, including required training materials.
9. Provide a schedule for commissioning activities with specific dates coordinated with overall construction schedule.
10. Define the process for completing pre-functional and startup checklists for systems, subsystems, and list of specific equipment requiring these checklists.
11. Include Step-by-Step procedures for Functional Testing of systems, subsystems, and equipment with descriptions for methods of verifying relevant data, recording the results obtained, and listing parties involved in performing and verifying tests.

B. Pre-Functional Checklists: CxA shall develop pre-functional checklists for all equipment to be commissioned. Pre-Functional Checklists shall be completed and signed by the Contractor, verifying that systems, subsystems, equipment, and associated controls are ready for testing. The Commissioning Agent may spot check Pre-Functional Checklists to verify accuracy and readiness for testing. Inaccurate or incomplete Pre-Functional Checklists shall be returned to the Contractor for correction and resubmission.

C. Site Visit Reports: CxA shall record test data, observations, and measurements on site visit forms. Updated Issues Log, photographs and other means appropriate for the application shall be included with Report.

D. Start-Up Reports: Contractor/Manufacturer created forms that document that factory start-up procedures have been followed for all equipment and systems to be commissioned. Provided by sub-contractors.

E. Functional Performance Testing: CxA shall develop functional performance test procedures for all equipment and systems to be commissioned. Site Visit Reports: CxA shall record test data, observations, and measurements on site visit forms. Photographs and other means appropriate for the application shall be included with data.

F. Test and Inspection Reports: CxA shall compile test and inspection reports and test and inspection certificates and include them in Systems Manual and commissioning report.

G. Commissioning Schedule: CxA shall review and provide input to the master project and construction schedules for commissioning activities.

H. Issues Log: CxA shall prepare and maintain an issues log that describes installation, and performance issues that are at variance with the Contract Documents. CxA will identify and track issues as they are encountered, documenting the status of unresolved and resolved issues.

1. Creating an Issues Log Entry:
    a. Identify the issue with unique numeric or alphanumeric identifier by which the issue may be tracked.
    b. Assign a descriptive title of the issue.
    c. Identify issue date.
    d. Identify test number of test being performed at the time of the observation, if applicable, for cross-reference.
    e. Identify system, subsystem, and equipment to which the issue applies.
    f. Identify location of system, subsystem, and equipment.
    g. Include information that may be helpful in diagnosing or evaluating the issue.
    h. Note recommended corrective action.
    i. Identify commissioning team member responsible for corrective action.
    j. Identify expected date of correction.
    k. Identify person documenting the issue.

2. Documenting Issue Resolution:
    a. Log date correction is completed or the issue is resolved.
    b. Describe corrective action or resolution taken. Include description of diagnostic steps taken to determine root cause of the issue, if any.
    c. Identify changes to the Contract Documents that may require action, if any.
    d. State that correction was completed and system, subsystem, and equipment are ready for retest, if applicable.
    e. Identify person(s) who corrected or resolved the issue.
    f. Identify person(s) documenting the issue resolution.

I. Commissioning Report: CxA shall document results of the commissioning process including performance of systems, subsystems, equipment, and issues. The commissioning report shall indicate whether systems, subsystems, and equipment have been completed and are performing according to the OPR, BoD and Contract Documents. The commissioning report shall include, but is not limited to, the following:
1. Discussion of performance of commissioned systems including any variance from OPR, BOD and the Contract Documents; record of conditions; and, if appropriate, recommendations for resolution. This report shall be used to evaluate systems, subsystems, and equipment and shall serve as a future reference document during OWNER occupancy and operation. It may also include a recommendation for accepting or rejecting systems, subsystems, and equipment.
2. Commissioning Plan.
3. Testing plans and reports.
4. Issues log.
5. Completed test checklists.
6. Listing of off-season test(s) not performed and a schedule for their completion.

J. Systems Manual: CxA shall gather required information and compile Systems Manual. Systems manual shall include, but is not limited to, the following:
1. As-built system narratives, schematics, and list of installed equipment
2. Operation and maintenance data

1.9 CXA SUBMITTALS

A. Commissioning Plan: CxA shall submit a draft commissioning plan. Deliver one copy to Contractor and one to OWNER. Present submittal in sufficient detail to evaluate data collection and arrangement process. One copy, with review comments, will be returned to the CxA for preparation of the final commissioning plan.

B. Pre-functional Checklists: CxA shall submit sample checklists and forms to Contractor and subcontractors for review, comment and approval. Contractor completed prefunctional checklists are required to be submitted for review and approved prior to proceeding with functional performance testing.

C. Functional Test Plan: CxA shall submit draft Functional Test Plan and checklists for comment. The final Functional Test Plan will be submitted and used for functional testing.

D. Site visit reports: CxA shall submit site visit reports as they are created.

E. Final Commissioning Report: CxA shall submit the draft commissioning report. One copy, with review comments, will be returned to the CxA for preparation of final submittal. The final report submittal must address previous review comments.

F. The CxA will provide appropriate contractors with a specific request for the type of submittal documentation the CxA requires facilitating the commissioning work. These requests will be integrated into the normal submittal process and protocol of the construction team. At minimum the request will include the manufacturer and model number, the manufacturer printed installation and detailed start-up procedures, sequences of operation, O&M data, performance data, any performance test procedures, control drawings and details. In addition, the factory checkout sheets or field technicians shall be submitted for review

1.10 COORDINATION

A. Scheduling: The Contractor shall work with the Commissioning Agent to incorporate the commissioning activities into the construction schedule. The Commissioning Agent will provide sufficient information (including, but not limited to, tasks, durations and predecessors) on commissioning activities to allow the Contractor to schedule commissioning activities. All parties shall address scheduling issues and make necessary notifications in a timely manner in order to expedite the project and the commissioning process. The Contractor shall update the Master Construction schedule as directed by the Owner.

B. Coordinating Meetings: CxA shall conduct coordination meetings of the commissioning team as needed to review progress on the commissioning plan, to discuss scheduling conflicts, and to discuss upcoming commissioning process activities.

C. Pretesting Meetings: CxA shall conduct pretest meetings with the commissioning team to review startup reports, coordinate controls sequence of operations, review pretest inspection results, review testing and balancing procedures, review testing personnel and instrumentation requirements, and manufacturers' authorized service representative services for each system, subsystem, equipment, and component to be tested.

D. Testing Coordination: CxA shall coordinate with the OWNER and Contractor to plan the sequence of testing activities to accommodate required quality-assurance and -control services with a minimum of delay and to avoid necessity of removing and replacing construction to accommodate testing and inspecting.

1. Schedule times for tests, inspections, obtaining samples, and similar activities.

E. Pipe Flushing Coordination: Discharge of pipe flush water to the sanitary sewer requires a nonroutine discharge permit, which the SLAC Water Programs Manager will procure on behalf of the project. This water will need to be contained and sampled prior to receiving discharge approval. Allow for 7 days to process non-routine discharge permit. If the wastewater is not approved for discharge to the sanitary sewer, disposal shall be coordinated with SLAC Waste Management Department.

F. The Subcontractors shall provide sufficient notice to the CxA (2 weeks minimum) regarding their completion schedule for the pre-functional checklists and startup of all equipment and systems. The contractor will use the Commissioning Readiness Letter (CxRL) in EXHIBIT A to notify the CxA that the systems are ready.

G. General

1. Functional testing is conducted after pre-functional testing and startup has been satisfactorily completed.
2. The BAS is sufficiently tested and approved by the CxA before it is used for TAB or to verify performance of other components or systems.

3. The air and water balancing is complete and approved before functional testing of airrelated or water-related equipment or systems.
4. Testing proceeds from components to subsystems to systems.
5. When the proper performance of all interacting, individual systems has been achieved, the interface or coordinated responses between systems is checked.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 SUBMITTALS

A. The CxA shall provide the Design Builder with a specific request for the type of submittal documentation required to facilitate the commissioning work. These requests shall be integrated into the submittal process and protocol of the project.

B. The submittal shall be reviewed and approved by the Design Builder's design team prior to being submittal to the CxA.

C. At minimum, the submittal request shall include:

1. Equipment manufacturer and model number.
2. Selection and operating data (Example: Flows, pressures, temperatures, fan curves, etc.).
3. The manufacturer's printed installation and detailed start-up procedures.
4. Full sequences of operation and control drawings.
5. O&M data, performance data, performance test procedures, details, and results of ownercontracted tests.
6. Installation and checkout materials that are shipped inside the equipment and the manufacturer field checkout sheet forms to be used by the factory or field technicians.

D. The CxA shall review and approve submittals related to the commissioned equipment for conformance to the Owner Project Requirements, Design documents and as it relates to the commissioning process, the functional performance of the equipment, completeness, and adequacy for developing test procedures.

E. The CxA's review is intended to aid in the development of functional testing procedures and to verify compliance with equipment specifications. The Commissioning authority shall notify the Owner and the DB of items missing or areas that are not in conformance with contract documents or Owner Project Requirements and which require resubmission.

F. The CxA may request additional design narrative from the DB A/E and Controls Contractor, depending on the completeness of the basis of design documentation and sequences provided with the design document Specifications.

G. These submittals to the CxA do not constitute compliance for O&M manual documentation. The O&M manuals are the responsibility of the Design Builder, though the CxA will review and approve them.

3.2 PRE-FUNCTIONAL CHECKLISTS AND FACTORY START UP REPORTS

A. The following procedures apply to all equipment to be commissioned.

B. Pre-functional Checklists are developed by the CxA and completed by the appropriate installing contractors for all major equipment and systems being commissioned before functional testing can begin. The checklist captures equipment nameplate and characteristics data, confirming the as-built status of the equipment or system. These checklists also ensure that the systems are complete and operational, so that the functional performance testing can be scheduled. The Contractor and vendors shall execute factory startup and provide the CxA with a copy of the signed and dated completed start-up checklists which will be submitted with the Pre-Functional checklists.

C. Startup and Initial Checkout Plan: The Contractor shall develop detailed startup plans for all equipment. The primary role of the Contractor in this process is to ensure that there is written documentation that each of the manufacturer recommended procedures have been followed and completed. Parties responsible for startup shall be identified in the Startup Plan and in the checklist forms.
   1. The full startup plan shall at a minimum consist of the following items:
      a. The Pre-Functional Checklists.
      b. The manufacturer's standard written startup procedures copied from the installation manuals with check boxes by each procedure and a signature block added by hand at the end.
      c. The manufacturer's normally used field checkout sheets.
      d. The Commissioning Agent will review/approve the full start-up plan.

D. Execution of Pre-functional Checklists and Startup.

   1. Pre-Functional checklists will be provided to the project site by the CxA.
   2. The contractor shall maintain a master copy of signed checklists.
   3. The installing contractors shall update the checklists as work is completed. Only individuals that have direct knowledge and witnessed that a line-item task on the prefunctional checklist was actually performed shall initial or check that item off.
   4. The CxA will periodically review the checklists for completeness and report on progress at the Cx meetings.

E. BAS Startup Testing, Adjusting, and Calibration
   1. Work and/or systems installed under this Division shall be fully functioning prior to Demonstration and Acceptance Phase. Contractor shall start, test, adjust, and calibrate all work and/or systems under this Contract, as described below:
      a. Inspect the installation of all devices. Review the manufacturer's installation instructions and validate that the device is installed in accordance.
      b. Verify proper electrical voltages and amperages, and verify that all circuits are free from faults.
      c. Verify integrity/safety of all electrical connections.
      d. Coordinate with TAB subcontractor to fine tune control settings that are determined from balancing and testing procedures. Record the following control settings as obtained from TAB contractor, and note any TAB deficiencies in the BAS, Prefunctional checklists and initiate an associated Action Item:
         1) Optimum duct static pressure setpoints for VAV air handling units.
         2) Minimum outside air damper settings for air handling units.
         3) Optimum differential pressure setpoints for variable speed pumping systems.
         4) Calibration parameters for flow control devices such as VAV boxes and flow measuring stations.
         5) BAS contractor shall provide access to the front end Building Automation System as a minimum to the TAB and CxA to facilitate

calibration. Connection for any given device shall local to it (i.e.: at the VAV box or at the thermostat). Shall be made at front end and shall allow querying and editing of parameters required for proper calibration and start up.

e. Test, calibrate, and set all digital and analog sensing, and actuating devices. Calibrate each instrumentation device by making a comparison between the BAS display and the reading at the device, using an instrument traceable to the National Bureau of Standards, which shall be at least twice as accurate as the device to be calibrated (e.g., if field device is +/-0.5% accurate, test equipment shall be +/0.25% accurate over same range). Record the measured value and displayed value for each device in the BAS Pre-functional Report.

f. Check each digital control point by making a comparison between the control command at the controller and the status of the controlled device. Check each digital input point by making a comparison of the state of the sensing device and the operator interface display. Record the results for each device in the BAS Prefunctional checklists.

g. Verify proper sequences by using the approved checklists to record results and submit with BAS Pre-functional checklists. Verify proper sequence and operation of all specified functions. There is inherent duplication between the functional performance testing of the Testing Contractor, and the thorough checking testing of the sequences by the BAS. Generally, the sequence checkouts indicated as the responsibility of the Testing Contractor under functional testing, must first be tested by the BAS under prefunctional testing.

h. Verify proper systems operation under emergency power. Cooperate and coordinate with Testing Contractor and CxA for comprehensive building power outage tests.

i. Verify all safety devices trip at appropriate conditions. Adjust setpoints accordingly.

j. Verify that all alarm thresholds for all analog devices are entered. Request direction from Owner as to alarm threshold parameters.

k. Tune all control loops to obtain the fastest stable response without hunting, offset or overshoot. Record tuning parameters and response test results for each control loop in the BAS Prefunctional Report. Except from a startup, maximum allowable variance from set point for controlled variables under normal load fluctuations shall be as follows. Within 2 minutes of any upset (for which the system has the capability to respond to) in the control loop, tolerances shall be maintained (exceptions noted):

1) Duct air temperature: ±1°F.
2) Space Temperature: ±2°F

3) Hot water temperature: ±2°F.
4) Duct pressure: ± 0.25” w.g.
5) Water pressure: ±1 psid
6) Air flow control: ±5% of setpoint velocity. For min OA flow loops being reset from CO_2, response to upset max time is one hour
7) Space Pressurization (on active control systems): ±0.02” w.g. with no door or window movements

F. For interface and DDC control panels:
1. Ensure devices are properly installed with adequate clearance for maintenance and clearly labeled in accordance with the record drawings
2. Ensure terminations are safe, secure and labeled in accordance with the record drawings
3. Check power supplies for proper voltage ranges and loading.
4. Ensure wiring and tubing are run in a neat and workman-like manner, either bound or enclosed in trough.
5. Check for adequate signal strength on communication networks.
6. Check for stand-alone performance of controllers by disconnecting the controller from the LAN. Verify the event is enunciated at operator interfaces. Verify that the controlling LAN reconfigures as specified in the event of a LAN disconnection.
7. Ensure that controller memory and control network through-put are adequate to support the extensive trending requirements. Reconfigure the system to provide a reliable and robust system as necessary.
8. Ensure all outputs and devices fail to their proper positions/states.
9. Ensure buffered and/ or volatile information is held through power outage.
10. With all system and communications operating normally, sample and record update/enunciation times for critical alarms fed from the panel to the operator interface.
11. Check for adequate grounding of all DDC panels and devices.

G. For Operator Interfaces:
1. Verify all elements on the graphics are functional and properly bound to physical devices and/or virtual points and that hot links or page jumps are functional and logical.
2. Output all specified system reports for review and approval.
3. Verify the alarm printing and logging is functional and per requirements
4. Verify trend archiving to disk and provide a sample to the CxA for review.
5. Verify paging/dial out alarm enunciation is functional.
6. Verify functionality of remote operator interfaces and that a robust connection can be established consistently.
7. Verify that required third party software applications required with the bid are installed and functional.
8. Verify proper interface with fire alarm system.

H. Submit Start-Up Test Report. Report shall be completed, submitted and approved prior to functional testing.

I. Deficiencies, Non-Conformance and Approval in Checklists and Startup.
1. The Contractor shall clearly list any outstanding items of the initial start-up and prefunctional procedures that were not completed successfully, at the bottom of the procedures form or on an attached sheet. The procedures form and any outstanding deficiencies are provided to the CxA within two days of test completion.
2. The CxA reviews the report and reports to the OWNER. The CxA shall work with the Contractor and vendors to correct and retest deficiencies or uncompleted items.

3.3 FUNCTIONAL PERFORMANCE TESTING

A. Common Elements for All Systems

1. Have the required submitted documentation convenient to testing area. Validate that all required documentation has been submitted and is per the contract requirements (very cursory review). CxA shall review the content of the documentation and validate that it is per contract documents.
2. CxA shall review the startup documentation at the start of functional performance testing. Review the startup tests and checklist documentation. CxA shall validate that startup is acceptably executed and complete. CxA shall ensure that any items indicated as outstanding in the checklists is entered as an Action Item and enter one if it is not. The checklists and start up tests/measurements shall be spot checked at the beginning of FPT to ensure accuracy. CxA shall complete a test that indicates he has reviewed the prefunctional checklists and finds them acceptable and note any outstanding items.
3. CxA shall check for and as applicable direct Contractor to demonstrate that access is sufficient to perform required maintenance.
4. CxA shall validate that all prerequisite work is complete and confirm via a test record that the CxA feels it is.
5. Specifically check labeling and ensure conformance to contract requirements.
6. Check proof indication, alarming on failure and restart/acknowledgement as applicable.
7. CxA shall observe operating conditions encountered at the start of FPT. CxA shall examine for normal functionality and record parameters as a test.
8. All dynamic systems powered by electricity shall be tested to simulate a power outage to ensure proper sequencing. Those on emergency power or uninterruptible power shall be tested on all sources.
9. CxA shall inspect the installation and compare it to contract requirements. Record the inspection as a test.
10. Capacities and adjusted and balanced conditions as applicable will generally be checked.
11. Verify all sequence modes and sequences of operation. CxA must initiate all modes and may not refer to or rely on a prefunctional test done by the BAS. Some examples of generic modes that apply to most systems include:

    a. Off Mode
    b. Failed Mode: Proof, safeties, power outage etc. See below for stress testing.
    c. Start Sequence in various modes
    d. Stop sequences in various modes

12. All adjusted, balanced, controlled systems shall be assessed to determine the optimal setting for the system as applicable. The optimal settings should be determined to establish reliable, efficient, safe and stable operation. CxA is responsible for placing systems in optimal condition for occupancy and not simply relying on initial design estimated settings. For adjustable setpoints, the team will document the conditions at the end of FPTs. The contractors will use RFIs / ACDs to confirm any deviations from the design.

13. Dynamic Graphics: The graphic for all components, systems, and areas sampled and required to be represented by a graphic shall be checked for adequacy and accuracy.

Furthermore, when setpoints are required to be adjustable, verify that they can be adjusted directly from the graphic screen.

14. All interfaces between two systems or equipment of different manufacturers must be checked for accuracy and functionality.

15. “Stress Testing”: CxA shall analyze systems to identify possible conditions where functionality may be compromised. CxA shall design non-destructive tests that will demonstrate either the automated response to the conditions or so that team can identify the best method for responding or fixing the condition. All tests and findings shall be documented.

B. Objectives and Scope. The objective of functional performance testing is to demonstrate that each system is operating according to the Contract Documents. Each system will be tested to verify that the system response is as designed. HVAC systems will be checked for conformance to the design sequences of operation and stable control, lighting control will be checked in each type of lighting area, security system cameras will be verified functional and able to see the correct areas. Proper system responses to such conditions as power failure, out of limit condition, equipment failure, etc. shall also be tested.

C. Early duct air leakage tests shall be performed as specified to ensure green and building code compliance. Make testing available for CxA to witness, and provide testing reports to CxA Pointto-point testing will be performed by controls contractor on all applicable systems, with results given to CxA prior to functional performance testing.

D. Development of Test Procedures: The test procedures are written by the CxA based upon the final operational sequences from available project documentation. The CxA shall develop specific test procedures and forms to verify and document proper operation of each system. Prior to execution, the CxA shall provide a copy of the test procedures to the Contractor who shall review the tests for feasibility, safety, equipment, and warranty protection. The test procedure checklists developed by the CxA shall include the following information:

1. System and equipment or component name(s).
2. Equipment location and ID number.
3. Date.
4. Project name.
5. Participating parties.
6. Reference to the specification section describing the test requirements, if applicable.
7. A copy of the specific sequence of operations.
8. Prerequisites for the test.
9. Special cautions, alarm limits, etc.
10. Specific step-by-step procedures to execute the test.
11. Acceptance criteria of proper performance with a Yes / No/NA check box.
12. A section for comments.

E. Test Methods.

1. Systems Functional Performance Testing shall be achieved by manual testing (i.e., persons manipulate the equipment and observe performance) and/or by monitoring the performance and analyzing the results using the control system's trend log capabilities or by standalone data loggers. The Contractor and Commissioning Agent shall determine which method is most appropriate for tests that do not have a method specified.

    a. Simulated Conditions: Simulating conditions (not by an overwritten value) shall be allowed, although timing the testing to experience actual conditions is encouraged wherever practical.
    b. Overwritten Values: Overwriting sensor values to simulate a condition, such as overwriting the outside air temperature reading in a control system to be something other than it really is, shall be allowed, but shall be used with caution and avoided when possible. Such testing methods often can only test a part of a system, as the interactions and responses of other systems will be erroneous or not applicable. Simulating a condition is preferable. e.g., for the above case, by heating the outside air sensor with a hair blower rather than overwriting the value or by altering the appropriate setpoint to see the desired response. Before simulating conditions or overwriting values, sensors, transducers, and devices shall have been calibrated.
    c. Simulated Signals: Using a signal generator which creates a simulated signal to test and calibrate transducers and DDC constants is generally recommended over using the sensor to act as the signal generator via simulated conditions or overwritten values.
    d. Altering Setpoints: Rather than overwriting sensor values, and when simulating conditions is difficult, altering setpoints to test a sequence is acceptable. . For example, to see the Air Conditioning compressor lockout initiate at an outside air temperature below 54 F (12 C), when the outside air temperature is above 54 F (12 C), temporarily change the lockout setpoint to be 4 F (2 C) above the current outside air temperature.
    e. Indirect Indicators: Relying on indirect indicators for responses or performance shall be allowed only after visually and directly verifying and documenting, over the range of the tested parameters, that the indirect readings through the control system represent actual conditions and responses. Much of this verification shall be completed during systems startup and initial checkout

2. Functional testing is performed by the contractors with the method and degree of testing as defined in this specification for each system. Each function and test shall be performed under conditions that simulate actual conditions as close as is practically possible. The Contractor executing the test shall provide all necessary materials, system modifications, etc. to produce the necessary flows, pressures, temperatures, etc.
necessary to execute the test according to the specified conditions. At completion of the test, the Contractor shall return all affected building equipment and systems to their pretest condition.

3. Multiple identical pieces of equipment may be functionally tested using a sampling strategy. The sampling strategy will be defined in these specifications with the commissioned systems list.

F. Coordination and Scheduling: See 1.10 COORDINATION section for coordination and scheduling details.

G. Problem Solving: The CxA will recommend solutions to problems found; however, the burden of responsibility to solve, correct and retest problems is with the Contractor and Owner's consultants.

3.4 OPERATION AND MAINTENANCE TRAINING REQUIREMENTS

A. Before the operation and maintenance training, CxA shall review training preparation for compliance with project documents.

B. Training is required per contract specifications. At a minimum, training is required for Mechanical systems, Lighting, and Controls systems.

C. The CxA requires submission of training records including attendance lists to verify appropriate people received the training.

3.5 GENERAL WARRANTY AND SEASONAL TESTING REQUIREMENTS

A. Coordinate MEP Contractor Call-back and Warranty Enforcement.

1. Perform review of commissioned systems:

a. Timed at approximately 2 months prior to the expiration of the project warranty period.
b. Project warranties have a typical duration of 12 months. Verify for this project.
c. The Warranty Period walk through inspection for the commissioned systems would then be typically scheduled at 10 months and should include representatives from
d. the Owner, Facilities Management, Design Builder, subcontractors, and manufacturers.

3.6 MEP Cx WARRANTY AND SEASONAL TESTING REQUIREMENTS

A. The Warranty Phase of a project can incorporate one or both of these two complimentary activities:

1. Seasonal Testing: Seasonal variation in operations or control strategies may require additional testing during peak cooling and heating seasons to verify systems performance. If the project finishes in the summer, testing of the heating systems may not be effective or conclusive, and can be postponed to colder weather if seasonal testing is included in the scope. The CxA coordinates this activity. Tests are executed and deficiencies corrected by the appropriate subcontractors, witnessed by Owner's staff and the CxA.

2. Warranty Testing: The CxA will perform an evaluation of the systems in the Cx scope to provide additional experienced review of operation prior to the expiration the warranty. The

CxA will request input from the Owner's operations staff and occupants about the performance of the building systems, as well as perform targeted evaluation of systems and controls. Additionally, open items on the Master Issues Log are often reviewed for completion, or significant issues observed during previous phases are evaluated to ensure they are still in a corrected state. A warranty walk review typically is performed onsite.

However, videoconference and BMS remote review by the CxA may be arranged in place of the onsite warranty walk.

B. These two activities often have overlapping scope and intent and will be performed during the same site evaluation(s), where possible.

C. The warranty period typically begins at the project's Substantial Completion or shortly thereafter.

3.7 COSTS OF COMMISSIONING WORK

A. The cost to the Contractor and Subcontractors to comply with the specified requirements and to support the work of the CxA shall be included in the Contractor's and Subcontractor's bid price.

B. It is the Contractor's responsibility to QC and pre-test all building equipment and systems. The CxA shall confirm function of each system. If a device, piece of equipment, sequence, or system fails a test, corrections shall be made immediately and retested.

C. If at any point in the Commissioning Functional Testing Process, should an issue or failure arise that requires significant time to correct or cannot be corrected immediately during the testing process, that results in a delay or prevents the CxA from completion of the functional testing, incurred costs shall be reimbursed by the Contractor. The associated costs of retesting are defined in 3.11 Exhibit A - Commissioning Readiness Letter (CxRL) .

3.8 COMMISSIONED SYSTEMS

System	Equipment	Level Note
HVAC System	Chiller (Water Cooled / Air Cooled)	5
	Cooling Towers	5
	HW Boilers	5
	Pumps	5
	Air Handling Units	5
	Computer Room Air Conditioners	5
	Lab Exhaust Fans	5
	Supply Fans	3
	Split Systems	3

	Fan Coil Units	3
	Variable Frequency Drives	3
	Exhaust Fans	3
	VAV Air Terminal Units	3
	Test Adjust Balance Report Values	3
Building Management System	Sequences of Operation, Monitored Points, and Alarms	5
	Metering/Monitoring Devices and Equipment	5
	Software Commissioning, GUI Presentation Commissioning, System Access Performance Criteria, Software Tools/Source Code Commissioning, Instrument Data Sheets, Middleware Commissioning, Internet Protocol Commissioning	5
Electrical System	~~Emergency Generator~~	~~5~~
	Automatic Transfer Switch (if applicable)	5
	UPS	5
	Lighting Systems & Controls	3
	Switchgear & Distribution Panels	2
Plumbing System	Domestic Water Heaters	5
	Thermostatic Mixing Valves	5
	HW Circulation Pump	5

	~~Emergency Eyewash~~ / Shower	5
System	Equipment	Level Note
Irrigation	Irrigation Controls	5
Renewable Energy	Photovoltaics	4
Security Electronics and Alarm Systems	Video Camera Surveillance	1
	Intercom System	1
	Paging System (if applicable)	1
	Door Controls	1
	Fire Alarm System	1
	Distributed Radio Antenna System (if applicable)	1
	Access Control System	1
Audio Visual Systems	Room Acoustics	2
	Sound Masking System	2
	Assisted Listening	2
	Video Projection	2
	Audio System	2
	Lighting and Lighting Controls	3

Testing Levels Defined:

Level 1 - The CxA will request and review equipment installation documentation, periodically observe and review the installation of building systems to verify that the operation meets the Owners Project Requirements.

Level 2 - The CxA will periodically observe and inspect the installation of equipment and systems and review project documentation (test reports). The CxA may spot check some of the system functions verify operational requirements are met.

Level 3 - The CxA will periodically observe and inspect the installation of equipment and systems and review project documentation (test reports) and will witness contractor performance testing of the system. Contractor shall test up to 20% of the system to prove operational requirements are met. The test sections shall be chosen at random by the CxA. Failure of any test section shall require retesting of that section and an additional test section equivalent in scope.

Level 4 - The CxA will periodically observe and inspect the installation of equipment and systems and review project documentation (test reports) and will witness contractor performance testing of the system.

Contractor shall test up to 50% of the system to prove operational requirements are met. The test sections shall be chosen at random by the CxA. Failure of any test section shall require retesting of that section and an additional test section equivalent in scope.

Level 5 - The CxA will periodically observe and inspect the installation of equipment and systems and review project documentation (test reports) and will witness contractor performance testing of the system. Contractor shall test up to 100% of the system to prove operational requirements are met. The test sections shall be chosen at random by the CxA. Failure of any test section shall require retesting of that section and an additional test section equivalent in scope.

* Testing percentages may vary based on testing success rate.

3.9 METHODS OF TESTING

A. HVAC Systems

1. CxA may witness specified duct air leakage testing during rough-in. Contractor to forward all duct air leakage reports to CxA for review.
2. CxA will visit the site during rough-in of ductwork, piping, and equipment to verify proper maintenance clearances and access are being maintained.
3. The CxA may witness contractor and/or factory start-up of equipment.
4. The TAB contactor shall re-measure up to 20% of the TAB report values for the CxA to observe and to verify balance values are within allowable design tolerances.
5. Stand-alone controls will be tested independent of item B below.

6. Contractor will demonstrate to the CxA that the operation of each system through all modes, alarms, and operating parameters meet the contract documents.

B. Building Management System

1. After receipt of the controls contractor's calibration and point to point reports by the CxA, the controls contactor will re-measure some of the points for the CxA to verify that the calibration and communication is correct. The points to be verified will be selected by the CxA.
2. Controls contractor shall provide an as-built shop drawing to the CxA for use in executing FPT.
3. All of the user graphics interfaces and displayed operating points will be demonstrated for the CxA by the controls contractor.
4. Controls contractor shall manipulate the system to demonstrate that it performs all of the specified modes of operation.
5. Points selected by the CxA will be trended for 1-2 weeks by the controls contractor to verify control operation and response. System to in auto without alarms.

C. Electrical Systems

1. During the installation the CxA will perform the following for the electrical systems:

   a. Periodically observe the installation of equipment.
   b. Review the completed Pre-functional Checklists (PFC).
   c. Verify the PFC's by observing the completed work and comparing to the values listed in the PFC.
   d. Review the factory authorized programming and checkout report of the lighting control panels and devices.
   e. The contractor is to provide NETA certified third party testing of the power distribution system and provide the CxA with a certified test report.
   f. CxA will review contractor provided as-builts for proper identification and labeling of all equipment, piping and devices.

2. To test the performance of the lighting control system the CxA will perform the following tasks:

   a. Witness the contractor testing each scene from each wall station.
   b. Verify these scenes match the design intent from the contract documents.
   c. Witness the contractor testing the integration to the other integrated systems such as audio visual and monitoring abilities from the BMS.
   d. Verify this integration allows control and/or monitoring from the other systems.
   e. Verify connectivity to the emergency lighting circuits.
   f. Witness the contractor testing the emergency lighting circuits.

3. Upon completion of the emergency power system, factory start-up and contractor pretesting, the CxA will witness a contractor test to verify complete system power loss and verify proper power provision of critical systems. The test will not be scheduled until all other systems dependent on emergency power have been tested and approved.

D. Plumbing

1. Domestic hot water will be tested by the CxA by measuring the hot water temperature at a percentage of the fixtures along with the time it takes to reach that temperature.
2. Contractor shall demonstrate domestic hot water boilers, pumps and controls through all modes of operation and alarms.
3. Contractor shall demonstrate to the CxA that the sanitary sewer and domestic booster pump operation through all modes and alarms meets approved sequence of operations.
4. After contractor has adjusted all fixtures for proper flush and sink fixture metering, the CxA will test plumbing fixtures for proper operation.
5. The contractor shall demonstrate the water management system to the CxA.
6. The CxA will test the compressed air and vacuum systems for proper operation.

E. Irrigation

1. The CxA will witness the contractor demonstration of the irrigation controller and coverage using the contractor's as-built drawings.

F. Security ACAMS (Access Control and Alarm Monitoring System), Video Surveillance Systems

1. Receive all installation and pre-functional test documentation from systems contractor(s).
2. Confirm all IP and MAC addresses have been requested from the owners IT department and installed by the contractor.
3. Verify cable and system component installation complies with the specifications and drawings.
4. Verify that the access control and video system software is the correct version and is programmed and operational.
5. Verify correct panel and door hardware power supplies and batteries are connected and operational.
6. Verify card reader operation at the door is per the specification.
7. Verify door alarms for forced, held open, closed, and reader reset functions are operational.
8. Verify intrusion alarm and duress alarm functions.
9. Verify camera field of view meets owner's requirements.
10. Verify camera recording and video storage meets specifications.

G. Communications

1. The CxA will witness the contractor test a percentage of the voice communication devices to verify operation is as specified.
2. The CxA will witness the contractor demonstrate that the assistive listening system operates as specified.
3. The CxA will witness the contractor demonstrate that the TV signal distribution system provides signal quality as specified.
4. The CxA will witness the contractor demonstrate the functionality of the AV system to verify all integrated equipment operates as specified.
5. The CxA will witness the contractor demonstrate that the operation of the paging system meets specifications.

H. Fire/Life Safety

1. The CxA will review the results of the contractor's test documents to verify that the system was tested and meets specification. Interaction with other systems such as HVAC will be specifically reviewed.

3.10 COST OF RE-TESTING AND COMMISSIONING READINESS LETTER (CxRL) – SYSTEMS FUNCTIONAL TESTING READINESS CERTIFICATION AND NOTIFICATION

A. If CxA arrives onsite, on the scheduled date for functional testing (as indicated on the CxRL, see 3.11 Exhibit A of this specification section) which cannot be completed due to systems readiness failure, systems technician no-show, or other circumstances not caused by CxA resulting in failed functional testing; it is understood that the CxA's client (listed on CxRL) will be invoiced for expenses incurred by CxA. The contractor also agrees to reimburse said client for incurred expenses. CxA expenses will be invoiced as noted in the CxRL.

B. It is the Contractor's responsibility to QC and pre-test all building equipment and systems. The CxA shall confirm function of each system. If a device, piece of equipment, sequence, or system fails a test, corrections shall be made immediately and retested. Corrections that cannot be corrected immediately or that delay completion of CxA testing shall be reimbursed by the Contractor.

3.11 Exhibit A: Commissioning Readiness Letter (CxRL) - Systems Functional Testing Readiness Certification and Notification

SLAC National Accelerator Laboratory / Large Scale Collaboration Center (SLAC) Menlo Park, CA

This letter shall serve as certification to 3QC that all applicable systems checked below have been fully tested to perform as specified in the Construction Documents, in accordance with 3QC's Functional Testing Checklists, and that all functional testing prerequisites as outlined in the Commissioning Specifications and Commissioning Plan have been completed and submitted to 3QC for review. 3QC is hereby officially notified to begin onsite functional testing of the following systems:

Systems Ready for Functional Testing by 3QC (Completed by Design Builder <u>as systems become available</u> and are ready for testing and meeting all criteria explained here within – <u>Check only the systems that are ready at this time</u>, use additional copies of this letter if needed as systems become ready.)		
Check Applicable System	Systems	Date DB is Requesting for CxA on-site Functional Testing *
☐	TAB Verification	
☐	BMS, BAS, DDC or EMS	
☐	HVAC Systems	
☐	Plumbing Systems	

☐	Emergency Power Systems	
☐	Lighting Control Systems	
☐	Renewable Energy Systems	
☐	Communication Systems	
☐	Electronic Safety and Security Systems	
☐	Irrigation	

* = Systems Technician Required – The contractor certifies that a systems technician familiar with and capable of operating each system to be commissioned will be available onsite throughout functional testing <u>by 3QC. For BMS, BAS, DDC or EMS systems this must be the commissioning technician/programmer.</u>

Failed Functional Testing – If the CxA arrives onsite, on the date(s) indicated above, for functional testing of identified system(s) and cannot be tested due to systems readiness failure, systems technician no-show, or other circumstances not caused by 3QC resulting in failed functional testing; it is understood that 3QC's client (listed below) will be invoiced for travel and expenses incurred by 3QC. The Contractor agrees to reimburse project Owner for incurred expenses as described in Specification Section 01 91 13, 1.4 COMPENSATION, B.

Signature of 3QC's Client or Representative Print Name Date

Signature of Design Build Representative Print Name Date

3.12 Exhibit B: Commissioning Activity Schedule

Commissioning Activity Schedule									
DB's Baseline Schedule Activity ID #	**Commissioning (Cx) Construction and Post-Construction Phases Activity**		**Building System: BECx**	**MEPCx**	**MBCx**	**Low Voltage Cx**	**Parallel Construction Site Activity**	**Duration of Cx Activity**	**CxA Required Site Visit**
	1	Cx Kick-off Meeting		•		•	Contractor mobilized onsite	1 day	•
	2	CxA provides electronic jobsite Cx Site Binder to DB; including PreFunctional Checklists		•		•	DB to receive Cx Site Binder from CxA during Cx Kick-off Meeting	-	
	3	CxA identifies submittals required for Cx		•		•	Design Builder provides a submittal list for CxA to review and highlight required submittals to be reviewed for Cx	-	
	4	CxA reviews submittals		•		•	A/E reviews submittals	TBD	
	5	CxA begins to receive O & M Manuals		•		•	Subcontractors provides O & M Manuals to Contractor	TBD	
	6	Site Verification; Mechanical equipment, update Cx Equipment List & update Cx Issues Log		•			Mechanical equipment set	TBD	•
	7	Low Voltage rough-in Site Observation & update Cx Issues Log as required				•		TBD	•

	8	CxA to start development of Cx Functional Test Checklists		•		•	Submittal reviews complete	-	
	9	CxA notified that permanent power is installed		•		•	Permanent power installed	-	
	10	CxA receives copies of the field Air Duct Leakage Testing Report		•			Air duct leakage testing completed	-	

Commissioning Activity Schedule

DB's Baseline Schedule Activity ID #	Commissioning (Cx) Construction and Post-Construction Phases Activity		Building System				Parallel Construction Site Activity	Duration of Cx Activity	CxA Required Site Visit
			BECx	MEPCx	MBCx	Low Voltage Cx			
	11	CxA conducts onsite functional testing procedures coordination meeting with DB, low voltage, and MEP subcontractors		•		•	1 month prior to Cx Functional Testing	1 day	•
	12	CxA issues Functional Test Checklists to DB, low voltage, and MEP subcontractors		•		•	1 Month prior to Cx Functional Testing	-	
	13	CxA Witnesses startup of HVAC equipment		•			HVAC contractor meets CxA onsite for startup	TBD	•
	14	CxA receives jobsite Pre-Functional Checklists from Cx Site Binder completed, signed, and dated		•		•	2 Weeks prior to Cx Functional Testing	-	

Commissioning Activity Schedule

DB's Baseline Schedule Activity ID #		Commissioning (Cx) Construction and Post-Construction Phases Activity	Building System: BECx	MEPCx	MBCx	Low Voltage Cx	Parallel Construction Site Activity	Duration of Cx Activity	CxA Required Site Visit
	15	Building LEED flush-out schedule received by CxA		•			DB coordinates LEED flush-out schedule with Cx functional testing activities	TBD	
	16	CxA receives field TAB Report reviewed by Engineer of Record		•			1 Week prior to Cx Functional Testing	-	
	17	CxA receives completed/signed Commissioning Readiness Letter (CxRL) indicating Commissioned Systems are ready for Cx Functional Testing		•		•	1 Week prior to Cx Functional Testing	-	
	18	CxA starts functional performance tests (FPTs) on TAB, BMS, HVAC, domestic hot water, lighting controls, communication,		•		•	DB sent CxA the following functional testing prerequisites: a. CxA received signed notification letter that TAB, BMS (including all	TBD	•

		security, and irrigation control systems					graphics), HVAC, domestic hot water, lighting controls, communication, security, and irrigation control systems are completed and ready for Cx functional testing. b. Pre-functional checklist completed and signed. c. CxA received TAB report		
	19	CxA verification of the TAB report air and water values		•			TAB subcontractor meets CxA onsite to verify TAB report	TBD	•
	20	CxA FPTs of BMS		•			Controls subcontractor meets CxA onsite to perform Functional Testing of BMS after BMS point-to-point & checkout is completed	TBD	•
	21	CxA FPTs of HVAC		•			Controls contractor and HVAC contractor meets CxA onsite to perform functional testing of HVAC	TBD	•
	22	CxA FPTs of domestic hot water		•			Plumbing subcontractor meets CxA onsite to perform functional testing of domestic hot water	TBD	•
Commissioning Activity Schedule									
	Commissioning (Cx)		Building System				Parallel		

DB's Baseline Schedule Activity ID #	Construction and Post-Construction Phases Activity		BECx	MEPCx	MBCx	Low Voltage Cx	Construction Site Activity	Duration of Cx Activity	CxA Required Site Visit
	23	CxA FPTs of lighting controls		•			Electrical lighting controls contractor and electrical contractor meets CxA onsite to perform functional testing of lighting controls	TBD	•
	24	CxA FPTs of irrigation controls		•			Landscaping contractor meets CxA onsite to perform functional testing of irrigation controls	TBD	•
	25	CxA FPTs of security systems				•	Security subcontractor meets CxA onsite to perform functional testing of security systems	TBD	•
	26	CxA FPTs of communication systems				•	Communication subcontractor meets CxA onsite to perform functional testing of communication systems	TBD	•
	27	CxA receives building systems owner training agendas and training schedule		•		•	Cx functional testing completed	-	
	28	CxA confirms that Cx issues are resolved. (Preferably all issues are resolved.)		•		•	DB receives responses addressing Cx issues	TBD	
	29	Cx Report		•		•	Cx systems testing completed	14 days	
	30	Cx Systems Manual		•		•	Cx systems O & M Manuals sent to CxA	14 days	
	31	Cx Warranty Review		•		•	10 months from date of owner's occupancy	1 day	•

SCHEDULE NOTES:

The following sequential priorities are required to be followed:

1. Equipment is not "temporarily" started (for heating or cooling), until pre-start checklist items and all manufacturer's pre-start procedures are completed, dirt, dust, and other environmental and building integrity issues have been addressed.
2. Functional performance testing does not begin until Pre-Functional, start-up and TAB is completed for a given system.
3. The controls systems and equipment under its control is not functionally tested until all points have been calibrated and Pre-Functional Checklists are completed.

END OF SECTION 01 91 13

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization	☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 07 84 00 – FIRESTOPPING

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Firestopping, in accordance with provisions of Contract Documents.

B. Completely coordinate with work of other trades.

1.2 QUALITY ASSURANCE

A. Installer Qualifications:
   1. Certified, licensed or approved by firestopping manufacturer, trained to install firestop products per specified requirements.
   2. Licensed by State or local authority, where applicable.
   3. Shown to have successfully completed not less than 5 comparable scale projects.

B. Provide firestop systems in compliance with following requirements:
   1. Obtain firestop system for each type of penetration and construction condition from a single firestop systems manufacturer.
   2. Firestop products and systems shall bear classification marking of qualified testing and inspection agency.
   3. Firestopping tests, performed by qualified, testing and inspection agency.
      a. UL or other agency, performing testing and follow up inspection services for firestop systems, acceptable to local authorities having jurisdiction.
   4. Existing applications for which no tested and listed classified system is available through a manufacturer:
      a. Provide Engineering Judgment or Equivalent Fire Resistance Rated Assembly (EFRRA) for submittal derived from similar UL system designs or other tests approved by local authorities having jurisdiction, prior to installation.
      b. Engineering judgment drawings must follow requirements set forth by International Firestop Council.
   5. Mold Resistance:
      a. Less than 1 per ASTM G21.
   6. Inspect applied firestopping systems in accordance with International Building Code (IBC) Chapter 17.
      a. See Section 01 45 23.

C. UL:
   1. UL 263, Fire Tests of Building Construction and Materials
   2. UL 723, Surface Burning Characteristics of Building Materials
   3. UL 1479, Fire Tests of Through Penetration Firestops
   4. UL 2079, Tests for Fire Resistance of Building Joint Systems

D. Underwriters Laboratories (UL) Fire Resistance Directory:
   1. Through Penetration Firestop Systems (XHEZ).
   2. Joint Systems (XHBN).

3. Fill, Void or Cavity Materials (XHHW).
4. Firestop Devices (XHJI).
5. Forming Materials (XHKU).
6. Wall Opening Protective Materials (CLIV).

E. ASTM International (ASTM):
1. ASTM E84 Surface Burning Characteristics of Building Materials
2. ASTM E119 Fire Tests of Building Construction and Materials
3. ASTM E136 Test Method for Behavior of Materials in a Vertical Tube Furnace at 750F
4. ASTM E814 Fire Tests of Through Penetration Fire Stops
5. ASTM E1399 Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems
6. ASTM E1966 Test Method for Fire Resistive Joint Systems
7. ASTM E2174 Standard Practice for On-site Inspection of Installed Fire Stops
8. ASTM E2307 Standard Test Method for Determining the Fire Endurance of Perimeter Fire Barrier Systems Using the Intermediate-Scale, Multi Story Test Apparatus (ISMA)
9. ASTM E2393 Standard Practice for On-site Inspection of Installed Fire Resistive Joint Systems and Perimeter Fire Barriers
10. ASTM G21 Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

F. National Fire Protection Association (NFPA):
1. NFPA 70: National Electrical Code
2. NFPA 101: Life Safety Code
3. NFPA 221: Standard for High Challenge Fire Walls, Fire Walls, and Fire Barrier Walls
4. NFPA 251: Fire Tests of Building Construction and Materials

G. Firestop Contractors International Association (FCIA):
1. MOP – FCIA Firestop Manual of Practice

H. International Firestop Council (IFC):
1. Recommended IFC Guidelines for Evaluating Firestop Engineering Judgments, latest revision.
2. Inspectors Field Pocket Guide, latest edition.

I. Identification Labels for Firestop Assemblies:
1. Follow guidelines set in Chapter 7 of International Building Code.

J. Pipe insulation shall not be removed, cut away or otherwise interrupted at wall penetrations or floor openings.
1. Provide products appropriately tested for the thickness and type of insulation utilized.

K. Cabling where frequent cable moves, additions, and changes are likely to occur in future:
1. Where cable trays are used:
   a. Utilize re-enterable products (e.g. removable intumescent pillows) specifically designed for retrofit.
2. Where cable trays are not used:
   a. Utilize fire rated cable pathway devices.
   b. Where not practical, re-enterable products designed for retrofit may be used.

L. Protect penetrations passing through fire resistance rated floor-to-ceiling assemblies contained within chase wall assemblies with products tested by being fully exposed to fire outside of chase wall.

1. Identify systems within UL Fire Resistance Directory with the words: Chase Wall Optional.

M. Fire-resistive Joint Sealant:
   1. Provide flexible fire resistive joint sealants to accommodate normal and thermal building movement without seal damage.
   2. Provide fire resistive joint sealants designed to accommodate a specific range of movement.
      a. Test in accordance with cyclic movement test criteria as outlined in: ASTM E1399, ASTM E1966 or UL 2079.
   3. Provide fire resistive joint systems subjected to an air leakage test.
      a. Conduct in accordance with UL 2079, with published L-Ratings for ambient and elevated temperatures.
   4. Coordinate firestopping with acoustical sealant requirements in Section 07 92 16.

N. Subject smoke wall containment systems to air leakage test.
   1. Conduct in accordance with UL 1479, with published L-Ratings for ambient and elevated temperatures.

O. System Description:
   1. Through Penetration Firestop Systems for protection of penetrations through following fire resistance rated assemblies, including both blank openings and openings containing penetrating items:
      a. Roof assemblies.
      b. Floor assemblies.
      c. Wall and partition assemblies.
      d. Existing, fire and smoke rated assemblies.
   2. Fire Resistive Joint Assemblies for linear voids where fire rated floor, roof, or wall assemblies abut one another, including following types of joints:
      a. Top and bottom of wall interface with overhead roof or floor structure:
         1) Coordinate with acoustical sealant specified in Section 07 92 16.
         2) Select products to maintain acoustical, smoke and fire ratings indicated.

1.3 SUBMITTALS

A. Product Data:
   1. Manufacturer's standard information indicating certification of products proposed for use on project.

B. Project Information:
   1. UL reports with illustration of systems, system numbers, temperature ratings, and products proposed for use on project.

C. Contract Closeout Information:
   1. Warranty.

1.4 WARRANTY

A. Written 5 year warranty guaranteeing quality of installation and meeting requirements of manufacturer’s written instructions and tested systems.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Firestopping:
1. Base:
a. Hilti Inc.
2. Optional:
a. 3M
b. Rectorseal
c. Specified Technologies, Inc.
d. Tremco, Inc.
e. United States Gypsum Company
f. W.R. Grace & Company

B. Forming Materials:
1. Base:
a. Hilti, Inc.
2. Optional:
a. Industrial Insulation Group.
b. Rock Wool Manufacturing
c. Roxul Inc.
d. Thermafiber

C. Other manufacturers desiring approval comply with Volume 2 of UL Building Materials Directory.

2.2 MATERIALS

A. Through Penetration Firestop Systems:
1. VOC content not to exceed 250 g/L
2. Base Products:
a. FS-ONE MAX Intumescent Firestop Sealant.
b. CFS-S SIL GG Elastomeric Firestop Sealant.
c. CFS-S SIL SL Elastomeric Firestop Sealant.
d. CP 620 Fire Foam.
e. CP 606 Flexible Firestop Sealant.

B. Fire resistive Joints:
1. VOC content not to exceed 250 g/L
2. Base Products:
a. CFS-SP WB Firestop Joint Spray.
b. CFS-S SIL GG Elastomeric Firestop Sealant.
c. CFS-S SIL SL Elastomeric Firestop Sealant.
d. CP 606 Flexible Firestop Sealant.

C. Firestop Devices:
1. Factory assembled collars lined with intumescent material sized to fit specific outside diameter of penetrating item.

2. Base Products:
   a. CP 680-P Cast-in-Place Firestop Device.
   b. CP 680-M Cast-in-Place Firestop Device.
   c. CP 681 Tub Box Kit.
   d. CFS-DID Firestop Device.

D. Intumescent Pads, Wall Opening Protective Materials:
1. Intumescent, non-curing pads or inserts for protection of electrical panels, switch and receptacle boxes, medical gas outlets and valve boxes and other items recessed in face of fire rated walls.
2. Base Product:
   a. CFS-P PA Firestop Putty Pad.
   b. CP 617 Firestop Putty Pad.
   c. Hilti Biox Insert.

E. Fire-rated Cable Pathways:
1. Steel raceway and intumescent pads with adjustable smoke seal sleeve.
2. Fire rating equal to rating of barrier device penetrates.
3. Pathway devices:
   a. Allow 0 to 100 percent fill of cables.
   b. Adjust automatically to cable additions or subtractions.
4. Size to accommodate quantity and size of electrical wires and data cables indicated plus 100 percent expansion.
5. Provide cable management devices with gang plates for single or multiple devices.
6. Base products:
   a. CP 653 BA Speed Sleeve.
   b. CFS-SL GP Gangplate.
   c. CFS-SL GP CAP Gangplate Cap.
   d. CFS-CC Firestop Cable Collar.
   e. CFS-SL SK Firestop Sleeve.
   f. CFS-SL RK Retrofit Sleeve.

F. Smoke and Acoustic Cable Pathways:
1. Non-rated steel raceway with adjustable smoke seal polyurethane sleeve for single cables and cable bundles.
2. Re-penetrable and self-closing.
3. Base product:
   a. CS-SL SA Smoke and Acoustic Sleeve

G. Firestop Putty:
1. Intumescent, non-hardening, water resistant putties containing no solvents, inorganic fibers or silicone compounds.
2. Provide firestop putty at, but not limited to, the gap between wire, cabling, or both, exiting an open end of conduit, where conduit penetrates one or both sides of a smoke or fire rated wall assembly.
3. Base products:
   a. CP 618 Firestop Putty Stick.
   b. CFS-PL Firestop Plug.

H. Wrap Strips:

1. Single component intumescent elastomeric strips faced on both sides with a plastic film:
2. Base Products:
    a. CP 643N Firestop Collar.
    b. CP 644 Firestop Collar.
    c. CP 648E/648S Wrap Strips.

I. Firestop Blocks and Plugs:
1. Non-curing, flexible intumescent device.
2. Re-enterable.
3. Base products:
    a. CFS-BL Fire Block.
    b. CFS-PL Firestop Plug.

J. Mortar:
1. Portland cement based dry-mix product formulated for mixing with water at Project site to form a non-shrinking, water-resistant, homogenous mortar.
2. Base product:
    a. CP 637 Firestop Mortar.

K. Silicone Sealants:
1. Moisture curing, single component, silicone elastomeric sealant for horizontal surfaces pourable or nonsag or vertical surface nonsag.
2. Base product:
    a. CFS-S SIL GG Elastomeric Firestop Sealant.
    b. CFS-S SIL SL Elastomeric Firestop Sealant.

L. Preformed Mineral Wool:
1. CP 767 Speed Strips
2. CP 777 Speed Plugs

M. Fire Sealant:
1. Single component latex or acrylic formulations that upon cure do not re-emulsify during exposure to moisture.
    a. CFS-S SIL GG Elastomeric Firestop Sealant.
    b. CFS-S SIL SL Elastomeric Firestop Sealant.
    c. CP 672 Firestop Joint Spray.
    d. CFS-SP WB Firestop Joint Spray.
2. VOC content of sealants shall be no greater than 250 g/L.

N. Composite Sheet:
1. Non-curing, re-penetrable material.
2. Base Products:
    a. CP 675T Firestop Board.
    b. CFS-BL FireBlock.

O. Forming Materials:
1. Materials listed as components in laboratory approved designs.
2. Mineral Wool:
    a. Base Product:SAF by Thermafiber, or
    b. Similar product specifically named as components in laboratory approved designs.

P. Acoustical Sealant:

1. Specified in Section 07 92 16.

2.3 THROUGH PENETRATION FIRESTOP SYSTEMS

A. General:
1. Schedules below identify requirements for acceptable through penetration firestop systems based on barrier type, fire resistive rating, and penetrant type. Each system must comply with building code and fire code as locally adopted and amended.
2. Requirements for single membrane penetrations and through penetration firestops are identical. Unless otherwise noted, penetrants which pass through a single membrane, shall be treated the same as if it passed through the entire fire resistive assembly.
3. Select each firestop system based on actual field conditions, including penetration type, shape, size, quantities and physical position within opening.
4. Refer to Plans for indication of the required ratings of fire resistive wall, floor, and roof assemblies.
5. Indicated ratings are minimum and may be exceeded.
6. Firestop Assemblies at Fire Rated Walls:
   a. The minimum Fire (F) Rating for Firestop assemblies in walls shall equal that of the wall, but not less than 1 hour.
   b. The minimum Temperature (T) Rating of Firestop assemblies in walls may equal zero.
   c. Smoke Barrier: In addition to (F) Rating, (L) Rating of maximum 5 CFM per SF .
   d. Non-rated walls and Smoke Partitions with no fire resistive requirement: Assembly with (L) rating.
7. Firestop assemblies at fire rated floors and roofs:
   a. Minimum Fire (F) and Temperature (T) Ratings of Firestop assemblies used in floors or roof shall equal hourly rating of floor or roof being penetrated, but not less than 1 hour.
      1) Exception 1: The T-rating may equal zero when portion of penetration, above or below floor, is contained within a wall.
      2) Exception 2: Firestops are not required for floor penetrations within a 2 hour rated shaft enclosure.

B. Voids in Wall with No Penetrations:
1. Fill with approved through penetration firestopping system.
2. Contractor’s option: Patch void in wall with like construction.

C. Penetrating Ducts with Dampers:
1. Utilize only firestop materials which are included in damper’s classification.
2. Do not install firestop systems that hamper performance of fire dampers.

D. Cable Trays and Similar Devices:
1. Provide re-enterable products specifically designed for removal and re-installation at openings within walls and floors designed to accommodate voice, data and video cabling.

E. Electrical panels and devices, medical gas outlets and valve boxes, film illuminators, and other items recessed in to face of rated walls:

1. Where electrical devices are placed on opposite sides of wall, and are less than 24 IN apart measured horizontally, install intumescent pads over back of devices in approved manner or maintain continuity of rated barrier within wall cavity surrounding recessed item.

2.4 FIRE RESISTIVE JOINT ASSEMBLIES

A. Where joint will be exposed to elements, fire resistive joint sealant must be approved by manufacturer for use in exterior applications and shall comply with ASTM C920.

B. Head of Wall Assemblies:
   1. General:
      a. Use at top of fire rated and smoke barrier walls and partitions where they abut floor and roof structures above.
      b. Select systems with D designation, rated for dynamic movement capability.
      c. Select systems that can accommodate deflection of structure above.
      d. Maximum Leakage for Fire resistive Joints in Smoke Barriers: 5 CFM or less per linear foot as tested in accordance with UL 2079.
      e. Seal non-fire rated sound control walls and smoke partitions with acoustical sealant as specified in Section 07 92 16.
   2. Minimum F and T ratings:
      a. The minimum fire rating for firestop assemblies in walls shall equal that of wall, but not less than 1 hour.
      b. The minimum temperature rating of firestop assemblies in walls may equal zero.
   3. Acceptable Systems:
      a. Metal stud and drywall partitions: Select system from UL HW-D-0000 Series.
      b. Concrete and Masonry Walls: Select system from UL HW-D-1000 Series.

PART 3 - EXECUTION

3.1 PREPARATION

A. Examine areas and conditions under which work is to be performed and identify conditions detrimental to proper or timely completion.

B. Surfaces to which firestop materials will be applied shall be free of dirt, grease, oil, scale, laitance, rust, release agents, water repellents, and any other substances that may inhibit optimum adhesion.

C. Provide masking and temporary covering to prevent soiling of adjacent surfaces by firestopping materials.

D. Do not proceed until unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install firestop systems in accordance with manufacturer’s instructions and conditions of testing and classification as specified in UL or other acceptable third party testing agency listing.

B. Penetrations through fire resistive floor assemblies shall be sealed with firestop system providing minimum Class 1 W-rating as tested in accordance with UL 1479 and ensure air and water resistant seal.

C. Protect materials from damage on surfaces subjected to traffic.

D. Identification Labels:
   1. Identify each firestop assembly as defined in Quality Assurance.
   2. Do not locate identification labels, tags, or both, on finished surfaces or where exposed to view by public.

3.3 FIELD QUALITY CONTROL

A. Maintain areas of work accessible until inspection by authorities having jurisdiction.

B. Where deficiencies are found, repair or replace assemblies to comply with requirements.

3.4 ADJUSTING AND CLEANING

A. Remove equipment, materials and debris, leaving area in undamaged, clean condition.

B. Clean surfaces adjacent to sealed openings free of excess materials and soiling as work progresses.

C. Perform patching and repair of firestopping systems damaged by other trades.

END OF SECTION 07 84 00

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization ☑ No ☐ Yes	
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 22 67 19 – PROCESS AND LABORATORY GAS PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Process Piping Systems, as indicated, in accordance with provisions of Contract Documents.

B. Work Included:
   1. Process piping, services and accessories.
   2. Rough piping and final connections to all equipment, other than process tools, whether furnished under this Section or not.
   3. Sleeves and supports; cutting, patching and framing of openings; and flashing of wall penetrations for work under this Section.

C. Systems and Accessories:
   1. Specialty gases as indicated on drawings.
   2. Owner furnished manifolds and gas racks.

D. Completely coordinate with work of other trades.

1.2 SUBMITTALS

A. Product Data:
   1. For piping and valves, submit list of suppliers with make and model number for materials, and manufacturer's printed data and illustrations of manufactured items identifying each size furnished.
   2. Submittals shall be approved by the Owner's representative before any purchase is made.

B. Contract Closeout Information:
   1. Operation and Maintenance Data for items requiring operational instructions or periodic maintenance such as: constant flow control valves, pressure relief valves, flow measurement devices, etc.
   2. Provide accurate as-built drawings and document any deviations from the original design.

1.3 QUALITY ASSURANCE

A. Valves: Manufacturer's name and pressure rating shall be marked on valve body.

B. Welding Materials and Procedures: Conform to ASME Code and applicable state labor regulations.

C. Welders Certification: In accordance with ANSI/AWS D1.1.

PART 2 - PRODUCTS

2.1 PIPE

A. CO2
1. General: Pipe and fittings cleaned for oxygen service or equal, per CGA pamphlet G-4.1. Use tubing filled with nitrogen purge and pipe ends capped. The interior surface shall be uniform, bright, smooth finish with particulate matter of no more than 0.038 g/in^2 (as called for in ASTM B280 specification).
2. Vacuum Insulated Transfer Hose with FNPT and MNPT ends.
3. Maximum allowable working pressures is 1240 PSI.
4. Tubing nominal length is 50 ft.
5. Use tube Manufacturer's tube support system.
6. Manufacturer: Cryoworks.

2.2 VALVES

1. Cryoworks Cryogenic ½" Globe Valve. C2041P-M11A.

2.3 LABORATORY CO2 MANIFOLD CONNECTIONS

1. Outlet tube fitting shall be per construction drawings.
2. Mount free-standing cylinder racks into concrete per manufacturer recommendations, submit anchoring submittal to SLAC for review and approval.

PART 3 - EXECUTION

3.1 PIPING

A. General:
1. Where different pipe materials connect, provide appropriate manufactured adapters or flanged connections using expanded PTFE, non-ink embossed "Gore-Tex" cut sheet gaskets; 1/8" thick Teflon® gasket. Torque to manufacturer's specifications.
2. Stubs: Install valves where indicated on the drawings and elsewhere to facilitate pipe cleaning, venting, drainage, flushing or testing.
3. Label all piping per ANSI/ASME A13.1 and in accordance with SLAC's Piping Identification Standard (SLAC-I-708-406-006-00).

3.2 CO2 INSTALLATION

A. Installation work shall conform to applicable codes and regulations. Work shall be done in a competent manner using the latest techniques of the trade.
1. Thread Fitting / Connections:

a. Only 100 percent pure PTFE Teflon tape is acceptable for all Process Gas Piping Systems not located in Radiologically Controlled Areas.

3.3 SYSTEMS PRESSURE TESTING

B. Pressure Test plans are required to prepare using SLAC's form and submit for approval from SLAC's Pressure Safety Manager. Pressure Test shall be recorded on SLAC's pressure test record form. Test using anything other than water must be approved in advance by the SLAC Pressure Safety Manager. Refer to ESH Manual CH 14 "Pressure Test Procedures" and Installation, Inspection, Maintenance and Repair Requirements"

C. Test piping as noted below. Piping shall meet test conditions with no leaks or loss in pressure. Repair or replace defective piping until tests are accomplished successfully. Use of oil pumped air or nitrogen is expressly forbidden. Nitrogen used for testing and purging operations shall be from a cryogenic source. Re-testing following piping repairs shall be performed for full specified time period.

System	Test Pressure	Test Medium	Test Time
Lab Specialty Gas	1.5x max operating pressure	Nitrogen	1 Hour

END OF SECTION 22 67 19

SECTION 230529 - HANGERS AND SUPPORTS FOR HVAC PIPING AND EQUIPMENT

SLAC BUILDING INSPECTION OFFICE
Construction Authorization (IFC)
BY: cnadler
DATE: 2/18/2026 PRS#: 26-009
Revised Authorization ☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Metal pipe hangers and supports.
2. Fastener systems.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Sustainable Design Submittals:

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Structural Performance: Hangers and supports for HVAC piping and equipment shall withstand the effects of gravity loads and stresses within limits and under conditions indicated according to ASCE/SEI 7.

1. Design supports for multiple pipes, including pipe stands, capable of supporting combined weight of supported systems, system contents, and test water.
2. Design seismic-restraint hangers and supports for piping.

2.2 METAL PIPE HANGERS AND SUPPORTS

A. Stainless Steel Pipe Hangers and Supports:

1. Description: MSS SP-58, Types 1 through 58, factory-fabricated components.
2. Padded Hangers: Hanger with fiberglass or other pipe insulation pad or cushion to support bearing surface of piping.
3. Hanger Rods: Continuous-thread rod, nuts, and washer made of stainless steel.

B. Copper Pipe and Tube Hangers:

1. Description: MSS SP-58, Types 1 through 58, copper-plated steel, factory-fabricated components.
2. Hanger Rods: Continuous-thread rod, nuts, and washer made of copper-plated steel or stainless steel.

2.3 FASTENER SYSTEMS

A. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

B. Mechanical-Expansion Anchors: Insert-wedge-type, zinc-coated or stainless steel anchors, for use in hardened portland cement concrete; with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

2.4 MATERIALS

A. Structural Steel: ASTM A36/A36M, carbon-steel plates, shapes, and bars; black and galvanized.

B. Stainless Steel: ASTM A240/A240M.

PART 3 - EXECUTION

3.1 APPLICATION

A. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb.

3.2 HANGER AND SUPPORT INSTALLATION

A. Metal Pipe-Hanger Installation: Comply with MSS SP-58. Install hangers, supports, clamps, and attachments as required to properly support piping from the building structure.

B. Framing System Installation: Arrange for grouping of parallel runs of piping, and support together on field-assembled strut systems.

C. Fastener System Installation:

1. Install powder-actuated fasteners for use in lightweight concrete or concrete slabs less than 4 inches thick in concrete after concrete is placed and completely cured. Use

operators that are licensed by powder-actuated tool manufacturer. Install fasteners according to powder-actuated tool manufacturer's operating manual.

2. Install mechanical-expansion anchors in concrete after concrete is placed and completely cured. Install fasteners according to manufacturer's written instructions.

D. Install hangers and supports complete with necessary attachments, inserts, bolts, rods, nuts, washers, and other accessories.

E. Install hangers and supports to allow controlled thermal and seismic movement of piping systems, to permit freedom of movement between pipe anchors, and to facilitate action of expansion joints, expansion loops, expansion bends, and similar units.

F. Install lateral bracing with pipe hangers and supports to prevent swaying.

G. Install building attachments within concrete slabs or attach to structural steel. Install additional attachments at concentrated loads, including valves, flanges, and strainers, NPS 2-1/2 and larger and at changes in direction of piping. Install concrete inserts before concrete is placed; fasten inserts to forms and install reinforcing bars through openings at top of inserts.

H. Load Distribution: Install hangers and supports so that piping live and dead loads and stresses from movement will not be transmitted to connected equipment.

I. Pipe Slopes: Install hangers and supports to provide indicated pipe slopes and to not exceed maximum pipe deflections allowed by ASME B31.9 for building services piping.

J. Insulated Piping:

1. Attach clamps and spacers to piping.

   a. Piping Operating above Ambient Air Temperature: Clamp may project through insulation.
   b. Piping Operating below Ambient Air Temperature: Use thermal-hanger shield insert with clamp sized to match OD of insert.
   c. Do not exceed pipe stress limits allowed by ASME B31.9 for building services piping.

2. Install MSS SP-58, Type 39, protection saddles if insulation without vapor barrier is indicated. Fill interior voids with insulation that matches adjoining insulation.

   a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.

3. Install MSS SP-58, Type 40, protective shields on cold piping with vapor barrier. Shields shall span an arc of 180 degrees.

   a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.

4. Shield Dimensions for Pipe: Not less than the following:

a. NPS 1/4 to NPS 3-1/2: 12 inches long and 0.048 inch thick.
b. NPS 4: 12 inches long and 0.06 inch thick.
c. NPS 5 and NPS 6: 18 inches long and 0.06 inch thick.
d. NPS 8 to NPS 14: 24 inches long and 0.075 inch thick.
e. NPS 16 to NPS 24: 24 inches long and 0.105 inch thick.

5. Pipes NPS 8 and Larger: Include wood or reinforced calcium-silicate-insulation inserts of length at least as long as protective shield.
6. Thermal-Hanger Shields: Install with insulation same thickness as piping insulation.

3.3 METAL FABRICATIONS

A. Fit exposed connections together to form hairline joints. Field weld connections that cannot be shop welded because of shipping size limitations.

B. Field Welding: Comply with AWS D1.1/D1.1M procedures for shielded, metal arc welding; appearance and quality of welds; and methods used in correcting welding work; and with the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.
4. Finish welds at exposed connections so no roughness shows after finishing and so contours of welded surfaces match adjacent contours.

3.4 ADJUSTING

A. Hanger Adjustments: Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe.

B. Trim excess length of continuous-thread hanger and support rods to 1-1/2 inches.

3.5 PAINTING

A. Touchup: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.

1. Apply paint by brush or spray to provide a minimum dry film thickness of 2.0 mils.

B. Touchup: Comply with requirements in Division 09 specification sections for cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal.

C. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A780/A780M.

3.6 HANGER AND SUPPORT SCHEDULE

A. Specific hanger and support requirements are in Sections specifying piping systems.

B. Comply with MSS SP-58 for pipe-hanger selections and applications that are not specified in piping system Sections.

C. Use hangers and supports with galvanized metallic coatings for piping that will not have field-applied finish.

D. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.

E. Use carbon-steel pipe hangers and supports metal trapeze pipe hangers and metal framing systems and attachments for general service applications.

F. Use copper-plated pipe hangers and copper or stainless steel attachments for copper piping and tubing.

G. Horizontal-Piping Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Unistrut P1000 Series as support for pipes or equivalent.
2. Adjustable, Steel Clevis Hangers (MSS Type 1): For suspension of noninsulated or insulated, stationary pipes NPS 1/2 to NPS 30.
3. Yoke-Type Pipe Clamps (MSS Type 2): For suspension of up to 1050 deg F, pipes NPS 4 to NPS 24, requiring up to 4 inches of insulation.
4. Carbon- or Alloy-Steel, Double-Bolt Pipe Clamps (MSS Type 3): For suspension of pipes NPS 3/4 to NPS 36, requiring clamp flexibility and up to 4 inches of insulation.
5. Steel Pipe Clamps (MSS Type 4): For suspension of cold and hot pipes NPS 1/2 to NPS 24 if little or no insulation is required.
6. Pipe Hangers (MSS Type 5): For suspension of pipes NPS 1/2 to NPS 4, to allow off-center closure for hanger installation before pipe erection.
7. Adjustable, Swivel Split- or Solid-Ring Hangers (MSS Type 6): For suspension of noninsulated, stationary pipes NPS 3/4 to NPS 8.
8. Adjustable, Steel Band Hangers (MSS Type 7): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
9. Adjustable Band Hangers (MSS Type 9): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
10. Adjustable, Swivel-Ring Band Hangers (MSS Type 10): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
11. Split Pipe Ring with or without Turnbuckle Hangers (MSS Type 11): For suspension of noninsulated, stationary pipes NPS 3/8 to NPS 8.

12. Extension Hinged or Two-Bolt Split Pipe Clamps (MSS Type 12): For suspension of noninsulated, stationary pipes NPS 3/8 to NPS 3.
13. U-Bolts (MSS Type 24): For support of heavy pipes NPS 1/2 to NPS 30.
14. Clips (MSS Type 26): For support of insulated pipes not subject to expansion or contraction.
15. Pipe Saddle Supports (MSS Type 36): For support of pipes NPS 4 to NPS 36, with steel-pipe base stanchion support and cast-iron floor flange or carbon-steel plate.
16. Pipe Stanchion Saddles (MSS Type 37): For support of pipes NPS 4 to NPS 36, with steel-pipe base stanchion support and cast-iron floor flange or carbon-steel plate, and with U-bolt to retain pipe.
17. Adjustable Pipe Saddle Supports (MSS Type 38): For stanchion-type support for pipes NPS 2-1/2 to NPS 36 if vertical adjustment is required, with steel-pipe base stanchion support and cast-iron floor flange.
18. Single-Pipe Rolls (MSS Type 41): For suspension of pipes NPS 1 to NPS 30, from two rods if longitudinal movement caused by expansion and contraction might occur.
19. Adjustable Roller Hangers (MSS Type 43): For suspension of pipes NPS 2-1/2 to NPS 24, from single rod if horizontal movement caused by expansion and contraction might occur.
20. Complete Pipe Rolls (MSS Type 44): For support of pipes NPS 2 to NPS 42 if longitudinal movement caused by expansion and contraction might occur but vertical adjustment is unnecessary.
21. Pipe Roll and Plate Units (MSS Type 45): For support of pipes NPS 2 to NPS 24 if small horizontal movement caused by expansion and contraction might occur and vertical adjustment is unnecessary.
22. Adjustable Pipe Roll and Base Units (MSS Type 46): For support of pipes NPS 2 to NPS 30 if vertical and lateral adjustment during installation might be required in addition to expansion and contraction.

H. Vertical-Piping Clamps: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Unistrut P1000 Series as support for pipes or equivalent.

2. Extension Pipe or Riser Clamps (MSS Type 8): For support of pipe risers NPS 3/4 to NPS 24.
3. Carbon- or Alloy-Steel Riser Clamps (MSS Type 42): For support of pipe risers NPS 3/4 to NPS 24 if longer ends are required for riser clamps.

I. Hanger-Rod Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel Turnbuckles (MSS Type 13): For adjustment up to 6 inches for heavy loads.
2. Steel Clevises (MSS Type 14): For 120 to 450 deg F piping installations.
3. Swivel Turnbuckles (MSS Type 15): For use with MSS Type 11, split pipe rings.
4. Malleable-Iron Sockets (MSS Type 16): For attaching hanger rods to various types of building attachments.
5. Steel Weldless Eye Nuts (MSS Type 17): For 120 to 450 deg F piping installations.

J. Building Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel or Malleable Concrete Inserts (MSS Type 18): For upper attachment to suspend pipe hangers from concrete ceiling.
2. Side-Beam or Channel Clamps (MSS Type 20): For attaching to bottom flange of beams, channels, or angles.
3. Center-Beam Clamps (MSS Type 21): For attaching to center of bottom flange of beams.
4. Welded Beam Attachments (MSS Type 22): For attaching to bottom of beams if loads are considerable and rod sizes are large.
5. C-Clamps (MSS Type 23): For structural shapes.
6. Top-Beam Clamps (MSS Type 25): For top of beams if hanger rod is required tangent to flange edge.
7. Side-Beam Clamps (MSS Type 27): For bottom of steel I-beams.
8. Steel-Beam Clamps with Eye Nuts (MSS Type 28): For attaching to bottom of steel I-beams for heavy loads.
9. Linked-Steel Clamps with Eye Nuts (MSS Type 29): For attaching to bottom of steel I-beams for heavy loads, with link extensions.
10. Malleable-Beam Clamps with Extension Pieces (MSS Type 30): For attaching to structural steel.
11. Welded-Steel Brackets: For support of pipes from below or for suspending from above by using clip and rod. Use one of the following for indicated loads:

    a. Light (MSS Type 31): 750 lb.
    b. Medium (MSS Type 32): 1500 lb.
    c. Heavy (MSS Type 33): 3000 lb.

12. Side-Beam Brackets (MSS Type 34): For sides of steel or wooden beams.
13. Plate Lugs (MSS Type 57): For attaching to steel beams if flexibility at beam is required.
14. Horizontal Travelers (MSS Type 58): For supporting piping systems subject to linear horizontal movement where headroom is limited.

K. Saddles and Shields: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel-Pipe-Covering Protection Saddles (MSS Type 39): To fill interior voids with insulation that matches adjoining insulation.
2. Protection Shields (MSS Type 40): Of length recommended in writing by manufacturer to prevent crushing insulation.

L. Spring Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Restraint-Control Devices (MSS Type 47): Where indicated to control piping movement.
2. Spring Cushions (MSS Type 48): For light loads if vertical movement does not exceed 1-1/4 inches.
3. Spring-Cushion Roll Hangers (MSS Type 49): For equipping Type 41, roll hanger with springs.

4. Spring Sway Braces (MSS Type 50): To retard sway, shock, vibration, or thermal expansion in piping systems.
5. Variable-Spring Hangers (MSS Type 51): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from hanger.
6. Variable-Spring Base Supports (MSS Type 52): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from base support.
7. Variable-Spring Trapeze Hangers (MSS Type 53): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from trapeze support.
8. Constant Supports: For critical piping stress and if necessary to avoid transfer of stress from one support to another support, critical terminal, or connected equipment. Include auxiliary stops for erection, hydrostatic test, and load-adjustment capability. These supports include the following types:

   a. Horizontal (MSS Type 54): Mounted horizontally.
   b. Vertical (MSS Type 55): Mounted vertically.
   c. Trapeze (MSS Type 56): Two vertical-type supports and one trapeze member.

M. Use powder-actuated fasteners or mechanical-expansion anchors instead of building attachments where required in concrete construction.

END OF SECTION 230529

SECTION 23 05 53 - IDENTIFICATION FOR HVAC PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Warning signs and labels.
2. Pipe labels.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Samples: For color, letter style, and graphic representation required for each identification material and device.

C. Equipment-Label Schedule: Include a listing of all equipment to be labeled with the proposed content for each label.

PART 2 - PRODUCTS

2.1 WARNING SIGNS AND LABELS

A. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, with predrilled holes for attachment hardware.

B. Letter and Background Color: As indicated for specific application under Part 3.

C. Maximum Temperature: Able to withstand temperatures of up to 160 deg F.

D. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

E. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances of up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.

F. Fasteners: Stainless steel rivets or self-taping screws.

G. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

H. Arc-Flash Warning Signs: Provide arc-flash warning signs in locations and with content in accordance with requirements of OSHA and NFPA70E.

I. Label Content: Include caution and warning information plus emergency notification instructions.

2.2 PIPE LABELS

A. General Requirements for Manufactured Pipe Labels: Preprinted, color coded, with lettering indicating service and showing flow direction in accordance with ASME A13.1.

B. Letter and Background Color: As indicated for specific application under Part 3.

C. Pretensioned Pipe Labels: Precoiled, semirigid plastic formed to cover full circumference of pipe and to attach to pipe without fasteners or adhesive.

D. Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.

E. Pipe Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings. Also include:

1. Pipe size.
2. Flow-Direction Arrows: Include flow-direction arrows on distribution piping. Arrows may be either integral with label or applied separately.
3. Lettering Size: Size letters in accordance with ASME A13.1 for piping.

2.3 DUCT LABELS

A. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, and having predrilled holes for attachment hardware.

B. Letter and Background Color: As indicated for specific application under Part 3.

C. Maximum Temperature: Able to withstand temperatures up to 160 deg F.

D. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

E. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances of up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.

F. Fasteners: Stainless steel rivets or self-tapping screws.

G. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

PART 3 - EXECUTION

3.1 PREPARATION

A. Clean piping and equipment surfaces of incompatible primers, paints, and encapsulants, as well as dirt, oil, grease, release agents, and other substances that could impair bond of identification devices.

3.2 INSTALLATION, GENERAL REQUIREMENTS

A. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.

B. Coordinate installation of identifying devices with locations of access panels and doors.

C. Install identifying devices before installing acoustical ceilings and similar concealment.

D. Locate identifying devices so that they are readily visible from the point of normal approach.

3.3 INSTALLATION OF PIPE LABELS

A. Install pipe labels showing service and flow direction with permanent adhesive on pipes.

B. Pipe-Label Locations: Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:

1. Within 3 ft. of each valve and control device.
2. At access doors, manholes, and similar access points that permit view of concealed piping.
3. Within 3 ft. of equipment items and other points of origination and termination.
4. Spaced at maximum intervals of 25 ft. along each run. Reduce intervals to 10 ft. in areas of congested piping, ductwork, and equipment.

C. Do not apply plastic pipe labels or plastic tapes directly to bare pipes conveying fluids at temperatures of 125 deg F or higher. Where these pipes are to remain uninsulated, use a short section of insulation or use stenciled labels.

D. Flow-Direction Arrows: Use arrows to indicate direction of flow in pipes, including pipes where flow is allowed in both directions.

E. Pipe-Label Color Schedule:

1. Chilled-Water Piping: White letters on an ANSI Z535.1 safety-green background.

END OF SECTION 230553

SECTION 23 07 19 - HVAC PIPING INSULATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes insulation for HVAC piping systems.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Sustainable Design Submittals:

C. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.

1. Detail application of protective shields, saddles, and inserts at hangers for each type of insulation and hanger.
2. Detail attachment and covering of heat tracing inside insulation.
3. Detail insulation application at pipe expansion joints for each type of insulation.
4. Detail insulation application at elbows, fittings, flanges, valves, and specialties for each type of insulation.
5. Detail removable insulation at piping specialties.

D. Samples: For each type of insulation and jacket indicated. Identify each Sample, describing product and intended use.

1.3 QUALITY ASSURANCE

A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or another craft training program certified by the Department of Labor, Bureau of Apprenticeship and Training.

B. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products in accordance with ASTM E84, by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.

1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.

1.4 COORDINATION

A. Coordinate sizes and locations of supports, hangers, and insulation shields specified in Section 230529 "Hangers and Supports for HVAC Piping and Equipment."

B. Coordinate clearance requirements with piping Installer for piping insulation application. Before preparing piping Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.

C. Coordinate installation and testing of heat tracing.

1.5 SCHEDULING

A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results.

B. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

PART 2 - PRODUCTS

2.1 INSULATION MATERIALS

A. Comply with requirements in Pipe insulation schedule below.

B. Products shall not contain asbestos, lead, mercury, or mercury compounds.

C. Products that come into contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested in accordance with ASTM C871.

D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable in accordance with ASTM C795.

E. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.

F. Insulated Pipe Shield Calcium insulation shall be used for each pipe at each support.

2.2 SEALANTS

A. Materials shall be as recommended by the insulation manufacturer and shall be compatible with insulation materials, jackets, and substrates.

B. Joint Sealants:
   1. Permanently flexible, elastomeric sealant.

a. Service Temperature Range: Minus 150 to plus 250 deg F.
b. Color: White or gray.

C. FSK and Metal Jacket Flashing Sealants:
1. Fire- and water-resistant, flexible, elastomeric sealant.
2. Service Temperature Range: Minus 40 to plus 250 deg F.
3. Color: Aluminum.

D. ASJ Flashing Sealants and PVDC and PVC Jacket Flashing Sealants:
1. Fire- and water-resistant, flexible, elastomeric sealant.
2. Service Temperature Range: Minus 40 to plus 250 deg F.
3. Color: White.

2.3 FACTORY-APPLIED JACKETS

A. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

1. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C1136, Type I.
2. ASJ-SSL: ASJ with self-sealing, pressure-sensitive, acrylic-based adhesive covered by a removable protective strip; complying with ASTM C1136, Type I.
3. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C1136, Type II.

2.4 TAPES

A. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C1136.
1. Width: 3 inches.
2. Thickness: 11.5 mils.
3. Adhesion: 90 ounces force/inch in width.
4. Elongation: 2 percent.
5. Tensile Strength: 40 lbf/inch in width.
6. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.

B. FSK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C1136.
1. Width: 3 inches.
2. Thickness: 6.5 mils.
3. Adhesion: 90 ounces force/inch in width.
4. Elongation: 2 percent.
5. Tensile Strength: 40 lbf/inch in width.
6. FSK Tape Disks and Squares: Precut disks or squares of FSK tape.

C. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor applications.
1. Width: 2 inches.

2. Thickness: 6 mils.
3. Adhesion: 64 ounces force/inch in width.
4. Elongation: 500 percent.
5. Tensile Strength: 18 lbf/inch in width.

D. Aluminum-Foil Tape: Vapor-retarder tape with acrylic adhesive.
1. Width: 2 inches.
2. Thickness: 3.7 mils.
3. Adhesion: 100 ounces force/inch in width.
4. Elongation: 5 percent.
5. Tensile Strength: 34 lbf/inch in width.

E. PVDC Tape for Indoor Applications: White vapor-retarder PVDC tape with acrylic adhesive.
1. Width: 3 inches.
2. Film Thickness: 2 mils.
3. Adhesive Thickness: 1.5 mils.
4. Elongation at Break: 120 percent.
5. Tensile Strength: 20 psi in width.

2.5 SECUREMENTS

A. Bands:
1. Stainless Steel: ASTM A240/A240M, Type 304 or Type 316; 0.015 inch thick, 1/2 inch wide with wing seal or closed seal.
2. Aluminum: ASTM B209, Alloy 3003, 3005, 3105, or 5005; Temper H-14, 0.020 inch thick, 1/2 inch wide with wing seal or closed seal.
3. Springs: Twin spring set constructed of stainless steel, with ends flat and slotted to accept metal bands. Spring size is determined by manufacturer for application.

B. Staples: Outward-clinching insulation staples, nominal 3/4 inch wide, stainless steel or Monel.

C. Wire: 0.080-inch nickel-copper alloy, 0.062-inch soft-annealed, stainless steel or 0.062-inch soft-annealed, galvanized steel.

PART 3 - EXECUTION

3.1 PREPARATION

A. Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

B. Clean and prepare surfaces to be insulated. Before insulating, apply a corrosion coating to insulated surfaces as follows:

1. Stainless Steel: Coat 300 series stainless steel with an epoxy primer 5 mils thick and an epoxy finish 5 mils thick if operating in a temperature range between 140 and 300 deg F.

Consult coating manufacturer for appropriate coating materials and application methods for operating temperature range.

2. Carbon Steel: Coat carbon steel operating at a service temperature of between 32 and 300 deg F with an epoxy coating. Consult coating manufacturer for appropriate coating materials and application methods for operating temperature range.

C. Coordinate insulation installation with the tradesman installing heat tracing. Comply with requirements for heat tracing that apply to insulation.

D. Mix insulating cements with clean potable water; if insulating cements are to be in contact with stainless steel surfaces, use demineralized water.

3.2 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of piping, including fittings, valves, and specialties.

B. Install insulation materials, forms, vapor barriers or retarders, jackets, and of thicknesses required for each item of pipe system, as specified in insulation system schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

D. Install insulation with longitudinal seams at top and bottom of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Do not weld brackets, clips, or other attachment devices to piping, fittings, and specialties.

G. Keep insulation materials dry during storage, application, and finishing. Replace insulation materials that get wet.

H. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

I. Install insulation with least number of joints practical.

J. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.

1. Install insulation continuously through hangers and around anchor attachments.
2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends attached to structure with vapor-barrier mastic.
3. Install insert materials and insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.

4. Cover inserts with jacket material matching adjacent pipe insulation. Install shields over jacket, arranged to protect jacket from tear or puncture by hanger, support, and shield.

K. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

L. Install insulation with factory-applied jackets as follows:

1. Draw jacket tight and smooth.
2. Cover circumferential joints with 3-inch-wide strips, of same material as insulation jacket. Secure strips with adhesive and outward-clinching staples along both edges of strip, spaced 4 inches o.c.
3. Overlap jacket longitudinal seams at least 1-1/2 inches. Install insulation with longitudinal seams at bottom of pipe. Clean and dry surface to receive self-sealing lap. Staple laps with outward-clinching staples along edge at [**2 inches**] [**4 inches**] o.c.

    a. For below-ambient services, apply vapor-barrier mastic over staples.

4. Cover joints and seams with tape, in accordance with insulation material manufacturer's written instructions, to maintain vapor seal.
5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints and at ends adjacent to pipe flanges and fittings.

M. Cut insulation in a manner to avoid compressing insulation more than 25 percent of its nominal thickness.

N. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

O. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches in similar fashion to butt joints.

P. For above-ambient services, do not install insulation to the following:

1. Vibration-control devices.
2. Testing agency labels and stamps.
3. Nameplates and data plates.

3.3 GENERAL PIPE INSULATION INSTALLATION

A. Requirements in this article generally apply to all insulation materials, except where more specific requirements are specified in various pipe insulation material installation articles.

B. Insulation Installation on Fittings, Valves, Strainers, Flanges, Mechanical Couplings, and Unions:

1. Install insulation over fittings, valves, strainers, flanges, mechanical couplings, unions, and other specialties with continuous thermal and vapor-retarder integrity unless otherwise indicated.

2. Insulate pipe elbows using preformed fitting insulation or mitered fittings made from same material and density as that of adjacent pipe insulation. Each piece shall be butted tightly against adjoining piece and bonded with adhesive. Fill joints, seams, voids, and irregular surfaces with insulating cement finished to a smooth, hard, and uniform contour that is uniform with adjoining pipe insulation.
3. Insulate tee fittings with preformed fitting insulation or sectional pipe insulation of same material and thickness as that used for adjacent pipe. Cut sectional pipe insulation to fit. Butt each section closely to the next and hold in place with tie wire. Bond pieces with adhesive.
4. Insulate valves using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as that used for adjacent pipe. Overlap adjoining pipe insulation by not less than 2 times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. For valves, insulate up to and including the bonnets, valve stuffing-box studs, bolts, and nuts. Fill joints, seams, and irregular surfaces with insulating cement.
5. Insulate strainers using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as that used for adjacent pipe. Overlap adjoining pipe insulation by not less than 2 times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Fill joints, seams, and irregular surfaces with insulating cement. Insulate strainers, so strainer basket flange or plug can be easily removed and replaced without damaging the insulation and jacket. Provide a removable reusable insulation cover. For below-ambient services, provide a design that maintains vapor barrier.
6. Insulate flanges, mechanical couplings, and unions using a section of oversized preformed pipe insulation to fit. Overlap adjoining pipe insulation by not less than 2 times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Stencil or label the outside insulation jacket of each union with the word "union" matching size and color of pipe labels.
7. Cover segmented insulated surfaces with a layer of finishing cement and coat with a mastic. Install vapor-barrier mastic for below-ambient services and a breather mastic for above-ambient services. Reinforce the mastic with reinforcing mesh. Trowel the mastic to a smooth and well-shaped contour.
8. For services not specified to receive a field-applied jacket, except for flexible elastomeric and polyolefin, install fitted PVC cover over elbows, tees, strainers, valves, flanges, and unions. Terminate ends with PVC end caps. Tape PVC covers adjoining insulation facing, using PVC tape.

C. Insulate instrument connections for thermometers, pressure gages, pressure temperature taps, test connections, flow meters, sensors, switches, and transmitters on insulated pipes. Shape insulation at these connections by tapering it to and around the connection with insulating cement and finish with finishing cement, mastic, and flashing sealant.

D. Install removable insulation covers at locations indicated. Installation shall conform to the following:

1. Make removable flange and union insulation from sectional pipe insulation of same thickness as that on adjoining pipe. Install same insulation jacket as that of adjoining pipe insulation.
2. When flange and union covers are made from sectional pipe insulation, extend insulation from flanges or union at least 2 times the insulation thickness over adjacent pipe insulation on each side of flange or union. Secure flange cover in place with stainless steel or aluminum bands. Select band material compatible with insulation and jacket.

3. Construct removable valve insulation covers in same manner as for flanges, except divide the two-part section on the vertical center line of valve body.
4. When covers are made from block insulation, make two halves, each consisting of mitered blocks wired to stainless steel fabric. Secure this wire frame, with its attached insulation, to flanges with tie wire. Extend insulation at least 2 inches over adjacent pipe insulation on each side of valve. Fill space between flange or union cover and pipe insulation with insulating cement. Finish cover assembly with insulating cement applied in two coats. After first coat is dry, apply and trowel second coat to a smooth finish.
5. Unless a PVC jacket is indicated in field-applied jacket schedules, finish exposed surfaces with a metal jacket.

3.4 FINISHES

A. Flexible Elastomeric Thermal Insulation: After adhesive has fully cured, apply two coats of insulation manufacturer's recommended protective coating.

B. Color: Final color as selected by Architect. Vary first and second coats to allow visual inspection of the completed Work.

C. Do not field paint aluminum or stainless steel jackets.

3.5 FIELD QUALITY CONTROL

A. Owner will engage a qualified testing agency to perform tests and inspections.

B. Engage a qualified testing agency to perform tests and inspections.

C. Perform tests and inspections.

D. Tests and Inspections: Inspect pipe, fittings, strainers, and valves, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to three locations of straight pipe, three locations of threaded fittings, three locations of welded fittings, two locations of threaded strainers, two locations of welded strainers, three locations of threaded valves, and three locations of flanged valves for each pipe service defined in the "Piping Insulation Schedule, General" Article.

E. All insulation applications will be considered defective if they do not pass tests and inspections.

F. Prepare test and inspection reports.

3.6 PIPING INSULATION SCHEDULE, GENERAL

A. Insulation conductivity and thickness per pipe size shall comply with schedules in this Section or with requirements of authorities having jurisdiction, whichever is more stringent.

B. Acceptable preformed pipe and tubular insulation materials and thicknesses are identified for each piping system and pipe size range. If more than one material is listed for a piping system, selection from materials listed is Contractor's option.

C. Items Not Insulated: Unless otherwise indicated, do not install insulation on the following:

1. Chrome-plated pipes and fittings unless there is a potential for personnel injury.

3.7 INDOOR PIPING INSULATION SCHEDULE

A. Chilled Water:

1. NPS 12 and Smaller: Insulation shall be one of the following:

a. Cellular Glass: Insulation to meet Title 24 requirements..
b. Flexible Elastomeric: Insulation to meet Title 24 requirements.
c. Mineral-Fiber, Preformed Pipe Insulation, Type I: Insulation to meet Title 24 requirements.
d. Polyolefin: Insulation to meet Title 24 requirements.

3.8 INDOOR, FIELD-APPLIED JACKET SCHEDULE

A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.

B. If more than one material is listed, selection from materials listed is Contractor's option.

C. Piping, Exposed:

1. None.
2. PVC: 20 mils thick.
3. Aluminum: 0.020 inch thick.
4. Painted Aluminum: 0.020 inch thick.

END OF SECTION 230719

SECTION 23 21 13 – HYDRONIC PIPING

PART 1 – GENERAL

1.1 SECTION INCLUDES

A. Pipe and Pipe Fittings.
B. Valves.
C. Chilled Water Piping System.

1.2 QUALITY ASSURANCE

A. Valves: Manufacturer's name and pressure rating marked on valve body. Remanufactured valves are <u>not</u> acceptable.

B. Welding Materials, Procedures, and Operators: Conform to ASME Section 9, ANSI/AWS D1.1, and applicable state labor regulations.

C. Pressure systems and piping shall be in compliance with Chapter 14 of SLAC's ES&H Manual. The contractor shall review this document and ensure all requirements are met.

1.3 REFERENCES (ANSI /ASME Latest edition)

A. ANSI/AWS D1.1 – Structural Welding Code.
B. ANSI/AWS A5.8 – Brazing Filler Metal.
C. ANSI B2.1 – Brazing Filler Metal.
D. ASME B16.1 - Cast Iron Pipe Flanges and Flanged Fittings, Class 25, 125, 250, and 800.
E. ASME B16.3 - Malleable Iron Threaded Fittings Class 150 and 300.
F. ASME B16.4 - Cast Iron Threaded Fittings, Class 125 and 250.
G. ASME B16.5 - Pipe Flanges and Flanged Fittings.
H. ASME B16.9 - Factory-Made Wrought Steel Butt Welding Fittings.
I. ASME B16.11 - Factory-Made Steel Socket Welding Fittings.
J. ASME B16.12 - Cast Iron Threaded Drainage Fittings.
K. ASME B16.18 - Cast Copper Alloy Solder Joint Pressure Fittings
L. ASME B16.21 - Nonmetallic Flat Gaskets for Pipes Flanges.
M. ASME B16.22 - Wrought Copper and Copper Alloy Solder Joint Pressure Fittings.
N. ASME B16.23 - Cast Copper Alloy Solder Joint Drainage Fittings (DWV).
O. ANSI B16.24 - Bronze Flanges and Flanged Fittings.
P. ANSI B16.28 – Wrought Steel Buttwelding Short Radius Elbows and Returns.
Q. ASME B16.29 - Wrought Copper and Wrought Copper Alloy Solder Joint Drainage Fittings
R. ANSI B31.2 - Fuel Gas Piping.
S. ANSI B36.10 - Wrought Steel and Wrought Iron Pipe.
T. ASME B18.2.1 - Square and Hex Bolts and Screws, Inch Series.

N. ASME B18.2.2 - Square and Hex Nuts, Inch Series.
O. ASME B31.3 - Process Piping - 2014 Edition.
P. ASME B31.9 - Building Services Piping - 2014 Edition.
Q. ASME Section 9 - Welding and Brazing Qualifications.
R. ASTM A 47 – Malleable Iron Castings.
S. ASTM A 53 - Pipe, Steel, Black and Hot-Dipped Zinc Coated, Welded and Seamless.
T. ASTM A 106 – Seamless Carbon Steel Pipe for High –Temperature Service.
U. ASTM A126 - Gray Cast Iron Castings for Valves, Flanges, and Pipe Fittings.
V. ASTM A181 - Forgings, Carbon Steel for General Purpose Piping.
W. ASTM A 197 – Cupola Malleable Iron.
X. ASTM A 234 - Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service.
Y. ASTM A 307 - Standard Specification for Carbon Steel Bolts and Studs, 60,000 PSI Tensile Strength.
Z. ASTM A733 - Standard Specification for Welded and Seamless Carbon Steel and Austenitic Stainless Steel Pipe Nipples.
AA. ASTM B32 - Standard Specification for Solder Metal.
BB. ASTM B88 - Seamless Copper Water Tube.
CC. ASTM B813 - Standard Specification for Liquid and Paste Fluxes for Soldering Applications of Copper and Copper Alloy Tube.
AA. ASTM D2105 - Standard Test Method for Longitudinal Tensile Properties of Fiberglass Pipe and tube.
BB. ASTM D2412 - Standard Test Method for Determination of External Loading Characteristics of Plastic Pipe by Parallel-Plate loading.
CC. ASTM E90-02 - Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions
DD. ASTM E413-87 – Classification for Rating Sound Insulation EE National Fire Protection Association (NFPA)
EE. SLAC ESH Manual – Chapter 14
FF. Sheet Metal and Air Conditioning Contractor National Association
GG. SMACNA Guidelines for Seismic Restraints of Mechanical Systems

1.4 SUBMITTALS

A. Submit product data. Include data on pipe materials, fittings, valves, and accessories.

B. Submittals shall meet the requirements of SLAC ESH Manual - Chapter 14.

C. Prior to procurement for approval:

1. Contact the pressure systems manager and submit catalogs and specification sheets.

2. Submit drawings and calculations if applicable.

D. Prior to installation:

1. Contact the pressure systems manager and submit installation procedures.

2. Submit installation manuals, or manufacturer's recommendations for installations.

E. Prior to pressure testing:

1. Contact the pressure systems manager and submit pressure test plan.

2. Submit pressure test layout.

3. Submit pressure test procedures.

4. Justification for pneumatic test

F. Prior to system operation:

1. Contact the pressure systems manager for inspection and registration.

a. A pressure system cannot be operated if not registered with the Pressure Systems Database.

2. Submit service manuals, and operating procedures.

1.5 QUALITY ASSURANCE

A. Valves: Manufacturer' name and pressure rating shall be marked on the valve body.

B. Welders' certification in accordance with ANSI/AWS D1.1

1.6 DELIVERY, STORAGE, AND HANDLING

A. Store and protect piping to prevent entrance of foreign matter into pipe and to prevent exterior corrosion.

B. Deliver and store valves in shipping containers with labeling in place.

1.7 COORDINATION DRAWINGS

C. CAD drawings to be provided to Coordinating Contractor for inclusion into composite coordination drawings.

PART 2 - PRODUCTS

2.1 MATERIALS SCHEDULE

Refer to the following subparts for complete material specifications for each pipe class specified below. All below grade nonmagnetic pipe shall have a tracer wire installed. All below grade fittings shall be epoxy coated and hardware shall be tar coated.

Piping System	Material Specification
Chilled Water Supply & Return (3" or less)	Pipe Class C1, Type L with 95-5 tin antimony solder (copper)

2.2 PIPE CLASS C1: COPPER TUBE AND FITTINGS (TYPE L)

A. Tubing: Copper tubing, hard drawn temper, Type L, as specified, ASTM B 88 soldered joints or annealed temper, type L, ASTM B88.

B. Fittings: Wrought copper, socket solder-type joint, ASTM B 88 and ANSI B16.22. Couplings shall be of the staked stop type unless full slip types are required for existing piping modifications.

C. Compression fittings shall be Swagelok, Parker, Hoke or equivalent manufacturer’s brand.

D. Flanges: Socket solder-type joint, 150 lb, plain (flat) face, cast bronze, ASTM B 62. Dimensions shall be in accordance with ANSI B16.24.

E. Unions: Socket solder-joint ends, cast bronze, ASTM B 62. Dimensions shall be in accordance with ANSI B16.18.

F. Solder: 95-5 tin antimony, or silver brazing alloy, as specified. Silver brazing alloy shall be ANSI/AWS A5.8, classification BCUP-5 containing 15% silver, 80% copper, 5% phosphorous such as Sil-Fos (Silvaloy 15) or Silvabrite 100 (0.4% Silver).

2.3 VALVE SCHEDULE.

Service	Type
Chilled water	Apollo Valves 1" 9A-105-01 Ball Valve Standard Configuration Bell & Gossett 3/4" NPT CB-3/4 Lead Free Circuit Setter Balance Valve

PART 3: EXECUTION

3.1 PREPARATION

A. Ream pipe and tube ends. Remove burrs. Bevel plain end pipe for butt welding.

B. Remove scale and dirt, on inside and outside, before assembly.

C. Prepare piping connections to equipment with flanges or unions.

3.2 PIPE AND TUBING INSTALLATION

A. Route piping in orderly manner and maintain gradient.

B. Install piping to conserve building space and not interfere with use of space.

C. Group piping whenever practical at common elevations.

D. Install piping to allow for expansion and contraction without stressing pipe, joints, or connected equipment.

E. Provide clearance for installation of insulation and access to valves and fittings.

F. Provide access where valves and fittings are not exposed.

G. Unless otherwise indicated, use wrought copper/bronze solder joint fittings complying with ANSI B16.22.1995

H. Route tubing to minimize the number of connections.

I. Swagelok brand compression fittings or equivalent fitting manufacturers such as Parker, Autoclave Engineers, Hoke, Ham-Let, and Imperial Eastman may be used.

1. Install all mechanical connectors in accordance with the manufacturer's instructions.

2. Fittings to be rated at a minimum of 150 percent design operating pressure.

3. Do not connect, mix, or interchange parts (caps, plugs, ferrules, bodies, etc.) of tube fittings made by different manufacturers (such as Parker to Swagelok). Improper fitting seal, DAMAGE, or INJURIES may result.

4. All selected copper tubing for use with a compression fitting must be annealed with a Rockwell hardness of less than 80 (Rb 80) for success in fitting makeup.

5. Metal tubing must always be softer than the compression fitting material. When the tubing and fitting are made of the same material, the tubing must be fully annealed.

J. Slope water piping and arrange to drain at low points.

K. Establish elevations of buried piping outside the building to ensure not less than 3 ft of cover.

L. Where pipe support members are welded to structural building framing, scrape, brush clean, and apply one coat of zinc rich primer to welding.

M. Prepare pipe, fittings, supports, and accessories not prefinished, ready for finish painting.

N. Establish invert elevations, slopes for drainage to ¼ inch per foot minimum. Maintain gradients.

O. Install bell and spigot pipe with bell end upstream.

P. Arrange refrigeration piping to return oil to compressor. Provide traps and loops in piping, and provide double risers as required. Slope horizontal piping 0.40 percent in direction of flow.

Q. Flood refrigerant piping system with nitrogen when brazing.

R. Follow ASHRAE 15 procedures for charging and purging of systems and for disposal of refrigerant.

3.3 PIPE JOINT CONNECTIONS

A. Threaded Pipe: Use Loctite 580 PST thread sealant, or equal, for general service applications, temperatures from –50 F to +400 F, metal threads, non- toxic, non-hardening. Do not use Teflon tape or any product that contains PTFE (TFE, FEP) in radiation areas.

B. Copper Tubing with Solder Joint Fittings: Use silver brazed joints for high pressure (>100 psi) or critical systems, including any piping located in or under concrete slab on ground, condensate piping located in underground conduits and manholes, and for attaching “Brazolet” fittings for any service.

1. Silver brazing alloy shall comply with ANSI/AWS A5.8, Class BCUP-5. classification BCUP-5 containing 15% silver, 80% copper, 5% phosphorous such as Sil-Fos (Silvaloy 15). Use care in silver brazing to prevent overheating of pipe and fittings. Disassemble solder type valves before silver brazing and keep bodies cool.

2. Make all other joints with 95-5 tin antimony, ASTM B32, Grade 5A. or Silvabrite 0.4% Silver solder.

C. Steel Pipe and Welding Fittings: Inert gas purge of pipe ID during welding is required. Weld procedure submittal is required before any pipe weld is allowed. Make joints by electric arc welding continuously around pipe. Welders shall be qualified according to ANSI B31.1.0

3.4 PIPE SLEEVES

A. On existing concrete construction, holes for new piping shall be made with power-driven circular cutters. No pipe sleeves are required.

B. On new concrete construction, provide pipe sleeves where piping passes through concrete floors, walls, or ceilings. Extend sleeves for the full thickness of the concrete with 1/2-inch clearance all around pipe for insulation.

C. On pipe penetrations below grade, caulk space between pipes and pipe sleeves with oakum and mastic and make watertight.

D. On all other floor and wall locations, secure sleeves to forms so they will not become displaced during pouring of concrete. Fill metal or fiber sleeves on decks with sand. Remove all sleeves from openings after removal of forms. Cut-in proper sized holes in concrete to replace sleeves crushed or knocked out of position during pouring of concrete. Caulk space around pipe with mastic and oakum.

3.5 PIPE SYSTEM PRESSURE TEST TABLE

System	Test Pressure	Duration	Test Medium
Chilled Water: (Pressurized)	150% the maximum system design pressure but not less than 100 psi, per chapter 14 section 2.4 SLAC ESH Test Pressure	1HR	Water

3.6 PIPE SYSTEM CLEANING and FLUSHING.

FINAL WATER QUALITY OBJECTIVE:

Project completion will require achievement all of the following water quality specifications:

PARAMETER	*CONTROL LIMIT*
Total Dissolved Solids	≤ 65 mg/L
Total Suspended Solids	≤ 5 mg/L
Oil & Grease	Not Detectable
Soluble Iron	≤ 0.5 mg/L Fe
Soluble Copper	≤ 0.1 mg/L Fe
Total Heterotrophic Bacteria Count	≤ 700 cfu/mL

VALIDATION & COMPLETION:

The cleaning phase will end once the following conditions are achieved:

1. Bulk water clarity is good to excellent, visually free of any obvious contaminants
2. T.S.S. concentration is less than 10 NTU in the bulk water stream
3. System strainers free of entrained debris
4. Differential pressures throughout the system are per design specifications
5. System flow rates and flow patterns are per design specifications
6. All the water quality specifications listed above have been achieved for two consecutive analyses.

WASTE WATER DISCHARGE:

Water discharged to local sanitary sewer during the project is expected to be within the local regulatory discharge limits, however as always, confirmation will be needed by SLAC. Generally speaking, at the intended feed rates, all intended treatment chemistry is readily biodegradable and complaint with the majority of Bay Area discharge regulations.

END OF SECTION 232100

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization	☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 26 00 10 – GENERAL ELECTRICAL REQUIREMENTS

PART 1 - GENERAL

1.01 SUMMARY

A. This Section covers general work of all Sections under Division 26.

B. Provide a complete working electrical installation with all equipment called for in proper operating condition. This document does not show or list every item to be provided. When an item(s) is not shown or listed and clearly necessary for proper operation of equipment, provide all items which will allow the system to function properly at no increase in Contract Price. It is the intent of the drawings and specifications that all systems be complete and ready for operation.

C. Use this Specification as a guide for workmanship and materials or construction.

D. Visit the site of proposed construction. Verify and inspect the existing site to determine conditions that affect this work.

E. Investigate and be apprised of the applicable codes, rules, and regulations as enforced by AHJs.

1.02 RELATED SECTIONS

A. Section 01 35 23 – Owner Safety Requirements.

B. Section 26 05 19 – Low Voltage Cables, Wires and Connectors.

C. Section 26 05 33 – Raceways.

D. Section 26 05 34 – Boxes for Electrical Systems.

E. Section 26 08 00 – Commissioning of Electrical Systems.

1.03 REFERENCES

A. The General Conditions, Supplementary Conditions, and applicable portions of Divisions 01 and 26 apply to the work of this Section as if printed herein.

1. IEEE Standard 242 – Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems.

2. NFPA-70E – Standard for Electrical Safety Requirements for Employee Workplaces.

3. IEEE Standard 1584 – Guide for Performing Arc-Flash Hazard Calculations.

4. SLAC Document DS-016-410-18 Drawing Deliverables for A&E Firms and Subcontractors.

1.04 DEFINITIONS

A. Following is a list of abbreviations generally used in Division 26:

1. ADA Americans with Disabilities Act.
2. AHJ Authority Having Jurisdiction.
3. ANSI American National Standards Institute.
4. APWA American Public Works Association.
5. ASTM American Society for Testing and Materials.
6. CBC California Building Code.
7. CFC California Fire Code.
8. DOE Department of Energy.
9. FCC Federal Communications Commission.
10. FM FM Global.
11. HVAC Heating, Ventilating and Air Conditioning.
12. IEC International Electrotechnical Commission.
13. IEEE Institute of Electrical and Electronics Engineers.
14. NETA National Electrical Testing Association.
15. NEMA National Electrical Manufacturers Association.
16. NFPA National Fire Protection Association.
17. OSHA Occupational Safety and Health Administration.
18. UL Underwriters Laboratories Inc.
19. USGBC United States Green Building Council.

B. Provide: To furnish and install, complete and ready for the intended use.

C. Furnish: Supply and deliver to the project site, ready for unpacking, assembly and installation.

D. Install: Includes unloading, unpacking, assembling, erecting, installation, finishing, protecting, cleaning, and similar operations at the project site to complete items of work furnished by others.

1.05 CONTENT

A. Electrical systems required for this work include labor, materials, equipment, and services necessary to complete installation of electrical work shown on drawings, specified herein or required for a complete operable facility and not specifically described in other Sections of these Specifications. All electrical installations and demolition must be performed by qualified electrical workers. See Section 01 35 23 for electrical worker certification and training requirements.

B. Demolition of existing electrical and control systems and equipment as shown on the contract plan. See SLAC ES&H Manual Chapter 8, Art. 7.2.13.

C. https://www-group.slac.stanford.edu/esh/eshmanual/references/electricalManual.pdf

D. Installation of new electrical and control equipment.

E. Feeders to Subcontractor and/or SLAC-furnished equipment and other equipment.

F. Electrical distribution and lighting.

G. Connection to utilization equipment identified on the one line diagrams.

H. Branch circuit wiring from the distribution panels to the existing loads that have been replaced.

I. Low voltage control, fire alarm and security systems.

J. Installation of grounding systems.

K. Installation of overcurrent protective systems and devices.

L. Installation of SCADA and metering devices.

1.06 GENERAL REQUIREMENTS

A. Provide the following in compliance with Division 01 requirements:

1. Product substitutions.

2. Product data.

3. Shop drawings.

4. Delivery, storage and handling.

5. Project/site conditions.

6. Sequencing and scheduling.

7. Closeout documentation.

8. Operation and maintenance documentation.

9. Record Drawings.

10. Construction waste management.

11. Cutting and patching.

12. Demolition.

B. The following requirements are in addition to requirements in Division 01:

1. Operation and Maintenance Documentation: Copies of certificates of code authority acceptance, and test data, product data, guarantees, warranties, and the like.

2. Shop Drawings: When requested by individual Sections provide shop drawings which include physical characteristics, electrical characteristics, device layout plan, wiring diagrams, and the like. Refer to individual Specification Sections for additional requirements for the shop drawings. During construction the subcontractors shall provide Field Construction Manager (FCM) with soft copy of redlined master set of drawing.

3. Closeout Documentation: Submit electrical code authority certification of inspection. Include documentation of onsite electrical testing that was performed. Provide final shop drawings. Furnish three (3) sets of hard copies and one memory stick with soft copies.

4. Record Drawings:

a. All trade contractors shall prepare a 3D CAD or Revit model to coordinate the architectural, mechanical, plumbing, IT, control and experimental equipment, structural and electrical work. The contractor shall be permitted to use Autodesk AutoCAD or Revit for this project. There shall be no cost differential in the software used.

b. Show changes and deviations from the Drawings. Include written Addendum, RFIs, Request for Change Order items.

c. Show exact routes of feeders 100 amp and larger, cable tray, surface metal raceway, conduits for signal systems 2 inches in diameter and larger.

d. Show exact location of switchgears, distribution panelboards, distribution switchboards, motor control centers, control panels, disconnects, VFDs, starters, and the like.

e. Make changes to drawings in electronic format. Obtain electronic copy from the Engineer of Record, use the same version of Revit, should Revit be used, to prepare record drawings as was used by the Engineer of Record. Provide electronic copy and hard copy to Engineer of Record for review.

f. Subcontractor shall make all redline record drawings available to SLAC personnel on site during construction. Subcontractor shall provide pdf files of redlines drawings to SLAC FCM upon request. In particular, single lines, panel schedules are important to be updated.

1.07 SUBMITTALS

A. Forward all submittals to SLAC for review and approval. Individual or incomplete submittals are not acceptable. Any substitutions shall be clearly noted and shown throughout submittal. Submittals not showing substitutions to specified equipment are at the subcontractor's risk of being rejected.

B. Identify each item by manufacturer, brand, trade name, number, size, rating, or by other data necessary to properly identify and check materials and equipment.

C. Identify each submittal item by reference to Specification Section paragraph in which item is specified, or by Drawing and Detail number.

D. Organize submittals in same sequence as they appear in Specification Sections, articles or paragraphs.

E. Equipment submittals of long lead items shall be provided are earliest possible date to ensure no delays to schedule.

F. Shop Drawings shall be performed per SLAC Design Specification DS-016-410-18 and be stamped with professional engineer's seal, show physical arrangement, construction details and finishes.

1. Drawings shall be drawn to scale and dimensioned where applicable.

2. Catalog cuts, calculations, and published material may be included to supplement scale drawings.

3. Internal wiring diagrams of equipment shall show wiring as actually furnished for the project with all optional items clearly identified as included or excluded. Clearly identify external wiring connections. Identify and obliterate superfluous material.

4. Weatherproof (WP): Indicates rating suitable for wet or damp location.

G. Submittal Deliverable

1. Prepare hard copy of the submittal material in accordance with the following:

2. Insert all literature in standard 3-ring binders for 8-1/2 inch by 11 inch pages with individual tabs. Do not staple literature on different products together.

3. Number all binders on the outside of the cover and indicate the Specification Section. Mark Binder No. 1 Project Manager's copy and No. 2 Engineer's copy. Both of these binders shall contain original manufacturer's literature.

4. Provide an index with binder. This index shall follow the same sequence as the Specifications.

5. Provide electronic copy of the submittal in PDF format. Submitted literature, catalog cuts, drawings and wiring diagrams shall be specifically applicable to this Project and shall not contain extraneous material or optional choices. Clearly mark on catalog cut to indicate the proposed item. Submittals shall include, but not be limited to those items listed in individual Sections.

6. Include all physical and performance data, including materials, manufacturer's names, model numbers, weights, sizes, capacities, performance curves, finishes, colors, accessories, calculations, and all other data required to completely describe equipment and to indicate complete compliance with Specifications and Drawings.

7. Include with complete submittals above, complete, large scale, dimensioned Shop Drawings, certified by manufacturer, of all major equipment and other equipment as directed by Engineer.

8. Electrical protective device and their associate time current curves are required for arc flash study. Provide submittal with minimum of 3 weeks.

H. Substitutions: In accordance with Division 1.

I. Resubmittals will be reviewed for compliance with comments made on the original submittal only and should be marked with a resubmittal number and dated.

J. Operating & Maintenance Instructions and Manuals:

1. Subsequent to final completions and testing operations, instruct the Owner's authorized representatives in operation, adjustment and maintenance of electrical equipment.

2. Submit three (3) copies of certificate, signed by Owner's Representatives, attesting to their having been instructed.

3. Before Owner's personnel assume operation of systems, submit three (3) sets of operating maintenance manuals. Bind data in vinyl covered loose-leaf binders with title index tabs identifying items therein to include as applicable:

a. Unit Substations.

b. Transformers.

c. Motor Control Centers.

d. Switchgears.

e. Switchboards.

f. Panelboards.

g. Enclosed breakers and safety switches.

h. Protective, metering and control equipment.

i. Lights and Lighting controls.

4. Provide one (1) full size print of Record Drawing One-Line Diagram, in metal frame with plexiglass front, and the electronic copy at each substation. Mount on the wall adjacent to each place of electrical installation.

5. Submit as-built drawings showing actual constructed conditions, in accordance with the provisions of Division 1.

1.08 QUALITY ASSURANCE

A. Materials and Systems:

1. Labels: Provide materials listed and labeled by Underwriters' Laboratories or testing firm acceptable to authority having jurisdiction, where listing service is normally provided for product.

2. Materials: Provide new and ship to jobsite in original manufacturer's containers or bundles.

B. Workmanship: Arrange work to obtain coordinated installation.

C. Code Compliance: Comply with SLAC’s applicable codes, laws, rules, regulations, and standards of applicable code-enforcing authorities in accordance with Division 1.

D. References and Standards: All materials and equipment shall comply with all SLAC applicable standards and requirements of the standards listed below. Nothing in the Drawings or Specifications shall be construed to permit Work not conforming to applicable laws, ordinances, rules, regulations. It is not the intent of Drawings or Specifications to repeat requirements of codes except where necessary for completeness or clarity.

1. American National Standards Institute (ANSI).
2. Association of Edison Illuminating Companies (AEIC).
3. Insulated Cable Engineers Association (ICEA).
4. Institute of Electrical and Electronics Engineers (IEEE).
5. National Electrical Code (NEC).
6. National Electrical Manufacturer's Association (NEMA).
7. National Fire Protection Association (NFPA).
8. International Electrical Testing Association (NETA).
9. California Building Code (CBC).
10. Underwriters' Laboratories, Inc. (UL).
11. California Code of Regulations, Title 24, Part 3 Section E, Title 8 and Title 17 Public Health (CCR).
12. State of California Low-Voltage Electrical Safety Orders (Federal OSHA).
13. State of California High-Voltage Electrical Safety Orders (Federal OSHA).
14. Codes and regulations noted in other Sections in Division 26, applicable State and Local Codes and Ordinances.
15. Seismic construction and restraints: In accordance with requirements of Title 17 (Public Health) and Title 24 (Building Standards Code) of California Code of Regulations.

1.09 SUBCONTRACTOR'S PROPOSAL

A. Bidder shall submit the following information with the proposal:

1. Preliminary schematic drawing indicating power distribution system (one-line diagram) for each K-substation and downstream equipment that is to be replaced.
2. Narrative of the work to be performed.
3. List of wiring materials proposed for systems which are applicable to this project, e.g., switchboards, panelboards, and the like.
4. Construction sequencing and phasing.
5. Any other information which the bidder considers pertinent in evaluating the proposal.

1.10 TECHNICAL DATA

A. Immediately after selection by SLAC, the successful bidder shall prepare and submit to Construction Manager the following applicable information:

1. Raceways.
2. Wires, cables, and connectors.
3. Boxes.
4. Wiring devices and device covers.
5. Circuit and motor disconnects.
6. Supporting devices.
7. Grounding to existing ground infrastructure.
8. Overcurrent protective devices.
9. Contactors and control devices.
10. Transformers.
11. Generators.
12. Uninterruptable Power Supplies (UPS).
13. Switchgears.
14. Switchboards.
15. Motor Control Centers (MCC).
16. VFDs.
17. Panelboards.
18. Metering and SCADA devices.
19. ATS, MTS, Docking Stations.
20. Communication wiring.

B. Shop drawings: Submit five sets of brochures, shop drawings and materials list. Provide electrical shop drawings at one time in a vinyl-covered, 3-ring loose-leaf binder.

1.11 CALCULATIONS

A. Preliminary calculations will be provided by the Engineer of Record for all items within this section.

B. Submit the following calculations once the exact feeder length and protection breaker models are known:

1. Electrical Equipment load data.

2. Fault duty calculations for entire electrical distribution system.

3. Voltage drop calculations from main service equipment to distribution and sub distribution centers.

4. Short circuit currents, Protective Devices Coordination and Arc Flash Study (if required).

5. Transient Study (Insulation Coordination Study) if required.

6. Cable pulling calculation.

7. Reference other electrical specification sections for additional calculations.

1.12 DELIVERY, STORAGE AND HANDLING

A. Protect from loss or damage. Replace lost or damaged materials and equipment with new at no increase in Contract Sum.

B. Subcontractor to follow manufacturer recommendations for storage and handling. Subcontractor to replace any equipment not properly stored and damaged from moisture, dust, rodents, dents, scratches. If needed the Subcontractor shall provide temporary space heaters to be used storage facility.

1.13 DRAWINGS AND COORDINATION WITH OTHER WORK

A. Drawings:

1. To avoid installation conflicts thoroughly examine the complete set of Contract Documents. For exact locations of building elements, refer to dimensioned architectural/structural drawings. Resolve conflicts with Construction Manager prior to installation.

2. For purposes of clarity, legibility, construction drawings are diagrammatic and the Contractor shall field verify dimensions and routing. Field measurements take precedence over dimensioned drawings.

3. Exact routing of wiring and locations of switchboards, panels, etc., shall be governed by structural conditions, obstructions and existing conditions. SLAC reserves right, at no increase in price, to make any reasonable change in locations of electrical items to group them into orderly relationships and/or increase their utility. Contractor shall verify SLACs requirements in this regard prior to roughing-in.

4. Mounting heights of brackets, outlets, etc., shall be as required to suit equipment served.

5. Intention is to indicate size, capacity, approximate location, direction and general relationship of one work phase to another, but not exact detail or arrangement.

6. Description of systems:

    a. Provide materials to provide functioning systems in compliance with performance requirements specified.

    b. Provide modifications required by reviewed shop drawings and field coordinated drawings.

B. Coordination:

1. Identify all conditions where physical space for installation, service, operations and maintenance is limited. If necessary, provide supplementary drawings. In the case of differences or disputes concerning coordination, interference or extent of Work between trades shall be decided by the Contractor whose decision shall be final.

2. Prior to installation of feeders to equipment requiring electrical connections, examine the manufacturer's shop drawings, wiring diagrams, product data, and installation instructions. Verify that the electrical characteristics detailed in the Contract Documents are consistent with the electrical characteristics of the actual equipment being installed. When inconsistencies occur, request clarification from Construction Manager.

3. Provide electrical equipment compatible with, acknowledge and accommodate the requirements of other trades. Resolve without additional cost to SLAC those details necessary to assure that the electrical system properly and completely function together when assembled and achieve required design criteria and performance and conform to requirements of governing codes and regulatory agencies.

4. Provide templates, information and instructions to other trades to properly locate holes and openings to be cut or provided for Electrical Work.

5. Lines of other services that are damaged as a result of this work shall promptly be repaired at no expense to the University to complete satisfaction of the University.

C. Large Scale Layout Drawings: In accordance with requirements of Division 1, prepare large scale detailed layout Drawings showing locations of equipment, conduit runs, panels, and all other elements of electrical systems where required by other Sections of this Division, plus sections of all congested areas to show relative position and spacing of affected elements. All symbols and designations used in preparing record Drawings shall match those used in

Contract Drawings.

D. Equipment Rough-In:

1. Rough-in locations shown on Electrical Drawings for equipment furnished by Owner and for equipment specified in other Divisions are approximate only.

2. Obtain exact rough-in locations from following sources:

3. From shop drawings for Contractor-furnished and installed equipment.

4. From equipment manufacturer shop drawings for Owner-furnished Contractor-installed equipment.

5. Verify electrical characteristics of equipment before starting rough-in. Where conflict exists between equipment and rough-in shown on Drawings obtain clarification from Engineer and provide as directed.

6. Unless otherwise shown or specified, provide direct raceway and conductor connections from building wiring system to equipment terminals for direct connected equipment which is Contractor furnished and Contractor installed, Owner furnished and Contractor installed.

7. In order to provide information for arc flash labels, the Subcontractor is required to furnish protective devices time-current curves and the latest field verified construction drawings.

8. Owner to provide arc flash warning labels to new electrical equipment prior to connecting the new equipment to an electrical power supply. After connection to the power supply equipment status shall be controlled through the lockout/tag out (LOTO) program. Restrict access to equipment per the arc flash label boundaries whenever electrical parts are exposed, whether or not they are energized.

1.14 CUTTING AND PATCHING

A. All cutting and patching required for work of this Division is included herein. Coordination between the Subcontractor and other trades is imperative. The Subcontractor shall bear the responsibility for the added expense of adjusting for improper holes, supports, etc. due to lack of coordination.

1.15 EXISTING UNDERGROUND UTILITIES AND SERVICES

A. Locations of existing utilities, where shown, established from best available information. Assume that this information is approximate. The Subcontractor shall verify exact location and depths before starting work. Should, during the course of the work, conditions are found to be different than indicated, notify the FCM (SLAC Field Construction Manager) immediately. Use extreme caution so as not to damage or break lines that are in use. If breakage does occur, Subcontractor shall be responsible for all damage and repairs resulting therefrom. All unusual conditions with respect to existing facilities shall be brought to the FCM's attention immediately.

PART 2 - PRODUCTS

2.01 MATERIALS FURNISHED

A. Material shall be new, and bear the label of Underwriter's Laboratories or other testing laboratory acceptable to authority having jurisdiction, where labeling exists for the class of equipment.

B. Provide equipment of one manufacturer, alike in appearance and function.

C. For equipment specified by manufacturer's number, include all accessories, controls, etc., listed in catalogue as standard with equipment. Furnish optional or additional accessories as specified.

D. Where no specific make of material or equipment is mentioned, use UL or CSA listed product of reputable manufacturer which conforms to requirements of system and other applicable specification sections:

1. Structural steel for supports: ASTM-A36.

2. Galvanize members installed in areas of high humidity or condensation.

3. Furnish other members with shop coat of rust inhibiting primer.

4. Shop fabricate for field assembly using bolts.

5. Minimize field welding.

6. Retouch primer and galvanizing after field welding.

7. Unless support is otherwise indicated where weight of equipment exceeds 400 pounds, submit engineering design and calculations signed and sealed by a registered Engineer licensed to practice Structural Engineering in the state in which the project is located.

8. Rain hoods and counter flashings not exposed to view:

    a. Sheet copper: Minimum 24 OZ.

    b. Fiberglass ASTM 5364, ASCE 7

9. Rain hoods and counter flashings exposed to view:

    a. Stainless steel: Minimum 20 GA.

E. Equipment and material damaged during transportation or installation shall be replaced. Provide a full time authorized representative to supervise Work specified in this Division; check all materials prior to installation for conformance with Drawings, Specifications, and approved Shop Drawings.

2.02 ENVIRONMENTAL CONDITIONS

A. General:

1. Provide NEMA 1 enclosures for electrical equipment unless otherwise indicated.

2. Conduit: See Section 26 05 33.

3. Cable: See Section 26 05 19.

4. Boxes and Fittings: See Section 26 05 34.

B. Damp and wet locations:

1. Exterior applications: Provide NEMA 3R enclosures for electrical equipment.

C. Corrosive environments:

1. Use NEMA 4X stainless steel or fiberglass enclosures in areas with corrosive atmospheres, including but not limited to:

a. All equipment and devices within the cooling tower yard.

b. Deionized water facility.

c. Waste management facility.

PART 3 - EXECUTION

3.01 GENERAL

A. Equipment shall be wired with approved type of conductors. No shared neutral shall be allowed. The phase sequence at SLAC is electrically counter-clockwise (CCW – ACB). The conductors phasing shall be color coded and marked as CW – ABC.

3.02 EXAMINATION

A. Construction Documents: Examine the entire set of Drawings to avoid conflicts with other systems. Determine exact route and installation of electrical wiring and equipment with conditions of construction.

B. Clarification: Should the Electrical Documents indicate a condition conflicting with the governing codes and regulations, refrain from installing that portion of work until clarified by Construction Manager.

3.03 PROTECTION

A. Provide covering and shielding for all equipment to protect from damage.

B. Protect nameplates on motors and similar equipment, to prevent defacing.

C. Repair, restore or replace damaged, corroded and rejected items.

D. Temporary power in construction environments must include GFCI protection for personnel for all 120-, 208-, 240-Volt applications where power is derived from a receptacle, temporary wiring installations, or a portable generator.
Note: Where spider boxes are used, they must be fed by a GFCI protected circuit. Spider Box receptacles must be GFCI-type.

E. 480-Volt Power supplied for construction environment must include UL 943C SFGFCI protection, either to power 480-Volt equipment or to supply a 120/208- or 120/240-Volt temporary power skid. SPGFCO units may be available from Facilities and Operations.

Note: The use of Assured Equipment Grounding Conductor Program in lieu of GFCI or SPGFCI protection is NOT permitted except in unusual cases with advanced written approval by the SLAC Electrical Safety Officer (ESO).

F. Engineered plans for temporary power must be submitted to the SLAC Building Inspection Office (BIO) for review and authorization prior to temporary power installation. Obtain SLAC BIO inspection prior to energizing temporary power circuits. Contact the SLAC FCM for assistance.

3.04 DEMOLITION

A. Scope:

1. Coordinate with SLAC so that work can be scheduled not to interrupt operations, normal activities, building access, access to different areas. SLAC will cooperate to the best of their ability to assist in a coordinated schedule, but will remain the final authority as to time of work permitted.

2. Existing Conditions: Determine the exact location of existing utilities and equipment before commencing work, compensate SLAC for damages caused by the failure to locate and preserve underground utilities. Replace damaged items with new material to match existing. Promptly notify SLAC if utilities are found which are not shown on Drawings.

B. Equipment:

1. Existing equipment and materials removed from existing construction and not indicated or required to be reused shall become the property of SLAC, if they so elect.

2. Present equipment and materials removed to SLAC representative, who shall select equipment and materials to retain.

3. Equipment and materials not retained shall become property of Subcontractor and shall be removed from site.

C. Execution:

1. Remove electrical equipment and devices and associated wiring from surfaces scheduled for demolition unless specifically shown as retained on Drawings.

2. Remove or relocate electrical boxes, conduit, wiring, equipment, and the like, as may be encountered in removed or remodeled areas in the existing construction affected by this work.

3. Remove abandoned wiring to leave site clean.

4. Keep outages to occupied areas to a minimum and prearrange outages with SLAC. Requests for outages shall state the specific dates and hours and the maximum durations, with the outages kept to these specific dates and hours and the maximum durations. Compensate SLAC for any damages resulting from unscheduled outages or for those not confined to the preapproved times.

5. Verify a location for storage of materials, supplies, tools, rubbish, and the like, prior to start of work.

6. Properly dispose of removed/replaced equipment containing hazardous materials.

3.05 INSTALLATION

A. Manufacturer's Directions: Follow the manufacturer’s directions for all items furnished, unless specified otherwise.

B. When requirements of the installation instructions conflict with the Contract Documents, request clarification from Construction Manager prior to proceeding with the installation.

C. Do not install electrical equipment in obvious passages, doorways, scuttles or crawl spaces which would impede or block the area passage's intended usage.

D. Where specified the electrical equipment shall be installed on housekeeping pad. The edge of housekeeping pad shall be not more 6 inches from the front of electrical equipment.

E. Shared neutral wires are not allowed in SLAC.

F. Equipment: Accurately set and leveled with supports neatly placed and properly fastened as shown and specified. Provide means of bringing in and installing equipment into position inside building or substation.

G. Each electrical equipment shall be provided with permanently installed phenolic or approved equal label or tag. Installation Subcontractor shall apply on finished floor 2-inch wide self-adhesive vinyl tape with black/yellow stripes to delineate work space required per NEC 110.26 and NEC 110.34.

H. Conduit Systems:

1. Install parallel with walls or structural elements: vertical runs plumb; horizontal runs level or parallel with structure as appropriate: groups racked together neatly with straight runs and bends both parallel and uniformly spaced.

2. For individual conduit runs directly fastened to the structure, use rod hangers manufactured by Caddy, UniStrut or B-line.

3. For multiple conduit runs, use UniStrut or B-line trapeze type conduit support designed for maximum deflection not greater than 1/8”.

4. All overhead conduits to be run outdoor or inside substations shall be of rigid-metal type (RMC).

5. The conduit system shall be installed as high as practicable to maintain adequate head room. The height shall be coordinated with other trades to achieve proper headroom.

6. Clearance: Do not obstruct spaces required by code in front of electrical equipment, access doors, etc.

7. Maximum number of bends in conduits shall be 3, up to 270 degrees total. If additional bends are required in conduit runs pull boxes shall be used.

I. Penetrations:

1. Pack space between conduit, sleeve in walls with non-combustible materials.

2. Penetrations through any damp-proofed/water-proofed surfaces by appropriate means shall maintain integrity of system penetrated. Install flashing, weatherproof sealants, and eschalant plates as needed.

3. Seal around penetrations with fireproofing material to maintain integrity of fire rating where occurs.

4. All electrical equipment, electrical rooms, substations, shall be rodent proof. No openings larger than ¼” shall be allowed.

5. Conduit sleeves shall be used in penetrations through concrete walls.

J. Hangers, Supports, Anchors and Chases:

1. Provide complete systems as required for installation of Electrical Work. Support and/or anchor all equipment and wiring systems to resist gravity and seismic forces in accordance with the requirements of the Building Code and in compliance with conditions stated in the Contract Documents.

2. Supporting components shall be constructed of metal only: no wood or combustible material will be permitted including supports for outlet boxes.

3. Hangers, anchors and supports for conduit runs: As specified.

4. Provide concrete inserts for attachment of hangers; subject to structural engineer's review.

5. Provide anchors for floor and wall mounted equipment.

6. Provide supports for wall mounted equipment.

3.06 PERFORMANCE

A. Excavating and Backfilling:

1. The Subcontractor shall obtain a SLAC approved excavation permit at least 10 days prior to commencement of work.

2. Provide all necessary shoring, sheeting and pumping.

B. Concrete: In accordance with the requirements of Division 3. The Subcontractor shall verify the equipment concrete pads are levelled and slopped without puddling.

C. Indoor Installation:

1. Arrange for necessary openings to allow entry of equipment.

2. Where equipment cannot be installed as structure is being erected, provide and arrange for building-in of boxes, sleeves or other devices to allow later installation.

3. Make all penetrations through roofs prior to installation of roofing. For penetrations required after installation of roofing:

4. In built-up roofing (BUR), provide all curbs, cants and base flashings.

5. In elastic sheet roofing (ESR), arrange and pay for flashing work by authorized roofer.

6. Install rain hoods and metal counter flashings as indicated and make all penetrations of electrical work through walls and roofs water and weathertight.
7. Furnish all clamps, waterproofing material and labor necessary.
8. Where metal flashings are applied over concrete, paint concrete with 1/8 IN of mastic cement first.
9. Set flashing in mastic cement, watertight.
10. Have repair and replacement of roof construction, damaged by this work, done in manner which will not nullify roof warranty.
11. Install equipment to permit easy access for normal maintenance.
12. Provide working spaces and dedicated electrical spaces required per NFPA 70, Article 110.26.
13. Maintain easy access to switches, motors, drives, pull boxes, receptacles, etc.
14. Relocate items which interfere with access.

D. Sleeves, Chases and Concrete Inserts:

1. Provide sleeves, chases, concrete inserts, anchor bolts, etc., before concrete is poured.
2. Sleeves, chases are prohibited in structural members, except where shown or approved in writing.

E. Cutting and Patching:

1. Provide cutting, fitting, repairing, patching and finishing of installed work.
2. Include installed work of other sections where it is necessary to disturb such work to permit installation of electrical work.
3. Repair or replace existing or new work disturbed.
4. Do not cut into existing services without first verifying with SLAC that service has been correctly identified.
5. Perform work that interrupts any service during premium time.
6. Include cutting into existing lines for new connection.
7. Cause least interference to normal operation of building.
8. Inform building engineering staff in advance of any shut off that will occur and give estimate of duration.

9. Begin work only after SLAC Area Manager is fully informed and has agreed to schedule of shut offs.

10. Before cutting, obtain approval of SLAC Structural Engineer.

11. Use only approved methods.

12. Cut all holes neatly and as approved by SLAC engineer possible to admit work.

13. Do not weaken walls or floors; locate holes in concrete to avoid structural members.

14. Locate openings and sleeves to permit neat installation of conduits and equipment.

15. Do not remove or damage fireproofing materials.

16. Install hangers, inserts, supports, and anchors prior to installation of fireproofing.

17. Repair or replace fireproofing removed or damaged, at no extra cost.

F. Painting: Include surface preparation, priming and finish coating for electrical cabinets, exposed conduit, pull and junction boxes.

G. A permanent, legible, single-line diagram of the switchgears, switchboards, integrated power centers shall be provided in a readily visible location within sight of equipment. This diagram shall clearly identify interlocks, isolation means, and all possible sources of voltage to the installation under normal or emergency conditions, including all equipment contained in each cubicle, and the marking on the switchgear shall cross-reference the diagram.

3.07 FIELD QUALITY CONTROL

A. Perform indicated tests to demonstrate workmanship, operation, and performance.

1. Conduct tests in presence of Architect and, if required inspectors of agencies having jurisdiction.

2. Arrange date of tests in advance with Architect, manufacturer and installer.

3. Give minimum of 72 hours notice to all inspectors.

4. Furnish or arrange for use of electrical energy, steam, water, diesel fuel, or gas required for tests.

5. Furnish all lubricating materials required for test.

6. Repair or replace equipment and systems found inoperative or defective and retest.

7. If equipment or system fails retest, replace it with products conforming with Contract

Documents.

8. Continue remedial measures and retests until satisfactory results are obtained.

9. Test equipment and systems as indicated for each item, unless otherwise recommended by manufacturer.

3.08 FINAL PERFORMANCE TEST

A. Tests:

1. Conduct tests of equipment and systems to demonstrate compliance with requirements specified in Division 26. Refer to individual Specification Sections for required tests. Document tests and include in Closeout Documents.

2. During site evaluations by Construction Manager provide an electrician with tools to remove and replace trims, covers, devices, and the like, so that a proper evaluation of the installation can be performed.

B. Prior to equipment testing or energization, it shall be cleaned from any dust & debris, wiped clean, and dried out.

C. Perform panel load balance, short circuit, and freedom from ground, and ground test (including ground fault protection where provided).

1. As part of putting systems in operation, provide tabulated results of load balance and voltage at each switchboard, panelboard and motor control center. Use true RMS measuring metering devices.

2. Provide written report that all circuits have been energized and no short circuits exist.

3. Provide neutral to ground resistance tests to prove neutral is grounded in only one location.

4. Provide ground test at service entrance and provide report on resistance to earth of the grounding electrode system.

3.09 CLEANING AND PAINTING

A. After other Work is accomplished, clean exposed conduit, panels (interiors and exteriors), fixtures, equipment and leave in satisfactory condition.

B. The horizontal tops of distribution transformers shall be provided with printed warning sign “Not for Storage” (NEC 450.9).

3.10 EQUIPMENT IDENTIFICATION

A. Refer to Section 260553 – Identification for Electrical Systems for equipment identification requirements.

3.11 SLAC PROCESS FOR SWITCHING OPERATIONS OF ALL ELECTRICAL EQUIPMENT

A. The following outlines the required processes for energy isolation of equipment and circuits, inspection and testing after construction installation, and energization of equipment and circuits. the outline is intended to provide guidance for the subcontractor to understand the expectation of the process requirements to most efficiently execute the process.

1. Subcontractor submits written request for switching electrical equipment for isolation including:

    a. Method of Procedure (MOP).

    b. Single line or drawing which identifies sequence of LOTO and Isolation Points.

2. Submit request a minimum of 3 days prior to need by date for equipment rated 480 VAC and above. Complexity (multiple trades or more than one point of LOTO) of LOTO requires 15 days.

3. Submit requests a minimum of 3 days prior to need by date for equipment rated 208 V and below. Complexity (multiple trades or more than one point of LOTO) of LOTO requires 15 days.

4. SLAC prepares the Electrical Work Plan (EWP) and Switching Orders based on the information provided by the Subcontractor's MOP.

5. SLAC qualified electrical personnel shall perform switching and apply LOTO (Lock-out/Tag-out) on disconnecting means upstream of isolated equipment.

6. Subcontractor's qualified electrical personnel shall apply the LOTO on disconnecting means upstream of isolated equipment after SLAC personnel applied own locks and tags. (Can be completed at the same time as item 3).

7. SLAC's qualified electricians shall perform zero-voltage verification.

8. Subcontractor's qualified electricians shall perform zero-voltage verification (Can be completed at the same time as item 4).

9. The Subcontractor performs installation work.

10. Prior to submitting request to energize equipment, the Subcontractor performs their own internal inspections and pre-energization tests to verify and ensure the system is complete and ready for the SLAC inspections and tests.

11. The Subcontractor submits written request for SLAC inspections and testing a minimum of 1 to 3 days prior to the need by date (see voltage levels in item 1. Above, for number of days prior). Complexity of the task may require more lead time.

12. Submit all applicable test reports, and torque charts, per project specifications.

13. SLAC will review and approve the report data.

14. If inspection or test reports are not approved, the Subcontractor will be required to address the items of concern, re-inspect and re-test, and resubmit reports as required.

15. SLAC will inspect and test Subcontractor's newly installed or modified electrical equipment prior to energization; during this activity the Subcontractor shall be present at all times to assist SLAC personnel and respond to questions and make corrections if necessary. See Section 01 45 32 Art. 1.02 for quality assurance requirements.

16. All required equipment ID and Danger labels shall be affixed.

17. After inspections and testing are completed with satisfactory results, but before new equipment initial energization, SLAC will schedule a formal readiness review.

18. SLAC will perform switching operations to energize the equipment; during this activity the Subcontractor shall be present at all times to assist SLAC personnel and respond to questions and make corrections if necessary.

19. SLAC performs voltage testing on fully energized system.

20. Test points will be identified by the Subcontractor and approved by SLAC.

21. During this activity the Subcontractor shall be present at all times to assist SLAC personnel and respond to questions and make corrections if necessary.

22. If necessary, the Subcontractor will make any adjustments to provide proper voltage (defined as 116 V-126 V for 120 V nominal systems, and proportionally equivalent for higher voltage systems).

END OF SECTION 26 00 10

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization ☑ No ☐ Yes	
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 26 05 09 – EQUIPMENT CONNECTIONS

PART 1 – GENERAL

1.01 SUMMARY

A. Connect equipment, whether furnished by Owner or other Divisions, electrically complete.

B. Power connections only in Division 26 for equipment provided by Owner or other Divisions. Equipment not set in place by Division 26.

C. Ground equipment with equipment grounding conductor. Consult with SLAC Electrical Engineer for connections wherever owner furnished equipment does not have a grounding point.

1.02 REGULATORY REQUIREMENTS

A. The following documents form part of the Specifications to the extent stated. Bring conflicts between Specifications, Drawings, and the referenced standards to the attention of the Project Manager, in writing, for resolution before taking any related action.

B. Conform to the latest adopted version of the National Electrical Code (NEC).

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 ELECTRICAL CHARACTERISTICS

A. Verify electrical characteristics of equipment prior to installation of conduits and wiring for equipment.

3.02 EQUIPMENT CIRCUIT WIRING

A. Connect feeder and branch circuits to equipment for complete system.

B. Connections to the electrical equipment shall be done per Manufacturer's recommendations. Appropriate hardware shall be used for indoor and outdoor installations, for copper and aluminum conductors. Spring washers shall not be reused.

C. Only one conductor per termination point is allowed unless this terminal is listed for more than one connection.

D. A minimum of Grade 5 zinc platted hardware shall be used in mechanical connections required bolt, nut, split washer and nut washer.

E. Barriers required to be placed in all service equipment such that no uninsulated, ungrounded service busbar or service terminal is exposed to inadvertent contact by persons or maintenance equipment while servicing load terminations (NEC 230.62(C).

3.03 BOLTED CONTACT TORQUE

A. Each bolted contact torque shall be checked per equipment manufacturer's provided data. Whenever torques are not specified, the subcontractor shall use typical values per latest NETA-ATS specifications. After torque verification each bolt shall be sealed with torque seal paint. The subcontractor shall document all torque-verified bolted connections in Torque Chart. Torque Chart shall be reviewed and approved by SLAC Engineer.

END OF SECTION 26 05 09

SECTION 26 05 19 - LOW VOLTAGE WIRES, CABLES, AND CONNECTORS

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:

1. Low Voltage (Less than 600V) Wires and cables.
2. Connectors.
3. Lugs and pads.
4. Service entrance cable.

1.02 SYSTEM DESCRIPTION

A. Provide wires, cables, connectors, lugs, and the like for a complete and operational electrical system.

1.03 SUBMITTALS

A. Provide product data for the following equipment:

1. Wires.
2. Cables.
3. Connectors.
4. Lugs.

B. Provide cable and wire testing report to SLAC (FCM) prior to energization readiness review and the project closeout documentation, see Project Closeout Requirements in Division 01. Provide a memory stick with PDF copies in addition to hard copies during each submittal.

1.04 PROJECT CONDITIONS

A. Wire and cable routing shown on Drawings is approximate unless dimensioned. The subcontractor is required to verify in field exact distances. Route wire and cable as required.

1.05 COORDINATION

A. Coordinate Work under provisions of Division 1.

B. Determine required separation between cable and other work.

C. Determine cable routing to avoid interference with other work.

D. Coordinate the size of switchgear/switchboard/panel board terminals to ensure they are adequately sized to accommodate specified use.

1.06 REGULATORY REQUIREMENTS

A. The following documents form part of the Specifications to the extent stated. Bring conflicts between Specifications, Drawings, and the referenced standards to the attention of the Project Manager, in writing, for resolution before taking any related action.

B. Conform to the latest adopted version of the National Electrical Code (NEC).

C. Furnish products listed by UL or other testing firm acceptable to AHJ.

PART 2 - PRODUCTS

2.01 GENERAL

A. Marking: Insulation type, voltage rating, size and listing label shall be printed with permanent markings repeating along entire length of conductor.

B. Provide all new wire and cable, manufactured within 12 months of delivery to site and continuously stored where protected from heat and weather.

C. Conductor packaging and reels: Clearly marked or tagged with Manufacturer's name, AWG, size, voltage rating, insulation type, agency listing and date of manufacture.

2.02 MANUFACTURERS

A. Wires and Cables: General Cable, Southwire, American or approved equal.

B. Connectors: Stranded conductors by Burndy, Ilsco, Thomas & Betts, Phoenix Contacts, Polaris or approved equal.

C. Splices:

1. Branch Circuit Splices (conductor sizes 8AWG or smaller): Wire Nuts.

2. Branch Circuit Splices (conductors sizes larger than 8AWG): Insulated multi-tap connectors.

3. Feeder Splices: Compression type splice with properly voltage rated heat or cold shrink or insulated multi-tap connectors with properly rated layers of 3M electrical insulated tape.

2.03 COPPER BUILDING WIRE

A. Description: Stranded, insulated and uninsulated, drawn copper current-carrying conductor with an overall insulation layer or jacket, or both, rated 600 V or less.

B. Standards:

1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.

2. UL Listed.

3. RoHS compliant.

4. Conductor and Cable Marking: Comply with wire and cable marking according to UL's "Wire and Cable Marking and Application Guide."

5. Conductors: Copper, complying with ASTM B 3 for bare annealed copper and with ASTM B 8 for stranded conductors.

C. Conductor: Copper

D. Insulation Voltage Rating: 600 volts.

E. Insulation: ANSI/NFPA 70

1. Use type THHN/THWN-2 for wire sizes 6AWG or smaller.

2. Use type XHWN-2, XHHW-2, or XLPE for wire sizes number 4AWG or larger.

F. Phase color to be consistent at feeder terminations; A--B-C, top to bottom, left to right, front to back. Contractor shall note that SLAC has a counterclockwise phase rotation, A, B, C.

1. Please note, SLAC is electrically connected counterclockwise, color coded or physically marked as ABC (clockwise).

G. Color Code Conductors as Follows:

PHASE	208 VOLT WYE	480 VOLT

A	Black	Brown
B	Red	Orange
C	Blue	Yellow
Neutral	White	Gray
Ground	Green	Green
Isolated Ground	Green w/yellow trace	N/A

H. Neutral conductors for single phase loads shall use color tracer color to match phase conductor color.

I. All instrumentation and control wires within equipment enclosures shall be 600V rated type SIS not smaller 14AWG.

2.04 PULL CORD

A. Empty branch circuit or system conduits: Provide mildew resistant polypropylene line, minimum 210 pound tensile strength. Greenlee Poly-Line or equal.

B. Empty feeder conduits or ducts: Provide mildew resistant polypropylene rope, minimum ¼ inch diameter. Durlaine or equal.

C. Refer to Section 26 05 33 for additional information.

2.05 CONNECTORS

A. A minimum of Grade 5 zinc platted hardware shall be used in mechanical connections required bolt, nut, split washer and nut washer.

B. Copper Pads: Drilled and tapped for multiple conductor terminals.

C. Lugs: Compression type for use with stranded branch circuit or control conductors; mechanical lugs not acceptable. Manufacturers: Ilsco, Panduit, Thomas & Betts, 3M, or approved equal.

D. Conductor Branch Circuits: Wire nuts with integral spring connectors for conductors 14AWG through 8AWG. Push-in type connectors where conductors are not required to be twisted together are not acceptable. Manufacturers: 3M, Ideal, Phoenix Contacts or approved equal.

2.06 LUGS AND PADS

A. Ampacity: Cross-sectional area of pad for multiple conductor terminations to match ampere rating of switchboard or panelboard bus or equipment line terminals.

PART 3 - EXECUTION

3.01 CONDUCTOR MATERIAL APPLICATIONS

A. Feeders: Copper; stranded for No. 12 AWG and larger.

B. Branch Circuits: Copper, stranded for No. 12 AWG and larger.

C. Use conductor not smaller than # 12 AWG for power and lighting circuits.

D. Use conductor not smaller than #14 AWG for control circuits.

3.02 CONDUCTOR INSULATION AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

A. In dry interior locations, use type THHN/THWN-2 or XLPE.

B. In wet or damp interior locations, use type THWN-2, THW or XHHW-2.

C. In exterior locations, including underground installations, use type THWN-2 or XHHW-2.

3.03 TEMPORARY CABLES

A. For 480V/277V and 208V/120VAC low power construction tools and portable lighting (30A and less) use multi-conductor grounded type SOOW, 600VAC, 12AWG and 10AWG water-resistant portable cords rated for outdoor installations. At supply end extension cords shall be plugged only in GFCI receptacles or furnished with GFCI (NEC 527.6(B)(2).

B. For temporary installations requiring more than 30A use 8AWG and larger, 90-degree C, 2000VAC rated, EPDM insulated Type W single conductor or multi-conductor grounded power cords.

C. All temporary cables shall be mechanically protected from damage. Installation method shall be approved by SLAC FCM and inspected by SLAC Electricians prior to energization.

3.04 INSTALLATION OF CONDUCTORS AND CABLES

A. Complete raceway installation between conductor and cable termination points according to Section 26 05 33 "Raceways and Boxes for Electrical Systems" prior to pulling conductors and cables.

B. Use wire, cable, conductor manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.

C. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips that will not damage cables or raceway.

D. Install exposed cables parallel and perpendicular to surfaces of exposed structural members, and follow surface contours where possible. Provide physical protection from damage.

E. Support cables according to Section 26 05 29 "Hangers and Supports for Electrical Systems."

F. Install products in accordance with manufacturer's instructions.

G. Pull all conductors into raceway at same time.

H. Install dedicated neutral wires only.

I. Install conductors used for low voltage control systems in conduit or approved raceway systems, unless conductors used have specific agency listing for direct installation in ceiling plenums.

J. Tape all connections made with non-insulated type connectors with insulating tape to 150 percent of the insulating value of the conductor insulation.

3.05 CONNECTIONS

A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A-486B.

B. Make splices, terminations, and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.

3.06 WIRE SPLICES, JOINTS, AND TERMINATION

A. Join and terminate wire, conductors, and cables in accordance with UL 486A, B, C, NEC and manufacturers' instructions.

B. Thoroughly clean wires before installing lugs and connectors.

C. Make splices, taps and terminations to carry full ampacity of conductors, without noticeable temperature rise.

D. Make splices and termination mechanically and electrically secure.

E. Make wire connections on conductors #8 AWG and smaller with wired nuts or spring type connector.

F. Make wire connections and splices on conductors #8 AWG and larger with solderless pressure connectors, insulated multi-tap connectors.

G. For temporary ground terminations, the use of split bolt type mechanical connectors is allowable when approved by SLAC.

H. For permanent ground terminations, use irreversible connectors.

I. Where determined that unsatisfactory splices or terminations have been installed, remove the devices and install approved devices at no additional cost.

J. Terminate wires in terminal cabinets, relay and contactor panels using terminal strip connectors.

K. Bundle spare conductors using nylon ties. Leave sufficient length to terminate anywhere in the panel or cabinet.

L. Use nylon cable ties for bundling and securing wire and cable as required to maintain harnessing.

M. Encapsulate below grade splices at outlet, pull and junction boxes using specified insulating resin kits. Make all splices watertight for exterior equipment and equipment in pump rooms.

N. Make up all splices and taps in accessible junction or outlet boxes with specified connectors. Use same color pigtails and wiretap as the feed conductor. Form conductor prior to cutting. Provide at least 6 inches of tail and neatly packed in box after splice is made up.

O. Do not use non-metallic sheathing cable.

P. Do not use AC (aluminum armored cable). Use of MC cable is limited to final connection to light fixtures or furniture partitions or end load.

Q. No. 8 AWG and smaller conductor connections:

1. Use spring wire connectors.

2. The integral insulator shall completely cover the stripped wires.

3. The number, size, and combination of conductors, as listed on the manufacturers packaging shall be strictly complied with.

4. Use manufacturer recommended hardware.

R. No. 6 AWG and larger conductor connections:

1. Use insulated multi-tap connectors for cable-to-cable connections.

2. Use manufacturer recommended hardware. Only one conductor per termination point is allowed unless this terminal is listed for more than one connection.

3. Terminate conductors from size 6 AWG to 750 Kcmil copper busses, safety switches, circuit breakers using bolted pressure or mechanical compression lugs in accordance with manufacturer's instructions.

4. Cable sizes 250 Kcmil and larger: Use not less than two clamping elements or compression indents per wire for connectors.

5. Insulate splices and joints with materials approved for the particular use, location, voltage and temperature. Insulate not less than that of the conductor insulation level being joined.

6. Use hydraulic crimping tool for making compression indents.

7. Apply oxide-inhibiting compound to conductors before joining, installing compression lugs or making terminations. Apply number of compression indents as directed by the manufacturer instructions.

3.07 IDENTIFICATION

A. Identify and color-code conductors and cables according to Section 26 05 53 "Identification for Electrical Systems."

B. Identify each spare conductor at each end with identity number and location of other end of conductor, and identify as spare conductor.

3.08 FIELD QUALITY CONTROL

A. Inspect wire and cable for physical damage and proper connection.

B. Measure tightness of bolted connections and compare torque measurements with manufacturer's recommended values.

C. Verify continuity of each branch circuit and power conductors.

D. Perform tests and inspections.

1. After installing conductors and cables and before electrical circuitry has been energized, test wiring conductors for compliance with requirements.

2. Perform each of the following visual and electrical tests:

3. Inspect exposed sections of conductor and cable for physical damage and correct connection according to the single-line diagram.

4. Test bolted connections for high resistance using one of the following:

5. A low-resistance ohmmeter.

6. Calibrated torque wrench.

7. Thermographic survey.

8. Inspect compression-applied connectors for correct cable match and indentation.

9. Inspect for correct identification.

10. Inspect cable jacket and condition.

11. Insulation-resistance test on each conductor for ground and adjacent conductors. Apply a potential of 500VDC for 300-V rated cable and 1000-VDC for 600-V rated cable for a one-minute duration. Include testing all new conductors sizes 4AWG or larger, including existing conductors modified by subcontractor. Unless otherwise noted on drawings for smaller size conductors.

12. Continuity test on each conductor and cable. Include testing all new conductors and existing conductors modified by subcontractor. Provide continuity test to confirm no cross neutrals, no shared neutrals, and circuit verification.

13. Uniform resistance of parallel conductors.

E. Initial Infrared Scanning: When required by SLAC Project Engineer after substantial completion, but before final acceptance, perform an infrared scan of each splice in conductors 3 AWG and larger. Remove box and equipment covers so splices are accessible to portable scanner. Correct deficiencies determined during the scan.

1. Instrument: Use an infrared scanning device designed to measure temperature or to detect significant deviations from normal values. Provide calibration record for device.

2. Record of Infrared Scanning: Prepare a certified report that identifies switches checked and that describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

F. Cables will be considered defective if they do not pass tests and inspections. Subcontractor to replace defective cables at no additional cost to SLAC.

G. Prepare test and inspection reports to record the following:

1. Procedures used.

2. Results that comply with requirements.

3. Results that do not comply with requirements, and corrective action taken to achieve compliance with requirements.

END OF SECTION 26 05 19

SECTION 26 05 26 - GROUNDING

PART 1 – GENERAL

1.01 SUMMARY

Section Includes:

1. Grounding materials.
2. Feeder and branch circuit grounding.
3. Raceway, cable tray and enclosure grounding.
4. Equipment grounding bonding.
5. Grounding Electrode System

1.02 SCOPE OF WORK

A. Provide grounding and bonding of electrical service, circuits, equipment, signal and communications systems.

B. Performance Requirements: Install equipment grounding bonding such that metallic structures, enclosures, raceways, junction boxes and other conductive items in close proximity with electrical circuits operate continuously at ground potential and provide a low impedance path for possible ground fault currents.

1.03 REFERENCED STANDARDS

IEEE Std. 80 – IEEE Guide for Safety in AC Substation Grounding

IEEE Std. 142 – IEEE Recommended Practice for Grounding in Industrial & Commercial Power Systems.

IEEE Std. 81 – IEEE Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Grounding System.

NETA-ATS-2017 – International Electrical Testing Association Acceptance Testing Specifications.

1.04 SUBMITTALS

A. Provide Shop drawings and product data for the grounding material.

B. Provide the following test reports for information: Grounding system test. Reference part 3 of herein specification Submit prior to energization and readiness review.

1.05 REGULATORY REQUIREMENTS

A. The following documents form part of the Specifications to the extent stated. Bring conflicts between Specifications, Drawings, and the referenced standards to the attention of the Project Manager, in writing, for resolution before taking any related action.

B. Conform to the latest adopted version of the National Electrical Code (NEC).

C. Furnish products listed by UL or other testing firm acceptable to AHJ.

D. Comply with UL 467 for grounding and bonding materials and equipment.

PART 2 - PRODUCTS

2.01 CONDUCTORS

A. Insulated Conductors: Copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction. Insulated conductors are required in wet locations as described by the NEC 250.

B. Bare Copper Conductors:

1. Solid Conductors: ASTM B 3.
2. Stranded Conductors: ASTM B 8.
3. Tinned Conductors: ASTM B 33.
4. Bonding Cable: ASTM B3, bare soft drawn stranded copper size to meet NEC 250 requirements.
5. Bonding Conductor: 4AWG or 6AWG, stranded conductor.
6. Bonding Jumper: Copper tape, braided conductors terminated with copper ferrules; 1-5/8 inches (41 mm) wide and 1/16 inch (1.6 mm) thick.
7. Tinned Bonding Jumper: Tinned-copper tape, braided conductors terminated with copper ferrules; 1-5/8 inches (41 mm) wide and 1/16 inch (1.6 mm) thick.

C. Grounding Bus: Predrilled rectangular bars of annealed copper, minimum dimensions of 2" height, 12" long minimum, ¼" thick, with 9/32-inch (7.14-mm) holes spaced 1-1/8 inches (28 mm) apart. Stand-off insulators for mounting shall comply with UL 891 for use in switchboards, 600 V and shall be Lexan or PVC, impulse tested at 5000 V. Leave 25% spare space or additional bus length. Exact dimensions to be determined to suit field conditions or as indicated on drawings.

2.02 CABLE TRAY & CONDUITS

A. All cable tray shall be bonded.

B. All conduits shall be bonded

2.03 CONNECTORS

A. Listed and labeled by an NRTL acceptable to authorities having jurisdiction for applications in which used and for specific types, sizes, and combinations of conductors and other items connected.

B. Grounding Connectors: Hydraulic compression tool applied connectors or exothermic welding process connectors or powder actuated compression tool applied connectors. Mechanical type of connectors are not acceptable.

C. Welded Connectors: Exothermic-welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.

D. Bus-Bar Connectors: Compression type, copper or copper alloy, with two hole wire terminals.

E. Ground Rod Clamps: Irreversible type, copper or copper alloy, terminal with hex head bolt.

F. Main and System bonding jumper. – Flexible tin-plated copper sized per NEC 250.28.

G. Manufacturers: Burndy Hyground Compression System, Erico/Cadweld, Amp Ampact Grounding System or approved equal.

2.04 COMPRESSION CONNECTORS

A. Manufacturers:

1. Thomas & Betts compression system.

2. Burndy "Hyground" compression system.

3. Approved equal.

B. Material: All copper.

2.05 EXOTHERMIC CONNECTIONS

A. Manufacturers:

1. Cadweld

2. Thermoweld
3. Burndyweld
4. Approved equal.

PART 3. EXECUTION

A. INSTALLATION

General: Install Products in accordance with manufacturer's instructions.

A. Grounding Conductors: Route along shortest and straightest paths possible unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.

B. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance.

C. Bonding to Structure: Bond directly to structural building steel, taking care not to penetrate any adjacent parts.

D. Bond supplemental grounding electrodes to grounding electrode system.

E. Bonding to Equipment Mounted on Vibration Isolation Hangers and Supports: Install bonding in flexible conduit or if exposed use braided copper jumper so vibration transmission is minimized to rigidly mounted equipment.

F. Raceways:

1. Ground metallic raceway systems. Bond to ground terminal with NEC code size jumper except where larger grounding conductor is included with circuit, use grounding bushing with lay-in lug trade size 1 inch and up.
2. Connect metal raceways, which terminate within an enclosure but without mechanical connection to the enclosure, by grounding bushings and ground wire to the grounding bus.
3. Where equipment supply conductors are in flexible metallic conduit, install insulated stranded copper equipment grounding conductor from outlet box to equipment frame.
4. Install equipment grounding conductor, NEC code size minimum unless noted on Drawings, in nonmetallic and metallic raceway systems.

G. Cable Tray

1. Ground metallic cable tray systems: Bond to grounding electrode system from the bonding loop with a bare 4/0AWG copper conductor.

2. Run bare 4/0 AWG copper conductor, unless otherwise noted on drawings, continuously along the length of the tray system. Bond sections of cable trays not mechanically continuous.

3. Secure bare copper conductor to the cable tray with grounding clamps, Burndy or approved equal.

H. Feeders and Branch Conduits:

1. Install continuous insulated equipment copper ground conductors in conduits and other circuits as indicated on the drawings.

2. Where installed in a continuous solid metallic raceway system and larger sizes that are not detailed, provide insulated equipment ground conductors for feeders and branch circuits sized in accordance with NEC Table 250-122.

I. Boxes, Cabinets, , Switchboards, Switchgears, Motor Control Centers, Transformer, Panelboards, and other electrical equipment enclosures:

1. Bond grounding conductors to enclosure with specified conductors and lugs. Install lugs only on thoroughly cleaned and stripped of paint contact surfaces.

2. Bond sections of service equipment enclosure to service ground bus.

J. Motors, Equipment and Appliances: Install code size equipment grounding conductor from source of power to equipment frame or manufacturer's designated ground terminal.

K. Receptacles: Connect ground terminal of receptacle to equipment ground system by 12AWG conductor bolted to outlet box except isolated grounds where noted. Self-grounding nature of receptacle devices does not eliminate conductor bolted to outlet box.

L. Ground Rods:

1. Drive rods until tops are 2 inches below finished floor or final grade unless otherwise indicated.

2. Interconnect ground rods with grounding electrode conductor below grade and as otherwise indicated. Make connections without exposing steel or damaging coating if any.

3. For grounding electrode system, install at least three rods spaced at least one-rod length from each other and located at least the same distance from other grounding electrodes, and connect to the service grounding electrode conductor.

M. Test Wells:

1. Ground rod driven through bottom of handhole. Handholes shall be at least 12 inches deep, with cover.

2. Test Wells: Install at least one test well for each service unless otherwise indicated. Install at the ground rod electrically closest to service entrance. Set top of test well flush with finished grade or floor.

N. Ground Ring

1. Install a grounding conductor, electrically connected to each building structure ground rod and to grounding electrode system, extending around the perimeter of the building or area indicated on grounding Drawings.

2. Install tinned-copper conductor not less than No. 4/0 AWG for ground ring and for taps to building steel.

3. Bury ground ring not less than 36 inches from building's foundation.

4. Bury ground ring at a depth of not less than 30 inches below earth's surface.

3.02 FIELD QUALITY CONTROL

A. Tests and Inspections:

1. After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements.

2. Inspect physical and mechanical condition. Verify tightness of bolted, electrical connections with a calibrated torque wrench according to manufacturer's written instructions. All bolted connections to ground shall be accessible.

3. Where non-NRTL equipment are connected, perform equipment ground/bonding test

4. Test completed grounding system at each location where a maximum ground-resistance level as specified in IEEE Std. 80 for substations and IEEE Std. 142 for industrial systems.

5. Make ground continuity positive throughout entire project.

6. Measure ground resistance no fewer than two full days after last trace of precipitation and without soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance.

7. Perform tests by fall-of-potential method according to IEEE 81. Perform test where new building service panels or switchgear, switchboard, motor control center, medium voltage transformer, integrated power center are installed. In addition, perform test where there is modifications of existing substation grounding system.

8. Where existing electrical service grounding system has been affected, perform point to point testing for resistivity per NETA/ATS.

B. Grounding system shall be considered defective if it does not pass tests and inspections.

C. Prepare test and inspection reports.

D. Excessive Ground Resistance: If resistance to ground exceeds specified value of 5 Ohm the Subcontractor shall notify Owner promptly. Include recommendations to reduce ground resistance.

END OF SECTION 26 05 26

SECTION 26 05 29 – SUPPORTING DEVICES

PART 1 - GENERAL

1.01 DESIGN RESPONSIBILITY

A. Design support and anchorage systems to resist all gravity and seismic forces in accordance with the requirements of the California Building Code.

1.02 SYSTEM DESCRIPTION

A. Safety factor of 4 required for every fastening device or support for electrical equipment installed. Support to withstand four times weight of equipment it supports. Provide seismic bracing per 2019 California Building Code (CBC) requirements for this building location.

B. The contractor shall review the existing building structural report to allow the final design of conduit supports to the building steel work. The structural calculations for the supporting conduit must take into account the existing conditions and to be performed per SLAC-I-720 0A24E-001-R006 - Seismic Design Specification for Buildings, Structures, Equipment, and Systems: 2020.

1.03 SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for the following:
   - Hangers.
   - Steel slotted support systems.
   - Nonmetallic support systems.
   - Trapeze hangers.
   - Clamps.
   - Turnbuckles.
   - Sockets.
   - Eye nuts.
   - Saddles.
   - Brackets.
2. Include rated capacities of furnished specialties and accessories.

B. Shop Drawings: Signed and sealed by a qualified professional engineer for fabrication and installation details for electrical hangers and support systems.

1. Trapeze hangers. Include product data for components.

2. Steel slotted-channel systems.

3. Nonmetallic slotted-channel systems.

4. Equipment supports.

5. Vibration Isolation Base Details: Detail fabrication, including anchorages and attachments to structure and to supported equipment. Include adjustable motor bases, rails, and frames for equipment mounting.

C. Delegated-Design Submittal: For hangers and supports for electrical systems.

1. Include design calculations and details of trapeze hangers.

2. Include design calculations for seismic restraints.

D. Coordination Drawings: Using input from installers of the items involved:

1. Structural members to which hangers and supports will be attached.

E. Seismic Qualification Certificates: For hangers and supports for electrical equipment and systems, accessories, and components, from manufacturer.

1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.

2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.

3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.

PART 2 - PRODUCTS

2.01 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer.

B. Seismic Performance: Hangers and supports shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

C. The term "withstand" means "the supported equipment and systems will remain in place without separation of any parts when subjected to the seismic forces specified.

D. Component Importance Factor: 1.0.

2.02 MATERIALS

2.02.1 CONCRETE FASTENERS

A. Drilled wedge expansion type concrete anchors, Phillips WS series, Ramset "Trubolt" or approved equal.

B. Provide powder driven concrete fasteners with washers by Remington, Ramset, or approved equal.

C. Drilled sleeve type expansion anchors, HILTI KWIK Bolt TZ series or approved equal.

2.02.2 CONDUIT STRAPS

A. Hot-dip galvanized, cast malleable iron, one hole type strap with cast clamp-backs and spacers as required.

B. OZ/Gedney "14-G" series strap and "141G" series spacer; Efcor "231" series strap and "131" series spacer; Thomas & Betts "1276" series strap and "1350" series spacer, or approved equal.

2.02.3 CONCRETE INSERTS

A. Pressed galvanized steel, spot insert, with oval slot capable of accepting support nuts of 1/4-inch to 1/2-inch diameter thread.

B. Unistrut No. M24 with "M2506" series nut; Superstrut No. 425 with "AB-102" series nut, Kinline No. 279 with "660" series nut, or equal.

2.02.4 DECK INSERTS

A. Steel plate 3/16-inch thick with threaded galvanized steel rod sized for load.

B. Superstrut No. C-475 series, Kinline No. 293 series, or equal.

2.02.5 CONSTRUCTION CHANNEL

A. 1-1/2 inch by 1-1/2 inch, 12-gauge galvanized steel channel with 9/16-inch diameter bolt holes, 1-7/8 inch on center, in the base of the channel.

B. B-Line by Cooper Industries, or equal.

2.02.6 THREADED ROD

A. Galvanized rod, sized for the load unless otherwise shown or specified.

2.02.7 CABLE TIES AND CLAMPS

A. Thomas and Betts Co. "Ty-Raps," Panduit "Pan-ty" or equal, one piece, nylon, reusable tape lashing ties. When exposed to UV rays and/or radiation, product used shall be rated for UV exposure.

2.02.8 DRYWALL MOUNTING BRACKET

A. Type MPLS by Caddy Fasteners for low voltage, Class 2 systems (TV antenna, phone jacks, and sound systems) unless otherwise noted on drawings.

2.02.9 BEAM CLAMPS

A. Beam clamps shall attach to both flanges of a beam, provide double-flange type clamps. Single-flange clamps are unacceptable.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Comply with NECA 1 and NECA 101 for application of hangers and supports for electrical equipment and systems unless requirements in this Section are stricter.

B. Maximum Support Spacing and Minimum Hanger Rod Size for Raceway: Space supports for EMTs, IMCs, and RMCs as scheduled in NECA 1, where its Table 1 lists maximum spacings that are less than those stated in NFPA 70. Minimum rod size shall be 1/4 inch (6 mm) in diameter.

C. Provide supporting devices as noted in other Sections of Division 26.

D. Fasten hanger rods, conduit clamps, outlet and junction boxes to building structure using precast inserts, expansion anchors, preset inserts or beam clamps.

E. Provide anchors, fasteners, and supports in accordance with NECA “Standard of Installation”.

F. Install products in accordance with manufacturer’s instructions.

G. Use adhesive or expansion type anchors for heavy duty load installations into concrete construction.

H. Use expansion anchors or preset inserts in solid masonry walls.

I. Use self-drilling anchors or expansion anchors on concrete surfaces.

J. Use plastic type anchors for light duty load installations into concrete, solid brick or masonry block construction. Light duty applications:

1. Metal surface raceways or multioutlet assemblies.
2. Outlet boxes.
3. Support EMT, 1 inch and smaller, using 1 or 2 hole strap.

K. Use plastic or hollow wall type anchors for installation of surface raceways or multioutlet assemblies into plaster wall construction

L. Support ¾ inch rigid conduit or IMC using 1 or 2 hole strap

M. Use sheet metal screws in sheet metal studs and wood screws in wood construction.

N. Do not fasten supports to piping, ductwork, mechanical equipment, or conduit.

O. Do not drill structural steel members unless first accepted in writing by the Owner.

P. Fabricate supports from structural steel or steel channel, rigidly welded or bolted to present a neat appearance. Use hexagon head bolts with spring lock washers under all nuts.

Q. Switchboards shall be supported and floor mounted.

R. Install surface-mounted cabinets, transformers and panelboards with a minimum of four anchors. Provide additional support backing in stud walls after to sheet rocking as required to adequately support cabinets and panels.

S. Bridge studs top and bottom with channels to support flush-mounted cabinets and panelboards in stud walls.

T. Anchor free-standing equipment on concrete pads where indicated.

W. Do not support conduit, cables, surface raceways, multioutlet assemblies or boxes to wallboard using plastic or hollow wall anchors. Provide independent supports to structural member for electrical luminaires, materials, or equipment installed in or on ceiling, walls or in void spaces or over furred or suspended ceilings.

X. Do not use other trade's fastening devices as supporting means for electrical equipment materials or fixtures.

Y. Do not use supports or fastening devices to support other than one particular item.

Z. Support conduits within 18 inches of outlets, boxes, panels, cabinets and deflections.

AA. Maximum distance between supports not to exceed 8 foot spacing.

BB. Securely suspend junction boxes, pull boxes or other conduit terminating housings located above suspended ceiling from the floor above or roof structure to prevent sagging and swaying.

CC. Provide seismic bracing per latest California Building Code (CBC) requirements.

END OF SECTION 26 05 29

SECTION 26 05 33 – RACEWAYS

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:

1. Raceways.
2. Conduit fittings.
3. Sleeves and chases.
4. Surface mounted raceways.

1.02 SYSTEM DESCRIPTION

A. Provide raceways, wires, cables, connector, boxes, devices, finish plates and the like for a complete and operational electrical system.

B. Electrical Connections: Connect equipment, electrically complete.

C. Supporting Devices: Safety factor of 4 required for every fastening device or support for electrical equipment installed. Support to withstand four times weight of equipment it supports. Provide seismic bracing per latest California Building Code (CBC) requirements for this building location.

D. The subcontractor shall review the existing building structural report to allow the final design of conduit supports to the building steel work. Subcontractor shall provide structural calculations for supporting conduit(s) must take into account the existing conditions.

1.03 SUBMITTALS

A. Submit For:

1. Raceways.
2. Conduit fittings.
3. Surface mounted raceways.

1.04 REGULATORY REQUIREMENTS

A. The following documents form part of the Specifications to the extent stated. Bring conflicts between Specifications, Drawings, and the referenced standards to the attention of the SLAC Project Manager, in writing, for resolution before taking any related action.

B. Conform to the latest adopted version of the National Electrical Code (NEC).

C. Furnish products listed by UL or other NRTL acceptable to AHJ.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Raceways: Allied Steel, Certainteed, Jones & Laughlin, Kraloy, Champion, or SLAC approved.

B. Conduit Fittings: 0-Z Gedney, Thomas & Betts, Crouse & Hinds, Champion or SLAC approved.

C. Surface Raceway System: Legrand Wiremold, Panduit, or SLAC approved.

2.02 CONDUITS

A. Rigid Metal Conduit (RMC):

1. Hot-dip galvanized after thread cutting.

2. Manufacture in conformance with UL 514 and ANSI C80.1.

3. Uniform finish coat with chromate for added protection.

B. Intermediate Metal Conduit (IMC):

1. Hot-dip galvanized after thread cutting.

2. Manufacture in conformance with UL 1242 and ANSI C80.6.

3. Uniform finish coat with chromate for added protection.

C. Electrical Metallic Tubing (EMT):

1. Hot-dip galvanized and chromate coated.

2. Manufacture in conformance with UL 797 and ANSI C80.3.

D. Flexible Metal Conduit (FMC):

1. Reduced wall flexible steel conduit.

2. Hot-dip galvanize steel strip prior to forming and joining.

3. Manufacture in conformance with Federal Specification A-A-55810.

E. Liquid-Tight Flexible Metal Conduit (LFMC):

1. Hot-dip galvanize steel strip prior to forming and joining.

2. PVC chemical resistant jacket extruded to core, up to 1 inch trade size.

3. PVC chemical resistant jacket, applied over core, up to 4 inch trade size.

F. PVC (RNC):

1. Class 40 heavy wall rigid conduit for installation in concrete duct banks.

2. Class 80 heavy wall rigid PVC where it would not be concrete encased.

3. Rated for use with 90C conductors.

4. Manufacture in conformance with UL 514 and NEMA TC-2.

G. Reinforced Thermosetting Resin Control (RTRC)

1. Champion Flame Shield® Fiberglass Phenolic Conduit or approve equal.

2. The conduit shall meet the requirements of NFPA130 and NFPA502. The conduit when installed together with the appropriately matched 2 hour fire-rated cable shall meet the requirements of the UL2196.

3. Iron Pipe Size (IPS) conduit shall be used with standard pipe clamps as the conduit is the same OD (outside diameter) as PVC and GRC conduits.

4. XW (Extra Heavy Wall) type conduit shall be used for Class I, Division 2 hazardous location applications or installations requiring severe impact strength. The XW conduit is allowed to be used in areas that are susceptible to damage per NEC Art. 355.10.F.

2.03 CONDUIT FITTINGS

A. Bushings:

1. Insulated Type for Threaded Rigid or Raceway Connectors without Factory Installed Plastic Throat Conductor Protection: Thomas & Betts 1222 Series or O-Z Gedney B Series.

2. Insulated Grounding Type for Threaded Rigid and Conduit Connectors: O-Z Gedney BLG Series.

B. Raceway Connectors and EMT Couplings:

1. Steel conductor and coupling bodies, with zinc electroplate or hot-dip galvanizing.

2. Connector locknuts are steel, with threading meeting ASTM tolerances. Locknuts are zinc electroplated or hot-dip galvanized.

3. Connector throats (EMT, flexible conduit, metal clad cable and cord-set connectors) have factory installed plastic inserts permanently installed. For normal cable or conductor exiting angles from the raceway (NEC bending radius), the cable jacket or conductor insulation bears only on the plastic throat insert.

4. Steel gland, Tomic or Breagle connectors and couplings.

5. For connecting FGC use stop-type couplings and terminal adapters with tapered threads and a straight shoulder so it can fit under the PVC sleeve of a PVC-coated steel fitting.

C. Expansion/Deflection Fittings:

1. EMT, O-Z Gedney Type TX.

2. RMC, O-Z Gedney Type AX, DX and AXDX, Crouse & Hinds XD.

3. FGC, Champion, double expansion joint with o-rings (back-to-back expansion joint).

PART 3 - EXECUTION

3.01 INSTALLATION

A. Finished Surfaces: Prevent cutting in connection with finished work. Make repairs in a manner approved by Construction Manager.

B. Fire rated walls: Each penetration through fire rated wall shall be caulked with 3M Firestop sealant or approved equal.

C. Exterior Building Wall: Each penetration through exterior building walls shall be weather proofed with 3M approved sealants. SLAC to review work for final approval.

D. Roof penetrations: Each penetrations through roof, provide flashing and weatherproofing. SLAC to review work for final approval.

E. Conduit:

1. Conduit Joints: Assemble conduit(s) in a continuous raceway and secure to boxes, panels, luminaires and equipment with fittings to maintain continuity. Provide watertight joints where embedded in concrete, below grade or in damp locations. Seal PVC conduit joints with solvent cement and metal conduit with metal thread primer. Rigid metallic conduit connections to be threaded, apply galvanized paint on threads, clean and tight (metal to metal). Threadless connections are not permitted for RMC.

2. Conduit Placement:

   a. Install continuous raceways for electrical and control wiring.

   b. Install conduits concealed in finished spaces, unless otherwise noted on design drawings.

   c. Install exposed conduits parallel or at right angles to building lines, tight to building surfaces and neatly offset into boxes.

   d. Do not install conduits or other electrical equipment in obvious passages, doorways, scuttles or crawl spaces which would impede or block the area passage's intended usage.

   e. Do not install conduits on surface of building exterior, across roof, on top of parapet walls, or across floors unless it is required per design drawings.

   f. Route raceway at least 6 inches from hot surfaces above 120F, including noninsulated steam lines, heat ducts, and the like.

3. Maximum Bends: Install code sized pull boxes to restrict maximum bends in a run of conduit to 270 degrees.

4. Conduit Terminations: Provide conduits shown on drawings which terminate without box, panel, cabinet or conduit fitting with not less than five full threads. Bushings and metal washer type sealer between bushing and conduit end.

5. Flexible Conduit: Install 12 inch minimum slack loop on flexible metallic conduit and PVC coated flexible metallic conduit.

6. Conduit Size: Provide conduit in minimum code permitted size for specified conductor size and quantity shown. Minimum trade size 3/4 inch.

7. Conduit Use Locations:

   a. Underground: PVC.

   b. Exterior Exposed locations, Wet Locations, Classified (NEC Art. 500) Locations, Interior Exposed locations up to 10’ above finished floor: RMC, IMC, or FGC Type

XW.

c. Corrosive environments (cooling towers, plating shop, DI plant, chemical hutches, labs or storage areas with corrosive materials): FGC Type XW.

d. Damp Locations: RMC and IMC up to 2 inches in diameter, FGC Type XW and IPS.

e. Interior locations and exposed 10 Ft above finished floor: EMT up to 3 inches in diameter, FGC type IPS.

f. Cast-In-Place Concrete and Masonry: RMC, IMC, and PVC.

g. Sharp Bends and Elbows: Use factory elbows for RMC, IMC, EMT, and FGC.

h. Install pull wire or nylon cord in empty raceways. Provide circuit numbering for each conduit where terminated into electrical equipment. Use printed permanent labeling.

i. Secure wire or cord at each end.

j. Elbow for Signal Systems: Use long radius factory elbows where linking sections of raceway for installation of signal cable.

k. Motors, recessed luminaires, low voltage transformers, pressure/temperature/flow instrumentation and equipment connections subject to movement or vibration, use flexible metallic conduit. The flexible conduit total length shall be limited by 6 Ft for luminaires and 3 Ft for other equipment.

l. Motors, low voltage transformers and other equipment connections subject to movement or vibration and subjected to any of the following conditions: exterior location, moist or humid atmosphere, water spray, oil or grease use PVC coated liquid tight flexible metallic conduit.

8. Branch Circuits: Homeruns for 20 amp branch circuits may be combined to a maximum of six current carrying conductors in a homerun raceway. Apply de-rating factors. Increase conductor size as needed.

9. Feeders: Do not combine feeder circuits.

10. Unless otherwise indicated, provide raceway systems for conductors.

F. Conduit Fittings:

1. Set screw type fittings are not allowed.

2. Use compression fittings in dry locations. Use raintight fittings in exterior and exposed locations for other than RMC type of conduit. Maximum size permitted in damp locations

and locations exposed to rain is 2 inches in diameter.

3. Use threaded type fittings in wet locations and damp or rain-exposed locations where conduit size is greater than 2 inches.

4. Use insulated hub and grounding bushings where conduit terminates to electrical equipment, pull boxes, or enclosures.

5. Use threaded type fittings in all hazardous locations.

6. Use PVC coated rigid steel conduit ells for underground power and telephone service entrance conduits. Use properly sized elbows for power service conduits and for telephone service conduits. Any buried conduit shall have a weather resistant warning tape buried at least 10-12 inches above the conduit trough out the whole conduit buried run.

7. Use insulated type bushings with ground provision at switchboards, panelboards, safety disconnect switches, junction boxes and the like that have feeders 6AWG and greater.

8. Provide bushing or EMT connector for conduits that do not terminate in box, enclosure, or the like.

9. Provide conduit listed expansion fittings at building expansion joints, separate structures, and at locations where conduit is exposed to thermal expansion and contraction.

10. Condulets and Conduit Bodies: Do not overfill condulets and conduit bodies. Refer to NEC Chapter 9, Table 1.

G. Sleeves and Chases - Floor, Ceiling and Wall Penetrations: Provide necessary rigid conduit sleeves, openings and chases where conduits or cables are required to pass through floors, ceiling or walls.

H. Building Seismic Joints: Conduit crossing building seismic joints: Provide listed conduit expansion/deflection fittings suitable for 4" expansion movement and 30° deflection movement. Subcontractor to verify in field locations of where conduits cross expansion joints.

END OF SECTION 26 05 33

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization	☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 26 05 34 - BOXES FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:

1. Junction and pull boxes.

1.02 SYSTEM DESCRIPTION

A. Outlet System: Provide electrical boxes and fittings for a complete installation. Include but not limited to outlet boxes, junction boxes, pull boxes, bushings, locknuts, and other necessary components.

1.03 SUBMITTALS

A. Provide shop drawings and product data for the following equipment:

1. Junction and pull boxes.

1.04 REGULATORY REQUIREMENTS

A. The following documents form part of the Specifications to the extent stated. Bring conflicts between Specifications, Drawings, and the referenced standards to the attention of the Project Manager, in writing, for resolution before taking any related action.

B. Conform to the latest adopted version of the National Electrical Code (NEC).

C. Furnish products listed by UL or other NRTL acceptable to AHJ.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Outlet Boxes: Bowers, Hubbell, or approved equal.

B. Weatherproof Outlet Boxes: Pass and Seymour, Bell, Allied Moulded, Carlon, or approved equal.

C. Junction and Pull Boxes: B-Line, Hoffman, Allied Moulded, or approved equal.

D. Box Extension Adapter for typical location: Pass and Seymour, Bell, Carlon, or approved equal.

E. Conduit Fittings: O-Z Gedney, Thomas & Betts, or approved.

F. Outlet Box Backing Pad: Lowry Outlet Box Pad, STI SSP Putty & Putty Pads or approved equal.

G. Acoustical Caulk: USG SHEETROCK Acoustical Sealant, Pecora AC-20 FTR, Johns Manville Firetemp CE, Tremco Acoustical Sealant, or approved

H. Outlet box insert: STI EP Powershield Electrical Box Inserts or approved equal.

2.02 JUNCTION AND PULL BOXES

A. Construction: Provide ANSI 49 gray enamel painted sheet steel junction and pull boxes, with screw-on covers; of the type shape and size, to suit each respective location and installation; with welded seams and equipped with steel nuts, bolts, screws and washers. Wherever splices are present, provide grounding/bonding. For corrosive location provide NEMA 4X boxes – stainless steel or fiberglass.

B. Location:

1. Install junction boxes in the accessible areas above “Drop” ceilings for drops into walls for receptacle outlets, wiring devices, electrical equipment, lights from overhead.

2. Install junction boxes and pull boxes to facilitate the installation of conductors and limiting the accumulated angular sum of bends between boxes, cabinets and appliances to 270 degrees.

2.03 BOX EXTENSION ADAPTER

A. Construction: Exterior die-cast aluminum or cast steel malleable. Interior extension rings shall be made of galvanized steel.

B. Location: Install over flush wall outlet boxes to permit flexible raceway extension from flush outlet to fixed or movable equipment. Bell 940 Series, Red Dot IHE4 Series.

2.04 CONDUIT FITTINGS

A. Requirements: Provide corrosion-resistant punched-steel box knockout closures, conduit locknuts and plastic conduit bushings of the type and size to suit each respective use and installation.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Locate boxes and conduit bodies so as to ensure accessibility of electrical wiring. Verify with design drawing that future installation, owner furnished equipment, furniture partitions, mechanical equipment, will not block accessibility

B. Recessed Boxes in Masonry Walls: Saw-cut opening for box in center of cell of masonry block, and install box flush with surface of wall.

C. Set metal floor boxes level and flush with finished floor surface.

D. Set nonmetallic floor boxes level. Trim after installation to fit flush with finished floor surface.

E. Round Boxes: Avoid using round boxes where conduit must enter through side of box, which would result in a difficult and insecure connection with a locknut or bushing on the rounded surface.

F. Octagonal boxes: Use for ceiling fan installations.

G. Anchoring: Secure boxes rigidly to the substrate upon which they are being mounted, or solidly embed boxes in concrete or masonry.

H. Special Application: Provide weatherproof outlets for locations exposed to weather or moisture. Outlet box insert: STI EP Powershield Electrical Box Inserts or approved equal.

I. Knockout Closures: Provide listed knockouts for dry or wet locations knockout closures to cap unused knockout holes where blanks have been removed.

J. Coordinate electrical device locations (switches, receptacles, and the like) to prevent mounting devices in mirrors, back splashes, behind cabinets, and the like.

K. Seal the back of boxes in acoustically rated assemblies with an outlet box backing pad and caulk gap between box and wall with non-hardening acoustical caulk.

L. Provide fire putty to recess boxes to maintain rating of wall.

M. Provide gasket seals for exterior boxes to maintain weatherproof rating.

N. All box covers shall be clearly marked in the field with printed labels identifying power source and circuit numbers.

END OF SECTION 26 05 34

SECTION 26 05 48 – VIBRATION AND SEISMIC CONTROLS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.01 DESCRIPTION

A. Selection, engineering, and incorporating all support, bracing, anchorage and seismic restraints for all equipment, conduits, pull boxes and other electrical systems.

B. Provide isolation for dry type transformers, and at electrical connections to rotating or vibrating equipment.

C. Provide seismic restraints for all electrical equipment.

D. Supports, bracing, anchorage and seismic restraints shall not reduce the vibration isolation capabilities of the system. Miscellaneous steel to make support compatible with equipment.

E. Safety factor of 4 required for every fastening device or support for electrical equipment installed. Support to withstand four times weight of equipment it supports. Provide seismic bracing per latest California Building Code (CBC) requirements for this building location.

F. The equipment and major components shall be suitable for and certified to meet all applicable seismic requirements of the latest California building code (CBC) through zone 4 application. Guidelines for the installation consistent with these requirements shall be provided by the switchgear manufacturer and be based upon testing of representative equipment. The test response spectrum shall be based upon a 5% minimum damping factor, cbc: a peak of 2.15g's, and a zpa of 0.86g's applied at the base of the equipment. The tests shall fully envelop this response spectrum for all equipment natural frequencies up to at least 35 hz. Guidelines for the installation consistent with these requirements shall be provided by the switchgear manufacturer and be based upon testing of representative equipment. The equipment manufacturer shall document the requirements necessary for proper seismic mounting of the equipment

G. The contractor shall review the existing building structural report to allow the final design of conduit supports to the building steel work. The structural calculations for the supporting conduit must take into account the existing conditions.

H. The following minimum mounting and installation guidelines shall be met, unless specifically modified by the above referenced standards.

I. The Subcontractor shall provide equipment anchorage details, coordinated with the equipment mounting provision, prepared and stamped by a licensed civil engineer in the state. Mounting recommendations shall be provided by the manufacturer based upon approved shake table tests used to verify the seismic design of the equipment.

J. The equipment manufacturer shall certify that the equipment can withstand, that is, function following the seismic event, including both vertical and lateral required response spectra as specified in above codes.

K. The equipment manufacturer shall document the requirements necessary for proper seismic mounting of the equipment. Seismic qualification shall be considered achieved when the capability of the equipment, meets or exceeds the specified response spectra.

1.02 QUALITY ASSURANCE

A. Vibration isolators and seismic restraints shall be of the same manufacturer.

B. Anchorage and Bracing: Anchor, support and brace conduit to resist seismic forces in accordance with requirements for anchorage bracing as specified.

1.03 STANDARDS

1.04 DESIGN RESPONSIBILITY

A. Design support and anchorage systems in accordance with procedures indicated herein and in Division 01 – General Requirements.

B. Contractor shall be responsible for the design of the support and bracing system for suspended pull boxes, conduit and cable trays. Pull boxes, conduit, and cable trays shall be supported and braced in accordance with structural seismic design criteria listed on structural drawings.

C. Seismic anchorage of all electrical systems for this project shall be in accordance with the California Building Code.

D. Seismic devices and anchorage shall in no way degrade the efficiency of vibration isolation systems.

E. Contractor shall submit the equipment anchor information to justify the use of the anchorage details in these drawings for review and acceptance by the design team.

F. Equipment anchorage design for each piece of equipment where the anchorage details on the drawings cannot be justified for use.

1. Equipment anchorage details that clearly show the method of attachment and bracing to the structure. These details shall be coordinated with the actual structural system on this project. ICBO report numbers shall be provided in the details for all connectors. Details shall be stamped and signed by a licensed Structural Engineer in the State of California.

1.05 SUBMITTALS

A. Product data and calculations to demonstrate compliance with the structural seismic design criteria requirements shown on structural drawings.

B. Manufacturer's product data sheets and installation instructions for each vibration isolator and seismic restraint.

C. Plan and elevation diagrams showing equipment, points of attachment, vibration isolators, seismic restraints, mounting methods, and hardware types and sizes.

D. Submit anchorage and bracing calculations stamped and signed by a California-registered Structural Engineer for all equipment as required by State, Federal or regulatory agencies. Calculations shall clearly show equipment weight, equipment center of gravity, location of attachment to the structure, and the seismic and gravity forces at each attachment location. Include designs for the attachment of the equipment or equipment support base to the structure, and the equipment to the support base.

E. Field inspection report.

1.06 FIELD INSPECTION

A. Upon completion of the installation, the manufacturer's local representative shall field inspect the installation and submit a report verifying the completeness and performance of the installation.

PART 2 - PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS

A. Amber-Booth, Mason Industries,

B. Vibration Eliminator Co.,

C. Vibration Mounting & Controls Inc., or Vibrex Vibration Control Systems.

D. SLAC Approved Manufacturer

2.02 VIBRATION ISOLATION AND SEISMIC RESTRAINTS

A. General:

1. Spring diameters shall be no less than 0.8 of the compressed height of the spring at rated load.

2. Springs shall have an additional minimum travel to solid equal to 50 percent of the rated deflection.

B. Mounting Method Type A:

1. Floor mounted spring isolators for seismic and restrained service.

2. Built in resilient limit stops shall limit upward, downward, and horizontal travel to a maximum of ¼ inch.

3. Trapped holes in top plate for bolting to equipment.

4. Mounting holes in bottom plate for bolting to concrete housekeeping pad.

5. Neoprene pad between bottom plate of isolator housing and bottom of spring isolator.

6. Mason Industries type SLR.

C. Mounting Method Type B:

1. Hanger rod neoprene isolators.

2. 45-degree slack seismic restraint cables.

3. Neoprene element with a projecting bushing to prevent steel to steel contact.

4. Steel retainer box encasing the neoprene element.

5. Rod shall be able to swing 15 degrees before contacting resilient bushing.

6. Mason Industries type HD neoprene hanger and type SCB seismic cable brace.

D. Mounting Method Type C:

1. Floor mounted bridge bearing neoprene mounts with all directional seismic capability.

2. Two separated and opposing molded bridge bearing neoprene elements contained in a ductile iron casting.

3. Mounting holes in bottom plate for bolting to concrete housekeeping pad.

4. Mason Industries type BR.

PART 3 - EXECUTION

3.01 GENERAL

A. Installation shall be in accordance with seismic restraint calculations and manufacturer’s installation instructions.

B. Verify that mounting methods provide the required vibration isolation and seismic restraint and that there are no vibration short circuits.

C. Conduit connected to rotating or vibrating equipment shall be flexible metal conduit or liquid tight flexible conduit.

END OF SECTION 26 05 48

SECTION 26 05 53 – Electrical Identifications

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:

1. Equipment identification labels.
2. Conductor identification numbers.
3. Circuit identification.
4. Conduit markers.

1.02 SYSTEM DESCRIPTION

A. Design Requirements:

1. Coordinate names, abbreviations and other designations with equipment specified in this or other Divisions of the Specification or identified on Drawings.
2. Fasten labels to equipment in a secure and permanent manner.
3. Mark underground utilities in conformance with ANSI standard Z535.1

1.03 REFERENCES

A. SLAC Specifications – Division 26. Electrical

B. ANSI standard Z535.1 Safety Colors for temporary marking and facility identification

1.04 SUBMITTALS

A. Submit shop drawings under provisions of Division 1.

B. Include data sheets for identification materials furnished.

C. Include schedule for nameplates and labels.

1.05 REGULATORY REQUIREMENTS

A. The following documents form part of the Specifications to the extent stated. Bring conflicts between Specifications, Drawings, and the referenced standards to the attention of the Project Manager, in writing, for resolution before taking any related action.

B. Conform to the latest adopted version of the National Electrical Code (NEC).

C. Furnish products listed by UL or NRTL acceptable to AHJ.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Engraved Labels: Lamicoid, or SLAC approved.

B. Conductor Numbers: Brady, or SLAC approved.

C. Underground Utilities Ribbon: Allen Systems, Inc., or SLAC approved.

2.02 ENGRAVED LABELS

A. Melamine plastic laminate, white with black core, 1/8-inch thick.

B. UV and weatherproof rated.

C. Engravers standard letter style, minimum 3/8-inch high capital letters for smaller equipment such as panel boards and control panels. Larger equipment, such as switchgear, MCC, switchboards, transformers, medium voltage equipment shall use minimum of 1-inch high capital letters.

D. Drill or punch labels for mechanical fastening. Use self-tapping stainless steel screws. Use proper size screw not to penetrate more than 1/8 exposure.

E. DYMO tape labels are not acceptable for hazardous locations and for end use equipment.

2.03 CONDUCTOR NUMBERS

A. Manufacturers' standard vinyl tubing cable and conductor markers. Preprinted black numbers on white or yellow field.

2.04 BRANCH CIRCUIT SCHEDULES

A. Provide branch circuit identification schedules, typewritten, clearly filled out, to identify load connected to each circuit and location of load. Numbers to correspond to numbers assigned to each circuit breaker pole position.

B. Provide circuit number for each phase (pole).

C. Provide two columns, odd numbers in left column, even numbers in right column, with 3-inch-wide line for typing connected load information.

2.05 RELAY PANEL SCHEDULE

A. Provide typewritten schedule to identify the incoming circuit, the controlled load, and the controlling devices for each relay.

2.06 CIRCUIT BREAKER IDENTIFICATION

A. Provide permanent identification number in or on panelboard dead-front adjacent to each circuit breaker pole position. Seton adhesive is approved, other adhesives by specific prior approval only.

B. Horizontal centerline of engraved numbers to correspond with centerline of circuit breaker pole position.

C. Provide circuit number for each phase (pole).

2.07 WIRE AND TERMINAL MARKERS

A. Silver Fox pre-printed non-shrink tubing wire labels or approved equal.

B. Thomas & Betts WSL, Brady B191 series, or approved equal.

PART 3 - EXECUTION

3.01 GRAPHICS

Coordinate names, abbreviations and designations used on Drawings with equipment labels.

3.02 INSTALLATION

A. Degrease and clean surfaces to receive nameplates and labels.

B. Install nameplates and labels parallel to equipment lines.

C. Secure nameplates to equipment fronts using screws, rivets, or adhesive. Secure nameplate to inside face of recessed panelboard doors in finished locations.

D. Do not use tape for any application, except for temporary labeling during construction phase.

E. Do not use hand written lettering or ID tags.

3.03 WIRE IDENTIFICATION

A. Provide wire markers on each conductor in panelboards, gutters, pull boxes, outlet and junction boxes, and at load connection. Identify with branch circuit or feeder number for power and lighting circuits, and with control wire number as indicated on equipment manufacturer's shop drawings for control wiring.

B. Provide conductor phase color coding as per SECTION 26 00 10 –General Electrical.

3.04 EQUIPMENT/SYSTEM IDENTIFICATION

A. Install an engraved label on each major electrical equipment indicating both equipment name and circuit serving equipment including but not limited to the following items:

1. Disconnect switches, identify item of equipment controlled.
2. Relays.
3. Contactors.
4. Control switches.
5. Selector switches.
6. Service disconnect and distribution switches, identify connected load.
7. Manual and automatic transfer switches.
8. Low voltage circuit breakers
9. Emergency and standby generators.
10. Uninterruptable power supplies (UPS).
11. Lighting invertors.
12. Central or master unit of each electrical system including communication/signal systems, unless the unit incorporates its own self-explanatory identification.
13. Medium voltage circuit breakers.
14. Medium voltage switches.
15. Medium voltage switchgears.
16. Transformers.
17. Main Switchboards.

18. Panelboards.
19. Motor Control Centers.
20. Integrated power centers (IPC)
21. Power distribution units (PDU)

3.05 LABEL COLORING

A. Label colors shall be as follows:

1.	Black	Normal Power
2.	Orange	Emergency and Legally Required Standby
3.	Red	Fire Alarm and Annunciation

3.06 NAMEPLATE ENGRAVING

A. Provide type "NP" nameplates of minimum letter height as noted below:

1. Switchgears, IPC, PDU, Panelboards, Switchboards and Motor Control Centers: 1/4-inch to identify equipment designation. 1/8-inch to identify voltage rating and source.
2. Individual Circuit Breakers, Switches and Motor Starters in Panelboards, Switchboards, and Motor Control Centers: 1/8-inch to identify circuit and load served, including location.
3. Individual Circuit Breakers, Enclosed Switches, and Motor Starters: 1/8-inch to identify load served.
4. Transformers: 1/4-inch to identify equipment designation. 1/8-inch to identify primary and secondary voltages, primary source, and secondary load and location.
5. Equipment Cabinets, Terminal Cabinets, Control Panels and other Cabinets enclosing apparatus: 3/8-inch to identify equipment and designation.
6. Receptacles: 1/2-inch clear P-touch or equal tape with 3/16-high black lettering to identify panelboard and circuit fed from. Attach to coverplate. Submit sample plate.
7. Junction and pull boxes, gutters, Wiremold wireways: 1”-inch clear P-touch or equal tape with 3/8”-high black lettering to identify panelboard and circuit fed from. Attach to coverplate. Submit sample plate. Labels shall be readable from the floor.

8. Conduits leaving and incoming MCC, IPC, switchboards, panelboards, junction and pull boxes: 1-inch U-line, weather resistant laser labels with ½” black lettering.

B. Provide type "LP" metal legend plates for attachment to panel mounted operator's devices such as push buttons, selector switches, etc.

C. Provide 2 inch high letters, red or white, porcelain enamel “Danger – High Voltage” warning sign on all asides of the high voltage section of unit substation.

3.07 BRASS TAGS

A. Provide brass tags for individual ground conductors to exposed ground bus indicating connection. For example: "Ufer", "Cold water bond".

B. Provide tags for all feeder cables in underground vaults and pull boxes. Provide tags for empty conduits in underground vault, pull boxes, and stubs.

3.08 PANELBOARD, IPC, AND SWITCHBOARD DIRECTORIES:

A. Provide typewritten directories arranged in numerical order showing number of room (use room numbering from field or request from SLAC space planning) in which each device served by each panelboard / switchboard / IPC circuit is located.

B. Mount directories in a 6 inch by 8 inch metal frame under a clear plastic cover inside each panelboard / switchboard door.

3.09 APPLICATION

A. Install engraved labels on the inside of flush panels. Labels shall be visible when door is opened. Install engraved label on outside of surface panel.

B. Install signs at locations detailed or, where not otherwise indicated, at location for best convenience of viewing without interference with operation and maintenance of equipment.

C. Where signs are to be applied to surfaces which require finish, install identification after completion of painting.

D. On the front of receptacle and switch finish plates provide label with the circuit that each device is connected to. Label is self-adhesive type with black letters and clear background, 18 point lettering size. In outdoor installations use Uline or SLAC approved equal UV and weatherresitant labels.

3.10 ELECTRICAL EQUIPMENT NOMENCLATURE

BUILDING #	VOLTAGE CLASS	EQUIPMENT TYPE	FLOOR, SECTOR, ROOM	SEQUENCE #
B950	12 – 12.47KV 41 – 4160V 23 – 2300V 6 – 600V 4 – 480V 2 – 208V/120V	TX – Transformer MSG – Main Switchgear MCC – Motor Control Center DP – Distribution Panel Or Switchboard PB – Panelboard CB – Enclosed Circuit Breaker BD - Breaker-Disconnect DS – Disconnect Switch GEN – Diesel Generator UPS – Uninterruptable Power Supply MTS - Manual Transfer Switch ATS - Automatic Transfer Switch IPC – Integrated Power Center PDU – Power Distribution Unit LP – Lighting Panelboard LCP – Lighting Control Panel INV – Lighting Inverter E___- Emergency Equipment S___- Standby Power Equipment		

END OF SECTION 26 05 53

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization ☑ No ☐ Yes	
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 26 08 00 - COMMISSIONING OF ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.01 SUMMARY

A. Section includes Definitions, warranties, test equipment requirements, and electrical commissioning requirements as required.

B. Related Sections:

– SECTION 01 91 13 General Commissioning Requirements.

– SECTION 26 00 10 General Electrical Requirements

– SECTION 26 08 05 Electrical Acceptance Testing

– Div. 26 Electrical Equipment Specific Sections.

1.02 WARRANTY

C. Manufacturer's Warranty:

1. Commissioning, inspecting, and testing shall not modify terms or time periods of mechanical equipment, systems, and controls warranties including related equipment and systems, and adjacent work.

2. Electrical system warranties shall start from date of SLAC Contract Administrator acceptance.

1.03 SUBMITTALS

D. Submittals as required by SECTION 26 00 10, General Electrical Requirements and Division 01, General Requirements.

E. In addition, submit the following:

1. Inspection and test reports.

2. Proposed functional performance checklist.

a. Provide functional performance checklist based on manufacturer startup procedure.

1) Include integration of all subsystems as a whole and complete system. Verify sequence of operation.

2) Checklist shall include all various configurations and modes of operation.

3. Bolted contact torque charts.

4. Certificates of completion of installation, prestart, and startup activities

5. O&M manuals

PART 2 - PRODUCTS

2.01 TEST EQUIPMENT

A. Provide test equipment required to perform startup, initial checkout and functional performance testing for the equipment being tested under Division 26, Electrical. Furnish two-way radios for each testing participant as needed.

B. Furnish special equipment, tools and instruments (specific to tested equipment and only available from vendor) required for testing. At conclusion of commissioning, turn equipment over to the Owner

C. Manufacturer: Furnish proprietary test equipment and software required by equipment manufacturer procedures for programming and/or start-up. Demonstrate its use, and assist in the commissioning process as needed. Proprietary test equipment (and software) to become the property of the Owner upon completion of the commissioning process.

D. Data logging equipment and software required to test equipment will be furnished by the Commissioning Authority during commissioning.

E. Testing equipment to be of sufficient quality and accuracy to test and/or measure system performance with the tolerances specified in the Specifications. All equipment shall be provided with dated proof of calibration and maintain recalibration if more than 12 months.

PART 3 - EXECUTION

3.01 GENERAL DOCUMENTATION REQUIREMENTS

With assistance from the installing contractors, the Commissioning Authority shall prepare Pre-Functional Checklists for commissioned components, equipment, and systems as required in SECTION 26 08 05 and equipment specific sections.

A. Red-lined Drawings:

1. Verify equipment, systems, instrumentation, wiring and components are shown correctly on redlined drawings.

2. Installation Subcontractor shall record the redlined drawing changes, as a result of Functional Testing and incorporate into the final as-built drawings.

B. Operation and Maintenance Data:

1. Submit a copy of O&M literature within 45 days of each submittal acceptance for use during the commissioning process..

2. The Commissioning Authority shall review a draft of the O&M literature once ~~for~~ conformance meets project requirements. Commissioning Authority will submit further comments to the Subcontractor for revising.

3. The Commissioning Authority to receive the final approved O&M literature once corrected by the Subcontractor.

C. Demonstration and Training:

1. Provide planning and scheduling of classroom training and onsite hands on training/demonstration as required by the specifications.

2. Submit complete training plan and schedule to the Commissioning Authority four weeks prior to training.

3. Submit training agenda for each training session to the Commissioning Authority one (1) week prior to the training session.

4. Notify the Commissioning Authority at least 72 hours in advance of scheduled test so that the Commissioning Authority and Owner's representative may observe testing. Submit copies of the test record to the Commissioning Authority, Owner, and Architect.

5. Engage a Factory-authorized service representative to train SLAC's maintenance personnel to adjust, operate, and maintain specific equipment.

6. Train SLAC's maintenance personnel on procedures and schedules for starting and stopping, trouble shooting, servicing, and maintaining equipment.

7. Review data in O&M Manuals.

3.02 SUBCONTRACTOR'S RESPONSIBILITIES

A. Perform commissioning tests at the direction of the Commissioning Authority.

B. Subcontractor redlines - reference specification SECTION 26 00 10.

C. Attend construction phase controls coordination meetings. Attend weekly commissioning meetings.

D. Participate in electrical systems, assemblies, equipment, and component maintenance orientation and inspection as directed by the Commissioning Authority.

E. Provide information requested by the Commissioning Authority for final commissioning documentation.

F. Include requirements for submittal data, operation and maintenance data, and training in each purchase order or sub-contract written.

G. Prepare preliminary schedule for electrical system orientation and inspections, operation and maintenance manual submissions, training sessions, equipment start-up and task completion for owner. Distribute preliminary schedule to commissioning team members.

H. Update schedule as required throughout the construction period.

I. During the startup and initial checkout process, execute the related portions of the pre-functional checklists for commissioned equipment.

J. Assist the Commissioning Authority in verification and functional performance tests. Remove any equipment covers necessary for SLAC to perform QC, Inspections and/or Tests. Have all medium voltage cables terminated but in a disconnected state,

K. Once SLAC electricians complete any applicable cable testing, subcontractor shall complete the cable connections and torqueing

L. Provide measuring instruments and logging devices to record test data, and provide data acquisition equipment to record data for the complete range of testing for the required test period.

M. Gather operation and maintenance literature on equipment, and assemble in binders as required by the specifications. Provide electronic version of this literature. Submit to Commissioning Authority 15 business days after submittal acceptance.

N. Coordinate with the Commissioning Authority the 48-hour advance notice so that the witnessing of equipment and system start-up and testing can to begin.

O. Participate in, and schedule vendors and subcontractors to participate in the training sessions. Notify SLAC in not less 30 days prior training date.

P. Provide written notification to the Commissioning Authority and Owner that work has been completed in accordance with the Contract Documents, and that the equipment, systems, and sub-system are operating as required.

Q. Obtain performance documentation from equipment supplier.

R. Provide training of the Owner's operating staff using expert qualified personnel.

S. Participate in initial energization readiness review meetings.

T. Equipment Suppliers

1. Submit requested submittal data, including detailed start-up procedures and specific responsibilities of SLAC, to keep warranties in force.

2. Assist in equipment testing per agreements with contractors.

3. Provide information requested by Commissioning Authority regarding equipment sequence of operation and testing procedures.

3.03 TESTING PREPARATION

A. Certify in writing to the Commissioning Authority that Electrical systems, subsystems, and equipment have been installed, started and, are operating according to the Contract Documents.

B. Certify in writing to the Commissioning Authority that Electrical instrumentation and control systems have been completed and that they are operating according to the Contract Documents.

C. Schedule pre-energization testing with sufficient time needed for SLAC review. Allow minimum of 10 business days.

D. Certify in writing that testing procedures have been completed and that testing reports have been submitted, discrepancies corrected, and corrective work approved.

E. Place systems, subsystems, and equipment into operating mode to be tested (e.g., normal shutdown, normal auto position, normal manual position, maintenance power (alternative feeder), and alarm conditions).

F. Inspect and verify the position of each device and interlock identified on commissioning checklists.

G. Testing Instrumentation: Install measuring instruments and logging devices to record test data as directed by the Commissioning Authority.

3.04 GENERAL TESTING REQUIREMENTS

A. Provide qualified technicians, appropriate instrumentation, and calibrated tools to perform commissioning test at the direction of the Commissioning Authority.

B. Scope of Electrical testing includes the entire Electrical installation, from the incoming power throughout all of the distribution system that is provided under this contract, including owner furnished equipment. Testing includes measuring, but is not limited to resistance, voltage, and amperage of system(s) and devices.

C. Reference general specification SECTION 26 00 10 and each applicable equipment specification sections.

D. Test operating modes, interlocks, control responses, and responses to abnormal or emergency/maintenance conditions, and verify proper response of each automation system controllers and sensors.

E. The Commissioning Authority along with the Electrical subcontractor and other contracted subcontractors to prepare detailed testing plans, procedures, and checklists for Electrical systems, subsystems, and equipment.

F. Develop Method of Procedure (MOP) for commissioning activities and submit it to SLAC in not less than 10 business days.

G. Tests shall be performed using design conditions whenever possible.

H. Simulated conditions may need to be imposed using an artificial load when it is not practical to test under design conditions. Before simulating conditions, calibrate testing instruments. Provide equipment to simulate loads. Set simulated conditions as directed by the Commissioning Authority and document simulated conditions and methods of simulation. After tests, return settings to normal operating conditions.

I. The Commissioning Authority may direct that set points and sensor values be altered when simulating conditions is not practical.

J. If tests cannot be completed because of a deficiency outside the scope of the Electrical system, document the deficiency and report it to SLAC. After deficiencies are resolved, reschedule tests. In test report include “as-found” and “as-left” conditions.

K. If the testing plan indicates specific seasonal testing, complete appropriate initial performance tests and documentation and schedule seasonal tests.

3.05 ELECTRICAL SYSTEMS, SUBSYSTEMS, AND EQUIPMENT TESTING PROCEDURES

A. Equipment Testing and Acceptance Procedures: Testing requirements are specified in individual Division 26, Electrical Sections. Provide submittals, test data, inspector record and certifications to the Commissioning Authority.

B. Electrical Instrumentation and Control System Testing: Field testing plans and testing requirements are specified in specific section for electrical distribution, instrumentation & control, and sequence of operations. Assist the Commissioning Authority with preparation of testing plans.

C. Emergency and Standby Generator Testing and Acceptance Procedures: Provide technicians, load banks, infrared cameras, instrumentation, tools and equipment to test performance of designated systems and devices at the direction of the Commissioning Authority.

D. Electrical Distribution System Testing: Provide technicians, load banks, infrared cameras, instrumentation, tools and equipment to test performance of designated systems and devices at the direction of the Commissioning Authority.

E. The work included in the commissioning process involves a complete and thorough evaluation of the operation and performance of components, systems and sub-systems within the Owner’s intended Project objective.

3.06 OPERATION AND MAINTENANCE MANUALS

A. The Operation and Maintenance Manuals to conform to Contract Documents requirements as stated in Division 26, Electrical.

3.07 TRAINING OF OWNER PERSONNEL

A. Electrical Subcontractor's training responsibilities:

1. Provide the Commissioning Authority with a training plan two weeks before the planned training.

2. Provide designated Owner’s personnel with comprehensive training in the understanding of the systems and the operation and maintenance of each major piece of commissioned electrical equipment or system.

3. Training starts with classroom sessions, , followed by hands on training on each piece of equipment, which illustrates the various modes of operation, including but not limited to startup, shutdown, control interlocks, fire/smoke alarm, power failure, power transfer scheme, etc.

4. During any demonstration, should the system fail to perform in accordance with the requirements of the O&M manual or sequence of operations, the system will be repaired or adjusted as necessary and the demonstration repeated.

5. The appropriate trade or manufacturer's representative provides the instructions on each major piece of equipment. This person may be the start-up technician for the piece of equipment, the installing contractor or manufacturer's representative. Practical building operating expertise as well as in-depth knowledge of modes of operation of the specific piece of equipment are required. More than one party may be required to execute the training.

6. The training sessions follows the outline in the Table of Contents of the operation and maintenance manual and illustrate whenever possible the use of the O&M manuals for reference.

7. The switchgear and other complex equipment hands on training shall be provided with temporary power and before energization with rated voltage.

8. If needed, schedule multiple training sessions for multiple groups of SLAC.

9. Training includes:

a. Use the printed installation, operation and maintenance instruction material included in the O&M manuals.

b. Include a review of the written O&M instructions emphasizing safe and proper operating requirements, preventative maintenance, special tools needed and spare parts inventory suggestions. The training includes start-up, operation in modes possible, shut down, seasonal changeover and any emergency procedures.

c. Discuss relevant health and safety issues and concerns.

d. Discuss warranties and guarantees.

e. Cover common troubleshooting problems and solutions.

f. Explain information included in the O&M manuals and the location of plans and manuals in the facility.

g. Discuss any peculiarities of equipment installation or operation.

10. Hands-on training includes start-up, operation in modes possible, including manual, shut down and any emergency procedures and preventative maintenance of pieces of equipment.

11. Fully explain and demonstrate the operation, function and overrides of any local packaged controls, not controlled by the central control system.

12. Schedule training after functional testing is complete, unless approved otherwise by SLAC.

END OF SECTION 26 08 00

SECTION 26 08 05 - ELECTRICAL ACCEPTANCE TESTING

PART 1 – GENERAL

1.01 SUMMARY

A. Section Includes:

1. Testing, evaluation and calibration of equipment provided, installed and connected in Division 26.

2. Evaluation of connection and normal operation of utilization equipment, provided in other Divisions, for installation and connection in Division 26.

1.02 RELATED SECTIONS

A. 014523 Testing and Inspection Services

B. 260010 General Electrical Requirements

C. 260800 Commissioning of Electrical

1.03 REFERENCES

A. Acceptance Testing Criteria: Adopted latest edition of Acceptance Testing Specifications for Electrical Power Distribution Equipment and Systems, published by NETA.

B. National Electrical Code (NEC), Applicable Codes, Standards and References:

1. National Electrical Manufacturer's Association (NEMA).
2. American Society for Testing and Materials (ASTM).
3. Institute of Electrical and Electronic Engineers (IEEE).
4. National Electrical Testing Association (NETA).
5. American National Standards Institute (ANSI).
6. State and local codes and ordinances.
7. Insulated Power Cable Engineers Association (IPCEA).
8. OSHA Part 1910; Subpart S, 1910.308.
9. National Fire Protection Association (NFPA).

1.04 SYSTEM DESCRIPTION

A. Performance Requirements:

1. Retain the services of a recognized independent NETA accredited testing firm for the purpose of performing inspections and tests as specified herein.

2. Independent testing firm to provide report directly to Construction Manager.

3. Testing firm shall provide all material, equipment, labor and technical supervision to perform tests and inspections.

4. It is the intent of these tests to assure that electrical equipment, Subcontractor or SLAC supplied, is operational within industry and manufacturer's tolerances and installed in accordance with design Specifications.

5. Tests and inspections determine suitability for energization.

6. Supply to the independent testing organization complete sets of SLAC approved shop drawings, coordination study setting of adjustable devices and other information requested by testing company.

7. In not later than 2 days prior to scheduled testing verify equipment to be tested has been properly stored and handled. Allow time to equipment cleaning and drying out.

B. Scope of Testing, Evaluation and Calibration:

1. Distribution substations (MV, transformer, LV).

2. Low voltage circuit breakers molded case type 225A and higher rated.

3. Low voltage circuit breakers with adjustable trip units.

4 Photovoltaic systems; Inverters

5 Switch Disconnects, Air Disconnect Switches (ADS),

6 Capacitor Banks

7 Static VAR compensator

8. Control Panels.

9. Cables and wires.

10. Grounding systems.

11. Motor control centers (MCC).

12. Integrated power centers (IPC).

13. Power distribution units (PDU)

14. Emergency systems
15. Standby generators
16. Automatic Transfer Switches (ATS)
17. Manual Transfer Switches (MTS) and Quick-Connect Docking Stations (QCDS)
18. Uninterruptable Power Supplies (UPS) and Lighting Inverters
19. Protection, control, instrumentation, metering and communication systems.
20. Switchboards
21. Panelboards
22. Busways and cable-busses.
23. Lighting systems.
24. Temporary power distribution equipment, Construction Power Skids, Shock Blocks

1.05 SUBMITTALS

A. Submit proof of qualifications of the testing firm, including copies of certifications required in Par. 1.03.A, with Division 26 submittals.

B. Submit Testing Plan what should include the following:

1. Proposed method of procedure
2. The list of equipment to be tested
3. Test forms with proposed test voltages and current set points
4. Test equipment to be used with current calibration sheets,
5. Proposed source of temp power including temp power to test equipment and control power to equipment to be tested.
6. Provide acceptance criteria for each testing procedure.

C. Test Reports:

1. Maintain written record of tests.
2. At completion of project, assemble and certify a final test report. Document testing and performance compliance with NETA-ATS recommended forms, parameters, and level of detail. Submit report to Construction Manager prior to final acceptance to include:
   - Include SLAC asset inventory number. Confirm with SLAC for numbering.

- Summary of project.
- Description of equipment tested.
- Visual inspection report.
- Description of tests.
- Test results.
- Conclusions and recommendations.
- As-found and As-left relay and meter settings files

1.06 QUALITY ASSURANCE

A. Qualifications of Testing Firm:

1. Corporately independent testing organization, which can function as an unbiased testing authority, professionally independent of the manufacturers, suppliers and installers of equipment or systems.
2. Independent organization as defined by a NETA Level II ETT certified testing agency.
3. Regularly engaged in the testing of electrical materials, devices, appliances, electrical installations and systems for the purpose of preventing injury to persons, malfunctioning of equipment, and damage to property.
4. Personnel engaged in testing practices for minimum of 5 years.
5. Use only qualified technicians, regularly employed by firm for testing services. Electrically unskilled employees are not permitted to perform testing or assistance of any kind. Electricians and line workers may assist, but may not perform testing or inspection services.
6. Submit proof of above qualifications with Bid Documents.
7. Personnel shall be trained in use proposed test equipment and have experience of this equipment operation minimum of 12 months. Provide proof of experience.
8. Testing Firm personnel shall be OSHA lock out and tag out trained

B. Certifications:

1. Comply with OSHA criteria for accreditation of testing laboratories, Title 29, Parts 1907, 1910 and 1936. Full membership in the NETA constitutes proof of such criteria.
2. The Lead, on site, Technician must be currently certified by NETA in Electrical Power Distribution System Testing.

3. Instruments used by testing firm to evaluate electrical performance must meet the NETA Specifications for Test Instruments.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 FIELD QUALITY CONTROL

A. Subcontractor's Responsibilities:

1. Verify/Confirm that all testing equipment and/or tools used for testing, are within the required manufacture’s calibration date

2. Perform routine insulation resistance, continuity and rotation tests for distribution and utilization equipment prior to and in addition to tests performed by the testing firm.

3. Notify the testing firm when equipment becomes available for acceptance testing. Coordinate acceptance testing with the Field Construction Manager. Alert all entities possibly impacted by the testing. Plan the testing execution such that the project schedule is unaffected.

3. Provide all field inspections and tests required by latest NETA-ATS.

4. Prior to testing, equipment shall be tested shall be cleared from dust and debris, wiped clean, and dried out.

B. Testing Firm's Responsibilities:

1. Notify Construction Manager prior to commencement of any testing.

2. Notify Contract Administrator about scheduled field testing not less than 3 days prior to it.

3. Report directly to Construction Manager any systems, material or installation found defective on the basis of acceptance tests.

4. Provide auxiliary portable power supply necessary for conducting tests, including temporary power to test equipment, tools, and control power for electrical equipment to be tested.

3.02 ADJUSTING

A. Final Settings: Testing firm responsible for implementing final settings and adjustments on protective devices and tap changes in accordance with power study.

END OF SECTION 26 08 05

SECTION 26 24 00 - PANELBOARDS

PART 1 - GENERAL

1.01 SUMMARY

Section includes Panelboards.

1.02 SYSTEM DESCRIPTION

Electrical Service System:

1. 480 volts, 3-phase, delta 3-wire,
2. 480/277 volts, 3-phase, wye, 4 wire,
3. 208/120 volts, 3-phase, wye, 4 wire,
4. 125 volts DC – for substation control power distribution, 2-pole, 2-wire.
5. or as specified on drawings.

1.03 DEFINITIONS

1. SVR: Suppressed voltage rating.
2. SPD: Surge Protective Device.

1.04 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Panelboards shall withstand the effects of earthquake motions determined according to SEI/ASCE 7.

   1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."

1.05 REFERENCE STANDARDS

- NEMA PB 1 – Panelboards.
- NEMA PB 1.1 – Instructions for Safe Installation, Operation and Maintenance of Panelboards Rated 600 Volts or Less.
- NEMA AB 1 – Molded Case Circuit Breakers.
- UL 50 – Enclosures for Electrical Equipment.
- UL 67 – Panelboards.
- UL 489 – Molded Case Circuit Breakers and Circuit Breaker Enclosures.
- UL 1449 – Standard for Surge Protective Device

1.06 SUBMITTALS

A. Product Data: For each type of panelboard, switching and overcurrent protective device, transient voltage suppression device, metering, networking, accessory, and component indicated. Include dimensions and manufacturers' technical data on features, performance, electrical characteristics, ratings, and finishes.

B. Shop Drawings: For each panelboard and related equipment.

1. Include dimensioned plans, elevations, sections, and details. Show tabulations of installed devices, equipment features, and ratings.

2. Detail enclosure types and details for types other than NEMA 250, Type 1.

a. Detail bus configuration, current, and voltage ratings.

b. Short-circuit current rating of panelboards and overcurrent protective devices.

c. Detail features, characteristics, ratings, and factory settings of individual overcurrent protective devices and auxiliary components.

d. Breaker type and type of trip unit.

3. Include wiring diagrams for power, signal, and control wiring.

4. Include time-current coordination curves for each type and rating of overcurrent protective device included in panelboards. Submit electronically and on translucent log-log graft paper; include selectable ranges for each type of overcurrent protective device.

C. Qualification Data: For qualified testing agency.

D. Seismic Qualification Certificates: Submit certification that panelboards, overcurrent protective devices, accessories, and components will withstand seismic forces defined in Division 26 SECTION "Seismic Controls for Electrical Systems." Include the following:

1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.

2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.

3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.

E. Field Quality-Control Reports:

1. Test procedures used.

2. Test results that comply with requirements.

3. Results of failed tests and corrective action taken to achieve test results that comply with requirements.

F. Panelboard Schedules: For installation in panelboards. Submit final versions.

G. Operation and Maintenance Data: For panelboards and components to include in emergency, operation, and maintenance manuals. Include the following:

1. Manufacturer's written instructions for testing and adjusting overcurrent protective devices.

2. Time-current curves, including selectable ranges for each type of overcurrent protective device that allows adjustments.

1.07 QUALITY ASSURANCE

A. Testing Agency Qualifications: Refer to specification SECTION 26 08 05 "Electrical Acceptance Testing".

B. Source Limitations: Obtain panelboards, overcurrent protective devices, components, and accessories from single source from single manufacturer.

C. Product Selection for Restricted Space: Drawings indicate maximum dimensions for panelboards including clearances between panelboards and adjacent surfaces and other items. Comply with indicated maximum dimensions.

D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

E. Comply with NEMA PB 1.

F. Comply with NFPA 70.

G. Comply with NECA 407, "Standard for Installing and Maintaining Panelboards."

1.08 DELIVERY, STORAGE, AND HANDLING

A. Remove loose packing and flammable materials from inside panelboards; install temporary electric heating, if needed, (250 W per panelboard) to prevent condensation.

B. Handle and prepare panelboards for installation according to NEMA PB 1.

C. Store in dry, clean, moisture free environment. Subcontractor is responsible for storage.

1.09 COORDINATION

A. Coordinate layout and installation of panelboards and components with other construction that penetrates walls or supported by them, including electrical and other types of equipment, raceways, piping, encumbrances to workspace clearance requirements, and adjacent surfaces. Coordinate incoming service entry with field installation. Maintain required workspace clearances and required clearances for equipment access doors and panels.

B. Coordinate sizes and locations of concrete bases with actual equipment provided. Cast anchor-bolt inserts into bases.

PART 2 - PRODUCTS

2.01 GENERAL REQUIREMENTS FOR PANELBOARDS

A. Fabricate and test panelboards according to IEEE 344 to withstand seismic forces defined in Division 26 Section 26 05 48 "Vibration and Seismic Controls for Electrical Systems."

B. Enclosures: Surface-mounted cabinets or recess as indicated on drawings.

1. Rated for environmental conditions at installed location. Indoor Dry and Clean Locations: NEMA 250, Type 1.
2. Panelboards located outdoors shall be rated NEMA 3R or as indicated on drawings.
3. Panelboards located in corrosive environment shall be rated NEMA 4X or as indicated on drawings.
4. Front: Secured to box with concealed trim clamps. For surface-mounted fronts, match box dimensions; for flush-mounted fronts, overlap box.
5. Hinged Front Cover: Entire front trim hinged to box and with standard door in door trim cover.
6. Gutter Extension and Barrier: Same gauge and finish as panelboard enclosure; integral with enclosure body. Arrange to isolate individual panel sections.
7. Finishes: Panels and Trim:
    a. Galvanized steel, factory finished immediately after cleaning and pretreating with manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat.
    b. Back Boxes: Galvanized steel.
8. Directory Card: Inside panelboard door, mounted in transparent card holder .

C. Incoming Mains Location: Top or bottom as indicated on the design drawings.

D. Phase, Neutral, and Ground Buses:

1. Material: Hard-drawn copper, 98 percent conductivity.
2. Equipment Ground Bus: Adequate for feeder and branch-circuit equipment grounding conductors; bonded to box, not isolated.
3. Bus bars shall be copper and ~~figured~~ rated at ~~on the basis of~~ 1000 amperes per square inch of cross-sectional area for copper, and 500 amperes per square inch for lugs and connections.

4. Isolated ground bus: Only if indicated on design drawings.

E. Conductor Connectors: Suitable for use with conductor material and sizes.

1. Material: Same as phase buses.
2. Main and Neutral Lugs: Mechanical type.
3. Ground Lugs and Bus-Configured Terminators: Mechanical type.

F. Future Devices: Mounting brackets, bus connections, filler plates, and necessary appurtenances required for future installation of devices.

G. Panelboard Short-Circuit Current Rating: Fully rated to interrupt symmetrical short-circuit current available at terminals per the Overcurrent Protective Device Coordination Study as specified, unless otherwise indicated.

H. Panelboard Short-Circuit Current Rating: Fully rated with a minimum integrated rating of 22,000 amperes RMS symmetrical at 240 volts for 240V maximum panelboards, 65,000 amperes RMS symmetrical at 277 volts for 480/277V panelboards, unless otherwise indicated

I. Built-in surge protective device (SPD) – UL 1449 Type 2 connected to the panelboard buses.

2.02 LIGHTING AND APPLIANCE BRANCH-CIRCUIT PANELBOARDS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Eaton Electrical,
2. ABB,
3. Siemens,
4. IEM,
5. SLAC approved.

B. Panelboards: NEMA PB 1, lighting and appliance branch-circuit type.

C. Mains: As indicated in design drawings.

D. Branch Overcurrent Protective Devices: Bolt-on thermal magnetic circuit breakers, replaceable without disturbing adjacent units.

1. Multiple pole units enclosed in a single housing or approved handle-ties to operate as a single unit.

E. Panelboards shall be provided with Modbus type, TCP/IP Ethernet, enabled integrated power (volts, amps, kw, kwh) meters.

F. Doors: Concealed hinges; secured with flush latch with tumbler lock; keyed alike.

2.03 DISCONNECTING AND OVERCURRENT PROTECTIVE DEVICES

A. Manufacturers: Same as panelboard manufacturer or approved.

B. Molded-Case Circuit Breaker (MCCB): Comply with UL 489, with interrupting capacity to meet available fault currents.

1. Thermal-Magnetic Circuit Breakers: (125 amps and less) Inverse time-current element for low-level overloads, and instantaneous magnetic trip element for short circuits.

2. For main feeder and branch circuit breakers larger than 125 amps, refer to specification SECTION 26 28 00.

3. GFCI Circuit Breakers: Single- and two-pole configurations with Class A ground-fault protection (6-mA trip).

4. Ground-Fault Equipment Protection (GFEP) Circuit Breakers: Class B ground-fault protection (30-mA trip).

5. Molded-Case Circuit-Breaker (MCCB) Features and Accessories:

a. Standard frame sizes, trip ratings, and number of poles.

b. Lugs: Mechanical style, suitable for number, size, trip ratings, and conductor materials.

c. Multipole units enclosed in a single housing or factory assembled to operate as a single unit.

d. Handle Padlocking Device: Fixed attachment, for locking circuit breaker handle in on or off position.

e. Handle Clamp: Loose attachment, for holding circuit-breaker handle in on position.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Receive, inspect, handle, and store panelboards according to NEMA PB 1.1.

B. Examine panelboards before installation. Reject panelboards that are damaged or rusted or have been subjected to water saturation.

C. Examine elements and surfaces to receive panelboards for compliance with installation tolerances and other conditions affecting performance of the Work.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 INSTALLATION

A. Install panelboards as directed by manufacturer's installation instructions.

B. Mount panelboard where the top most circuit breaker handle shall not be higher than 6-foot 7-inches from the floor (NEC 240.24.A).

C. Install panelboards surface or flush mounted in accessible locations as indicated on drawings. Maintain or exceed minimum clearances required by code.

D. Where flush panels are installed, verify available recessing depth and coordinate wall framing with other divisions.

E. Feeder conductors to enter directly in line with lug terminals wherever practicable. Feeder conductors, except ground and neutral, not to exceed 45 degree deflection from raceway entry to feeder phase lugs.

F. Where panels are installed flush, provide one spare conduit of each size from panel to accessible space above.

G. Where panels are installed flush in fire rated walls maintain fire rating of wall.

H. Surface panels to have metal trim covers with no sharp edges or corners. Surface panel enclosure finish to match trim cover.

I. Where two or more panels are installed side-by-side, provide covers of same size with each trim independently removable without disturbing the other sections.

J. Install overcurrent protective devices and controllers not already factory installed.

J. Provide communication conduit and wiring for lighting panelboards with integrated energy meters.

K. Markings on breaker panel dead-fronts should indicate the circuit number for each pole circuit.

K. Install listed filler plates in unused spaces.

3.03 IDENTIFICATION

A. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs complying with SECTION 26 05 53.

B. Create a directory to indicate installed circuit loads; incorporate Owner's final room designations. Obtain approval before installing. Use a computer or typewriter to create directory; handwritten directories are not acceptable.

C. Panelboard Nameplates: Label each panelboard with a nameplate complying with requirements for identification specified in SECTION 26 05 53.

1. Device Nameplates: Label each branch circuit device in distribution panelboards with a nameplate com Melamine plastic laminate, white with black core, 1/8-inch thick.
2. UV and weatherproof rated.

3. Engravers standard letter style, minimum 3/8-inch high capital letters.
4. Drill or punch labels for mechanical fastening. Use self-tapping stainless steel screws. Use proper size screw not to penetrate more than 1/8 exposure.

3.04 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

B. Perform tests and inspections.

C. Acceptance Testing Preparation:

1. Test insulation resistance for each panelboard bus, component, connecting supply, feeder, and control circuit.
2. Test continuity of each circuit.

D. Tests and Inspections:

1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
2. Verify neutral wiring does not have cross connections and neutral conductors are not being shared in non-multiwire circuits (absolutely no shared neutrals in 120VAC circuits)
3. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
4. All open penetrations larger than 3/8” shall be sealed with KO plugs or other approved means
5. All raceways penetrating (connecting) to the Panelboard shall be sealed with Duct Seal or other approved means

E. Panelboards will be considered defective if they do not pass tests and inspections.

F. Prepare test and inspection reports, including a certified report that identifies each panelboard(s) and describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

3.05 ADJUSTING

A. Adjust moving parts and operable component to function smoothly and lubricate as recommended by manufacturer.

B. Set field-adjustable circuit-breaker trip ranges as indicated by provided coordination study.

3.06 PROTECTION

A. Temporary Heating: Apply temporary heat as needed to maintain temperature according to manufacturer's written instructions.

1.07 CLEANING

A. Thoroughly clean the exterior and the interior of each panelboard in accordance with manufacturer's installation instructions.

B. Vacuum construction dust, dirt, and debris out of each panelboard.

C. Where enclosure finish is damaged, touch up finish with matching paint in accordance with manufacturer's specifications and installation instructions.

END OF SECTION 26 24 00

SECTION 26 27 26 – WIRING DEVICES

PART 1 - GENERAL

1.01 SUMMARY

A. Furnish labor, materials, tools, equipment, and services for Wiring Devices, as indicated, in accordance with provisions of Contract Documents.

B. Completely coordinate with work of other trades.

1.02 QUALITY ASSURANCE

A. Provide wiring devices conforming to the following standards

1. Underwriter's Laboratories (UL).

   a. UL 20 - General-Use Snap Switches

   b. UL 498 – Standard for Attachment Plugs and Receptacles.

   c. UL 514D – Cover Plates for Flush-Mounted Wiring Devices.

   d. UL 943 – Standard for Safety for Ground-Fault Circuit-Interrupters

2. National Electric Manufacturers Association (NEMA).

   a. WD-1 – General Color Requirements for Wiring Devices.

   b. WD-6 – Wiring Devices – Dimensional Requirements.

3. US Federal Specifications.

   a. Fed Spec receptacles (WC-596F).

   b. Fed Spec device plates (W-P-455).

1.03 SUBMITTALS

A. Product Data: Technical data on each type of device.

PART 2 - PRODUCTS

2.01 APPROVED MANUFACTURERS

A. Wiring Devices:

1. Hubbell

2. Cooper (Eaton)

3. Meltric (Marechal Electric)

4. Leviton Manufacturing

5. Pass & Seymour (Legrand)

6. WattStopper (Legrand) for lighting sensor switches.

B. All wiring devices, (switches, receptacles, etc) shall be provided by the same manufacturer.

2.02 MATERIALS

A. Application of Duplex and Single Receptacles

1. Flush, grounding convenience outlets for side wiring, or side and back wiring.

2. Listed per UL 498 for general use and certified by UL to Fed Spec WC-596F, and shall be visibly marked with the "UL-FS" mark to confirm certification.

3. Constructed with impact resistant nylon or polyester face and body.

4. Per NEC article 406.8, receptacles that are installed in damp or wet locations need to be 'Weather-Resistant' (WP)'

5. Receptacles installed in the building offices and public use areas shall be of commercial grade. Receptacles installed in shop areas and laboratories shall be industrial grade.

6. Receptacles installed in construction areas, Klystron Gallery, and Accelerator Housing shall be industrial grade – heavy duty.

7. Receptacles installed in recreational areas accessible by general public shall be temper resistant.

B. Recommended Receptacles:

1. Specification grade for general use.

a. 20A, 125V, 2 pole, 3-wire self-grounding, duplex, specification grade, NEMA 5-20R; Hubbell HBL5362.

b. 20A, 125V, 2 pole, 3-wire self-grounding, single, specification grade, NEMA 5-20R; Hubbell HBL5361.

2. Ground Fault Circuit Interrupter (GFCI) type receptacle:

a. With built-in ground fault interruption, 4 to 6 mA trip sensitivity, 0.025 second trip time, 10,000 A maximum interrupting capacity, self-testing, indicator and reset. 20A, 125V, 3-wire duplex, specification grade, NEMA 5-20R; Hubbell GFRST20.

3. Tamper resistant (safety-type) receptacles.

   a. NEC Article 406.12 compliant. 20A, 125V, hospital grade, 2 pole, 3-wire, grounding, duplex, safety type: NEMA 5-20R; Hubbell HBLSLG-63H.

4. USB charger duplex receptacles:

   a. Two USB Type 2.0 ports 3 amp, 5 volts DC. Complies with battery charging specification USB BC1.2. Compatible with USB 1.1/2.0/3.0 devices, including Apple products. Complies with Part 15 of the FCC rules. 20A, 125V, 3-wire duplex, hospital grade, NEMA 5-20R; Hubbell USB8300.

5. Special Purpose Receptacles:

   a. Straight Blade:

      1) NEMA 6-20R receptacle: 20A, 250V, 2 pole, 3 wire grounding, side and back wired, single; Hubbell HBL5461.

      2) NEMA 14-20R receptacle: 20A, 125/250V, 3 pole, 4 wire, grounding, single; Hubbell HBL8410.

      3) NEMA 15-20R receptacle: 20A, 250V, 3 pole, 4 wire, grounding, single, Hubbell HBL8420.

      4) NEMA 5-30R receptacle: 30A, 125V, 2 pole, 3 wire grounding, single, Hubbell HBL9308.

      5) NEMA 6-30R receptacle: 30A, 250V, 2 pole, 3 wire grounding, single, Hubbell HBL9330.

      6) NEMA 10-30R receptacle: 30A, 125/250V, 3 pole, 3 wire, single, Hubbell HBL9350.

      7) NEMA 14-30R receptacle: 30A, 125/250V, 3 pole, 4 wire, grounding, single, Hubbell HBL9430A.

      8) NEMA 15-30R receptacle: 30A, 250V, 3 pole, 4 wire, grounding, single, Hubbell HBL8430A.

      9) NEMA 6-50R receptacle: 50A, 250V, 2 pole, 3 wire, grounding, single, Hubbell HBL9367.

      10) NEMA 10-50R receptacle: 50A, 125/250V, 3 pole, 3 wire, Hubbell HBL7962.

      11) NEMA 14-50R receptacle: 50A, 125/250V, 3 pole, 4 wire, grounding, single, Hubbell HBL9450A.

      12) NEMA 15-50R receptacle: 50A, 250V, 3 pole, 4 wire, grounding, single, Hubbell HBL8450A.

   b. Electric cart cord reel charging station Retractable cord reel, KH Industries, RTF Series, Model #: RTFQ3L-WW-B12O or approved equal.

c. Twist-Lock:

1) NEMA L5-20R receptacle: 20A, 125V, 2 pole, 3 wire, grounding, single; Hubbell HBL2310.

2) NEMA L6-20R receptacle: 20A, 250V, 2 pole, 3 wire, grounding, single, Hubbell HBL2320.

3) NEMA L14-20R receptacle: 20A, 125/250V, 3 pole, 4 wire, grounding, single, Hubbell HBL2410.

4) NEMA L15-20R receptacle: 20A, 250V, 3 pole, 4 wire, grounding, single, Hubbell HBL2420.

5) NEMA L5-30R receptacle: 30A, 125V, 2 pole, 3 wire, grounding, single, Hubbell HBL2610.

6) NEMA L6-30R receptacle: 30A, 250V, 2 pole, 3 wire, grounding, single, Hubbell 2620.

7) NEMA L14-30R receptacle: 30A, 125/250V, 3 pole, 4 wire, grounding, Hubbell HBL2710.

8) NEMA L21-30R receptacle: 30A, 250V, 3 pole, 4 wire, grounding, single, Hubbell HBL2810.

9) Hubbell CS6369 Locking Receptacle, 50A, 125/250V, 3 Pole, 4 Wire, grounding.

10) Hubbell CS8269 Locking Receptacle, 50A, 250V, 2 Pole, 3 Wire, grounding.

11) Hubbell CS8369 Locking Receptacle, 50A, 250V, 3 Pole, 4 Wire, grounding.

12) Hubbell CS8469 AC Receptacle CA STD 50A, 480V, 2 pole, 3 wire, grounding.

13) Hubbell CS8169 AC Receptacle CA STD 50A, 480V, 3 pole, 4 wire, grounding.

d. Switch Rated:

1) MELTRIC DS20: configurable, 20A, 600V, 3 pole, 4 wire, grounding, NEMA 3R

2) MELTRIC DSN20: configurable, 20A, 600V, 3 pole, 4 wire, grounding, NEMA 4X for corrosive environment.

3) MELTRIC DS30: configurable, 30A, 600V, 3 pole, 4 wire, grounding, NEMA 3R.

4) MELTRIC DSN30: configurable, 30A, 600V, 3 pole, 4 wire, grounding, NEMA 4X for corrosive environment.

5) MELTRIC DS60: configurable, 60A, 600V, 3 pole, 4 wire, grounding, NEMA 3R.

6) MELTRIC DSN60: configurable, 60A, 600V, 3 pole, 4 wire, grounding, NEMA 4X for corrosive environment.

C. Light Switches:

1. Flush, grounding light switches for side wiring, or side and back wiring.

2. Listed per UL 20 for general use and visibly marked with the "UL-FS" mark to confirm certification.

3. Constructed with impact resistant nylon or polyester face and body.

4. Per NEC article 404.4, switches that are installed in damp or wet locations need to be enclosed in weather-proof enclosure or cabinet that complies to NEC 312.2.

5. Light switches installed in the building offices and public use areas shall be of commercial grade:

    a. Wall Switch Box Occupancy Sensor, 1050 Sq. Ft Passive Infrared, Ultrasonic, Legrand – WattStopper DW-200 or approved equal.

    b. 20-Amp, Leviton Decora Plus or approved equal rocker single-pole AC quiet switch, 120/277-volt, commercial grade, back and side wired, with grounding.

6. Light switches installed in shop areas, laboratories, and construction zones shall be 20-Amp, toggle single-pole ac quiet switch, 120/277-volt, extra heavy duty spec grade, with self-grounding.

7. For switches operated by commercial or industrial lighting control systems refer to Section 26 09 43 “Digital Lighting Control System”.

D. Finish:

1. Device color and faceplate color shall match architectural drawings and specs.

E. Device Plates:

1. Device plates for concealed wiring: Same manufacturer as devices to suit device covered; single, or ganged, in one piece with beveled edges that match faces of plates.

2. Flush, brushed-finish, type 304 stainless steel, where indicated.

F. Labeling:

1. General:

    a. Labeling of device plates is required. Provide clear adhesive labels. Labels for outdoor installation shall be UV rated with not less than ½” bold font.

    b. Label inside of all device plates or box with panelboard and circuit number supplying the devices.

    c. Provide label for each receptacle and light switch. Indicate circuit number and panel name.

    d. Provide additional label for cord drop receptacles at end device.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Locate devices as indicated as indicated on design drawings and as scheduled in Section 26 00 10 “General Electrical”.

B. Receptacles in general use areas provide as indicated below:

1. GFCI: Exterior, basement, tunnels, damp locations, bathrooms, showers, six feet within sink, fountains, Bathrooms, Garages and Accessory Buildings, Outdoors, Crawl Spaces, Basements, Kitchens, Lounges.

2. Stainless Steel Plates: Shop areas, Garages, Chemical storage handling areas, Cleanrooms, cold room, wet labs, accelerator housing areas, commercial kitchens, and other damp locations or corrosive environment.

3. In Use Cover: Exterior locations

4. Tamper Resistance: Lobbies

5. Twistlock: Data center server and networking rows, high bay lights, ceiling mount receptacles.

C. Center outlets and light switches with regard to paneling, furring, trim, tile, etc.

D. Install minimum of 4 (four) square box for wiring devices. Refer to 260534 ”Boxes for Electrical Systems”.

E. Use back boxes for wiring devices meeting NEC 314.16, box fill.

F. Use FS type box for surface mount devices, refer to 260534 “Boxes for Electrical Systems”.

G. Refer to architectural design drawings for mounting heights of wiring devices.

H. Where several outlets or light switches occur in a room, symmetrically arrange them.

I. Any light switch of outlet which is improperly located must be corrected at Subcontractor's expense.

J. Set wiring devices plumb or horizontal and extending to finished surface of wall, ceiling or floor as case may be without projecting beyond same.

K. Install light switches and receptacles indicated on wood trim, cases or other fixtures symmetrically. Where necessary, set with long dimension of plate horizontal, or gang in tandem.

L. Any receptacle installed within 6 feet in any direction of the edge of a sink shall be GFCI protected by a local GFCI receptacle (not by means of a GFCI breaker).

M. All 15 and 20 amp, 125 and 250 volt non-locking receptacles exposed to damp or wet locations shall be listed as Weather-Resistant type and shall be GFCI protected

N. Receptacles supplying power to a drinking water fountain shall be GFCI protected

O. All protected downstream devices shall be labeled as protected by upstream GFCI receptacle.

END OF SECTION 26 27 26

SECTION 26 28 00 – LOW VOLTAGE OVERCURRENT PROTECTIVE DEVICES

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:

1. Circuit breakers.
2. Fuses

1.02 SUBMITTALS

A. Product Data:

1. Technical data on each type of device including:
   a. Outline drawings with dimensions.
   b. Ratings for voltage, amperage and maximum interrupting ratings.
   c. Trip unit functions and adjustments (adjustability, range, ground, Long time/short time/instantaneous)
   d. Accessories (key interlocks, limit switch, PPS interlock, shunt trips, undervoltage, aux contacts, etc)
   e. Wiring diagrams.
   f. Soft copy of time/current characteristic trip curves (and Ip & I^2t let through curves for current limiting circuit breakers) for each type of circuit breaker.
2. Submit with associated switchboard or other assembly.
3. Subcontractor shall provide at earliest phase of project for SLAC short circuit/coordination/ and arc flash study. Submittal is required for arc flash labels and energization of electrical equipment.
4. Coordination with switchgear, switchboard, and other associated assembly specification sections

B. Contract Closeout Information:

1. Operating and maintenance data.
   a. Include instructions for circuit breaker mounting, trip unit functions and adjustments, trouble shooting, accessories and wiring diagrams.
   b. Include required maintenance.

1.03 QUALITY ASSURANCE

A. System standards:

1. NEMA AB 1 - (National Electrical Manufacturers Association) Molded Case Circuit Breakers and Molded Case Switches

2. UL 489 - (Underwriters Laboratories Inc.) Molded Case Circuit Breakers and Circuit Breaker Enclosures

3. UL 943 - Standard for Ground Fault Circuit Interrupters

4. CSA C22.2 No. 5.1 - M91 - (Canadian Standard Association) Molded Case Circuit Breakers

5. Federal Specification W-C-375B/GEN - Circuit Breakers, Molded Case; Branch Circuit and Service

6. All power circuit breakers shall be constructed and tested in accordance with ANSI C37.13, C37.16, C37.17, C37.50, UL 1066 and NEMA SG-3 standard.

7. IEEE 141(Red Book) – Recommended Practice for Electric Power Distribution for Industrial Plants.

8. IEEE 399 (Brown Book) – Recommended Practice for Industrial and Commercial Power Systems Analysis.

9. NFPA 70E – Standard for Electrical Safety Requirements for Employee Workplaces.

10. IEEE 519 – Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems

PART 2 - PRODUCTS

2.01 APPROVED MANUFACTURERS

A. Overcurrent protective devices (up to 1000V).

1. Eaton Electrical

2. ABB

3. Siemens

4. SLAC Approved

B. Fuses (up to 1000V):

1. Bussmann.

2. Mersen.

3. Brush.

4. Littlefuse.

5. S&C.

6. SLAC Approved.

2.02 CIRCUIT BREAKERS

A. Power Circuit Breakers:

1. Provide circuit breakers as required by other specifications and drawings. Provide special features as indicated including but not limited to:

   a. Drawout construction.
   b. Electrical operation.
   c. Key interlock for main-tie-main or main-tie-tie-main arrangements for SLAC approved open transition scheme. Docking station or temporary power source manual transfer switch scheme.
   d. Ground fault protection.
   e. LSIG adjustable digital trip units.
   f. Modbus remote status/alarms for SCADA integration.
   g. Aux contacts for open/close breaker status.

2. Provide lugs rated for 75 degree C wire minimum.

3. Subcontractor shall review one-line diagrams and confirm that circuit breakers have adequate lugs to accommodate size and quantity of conductors indicated on one line diagrams, panel and motor control schedules.

4. Lugs shall be UL Listed to accept stranded copper conductors.

5. Circuit breakers shall be capable of accepting bus connections.

6. Overcurrent devices shall be fully rated for available fault current unless otherwise specifically indicated.

7. Power circuit breakers shall be provided with arc reducing maintenance switch (ARMS) with local status indicator, which will allow a worker to set the circuit breaker trip unit to “no intentional delay” to reduce the clearing time while the worker is working within an arc-flash boundary, and then to set the trip unit back to a normal setting after the potentially hazardous work is complete.

B. Molded Case Circuit Breakers (MCCB)

1. Provide circuit breakers as required by other specifications and drawings. Provide special features as indicated including but not limited to:

a. Constructed of glass reinforced insulating material. Current carrying components shall be completely isolated from handle and accessory mounting area.

b. Provide over center, trip free, toggle operating mechanism, which shall provide quick-make, quick-break contact action. Provide common tripping of two and three pole circuit breakers.

c. Circuit breaker handle shall reside in a tripped position between ON and OFF to provide local trip indication. Circuit breaker escutcheon shall be clearly marked ON and OFF in addition to providing International I/O markings.

d. Maximum ampere rating and UL, IEC, or other certification standards with applicable voltage systems and corresponding interrupting ratings shall be clearly marked on face of circuit breaker.

e. Provide each circuit breaker with push-to-trip button, located on face of circuit breaker to mechanically operate circuit breaker tripping mechanism for maintenance and testing purposes.

f. Provide factory seal with date code on face of circuit breaker.

g. Provide circuit breakers equipped with UL Listed electrical accessories as noted on associated schedule or drawing.

h. Provide circuit breaker handle accessories with provisions for locking handle in ON and OFF position as noted on associated schedule or drawing.

i. Terminal lugs shall be rated not less than 75 degrees C.

j. Terminal lugs shall be rated for and capable of terminating number and size of cables indicated on drawings.

k. Provide circuit breakers UL Listed for reverse connection without restrictive line and load markings and suitable for mounting in any position.

l. Provide circuit breakers UL Listed to accept field installable/removable mechanical type or compression type lugs. Provide lug body bolted in place; snap in design not acceptable.

C. Thermal-Magnetic Circuit Breakers

1. Shall be used only as follows unless otherwise indicated:

a. Main, feeder and branch circuit breakers rated 125 amps and less in lighting and appliance panelboards as defined in Section 26 24 00.

b. Main, feeder and branch circuit breakers rated 125 amps and less in distribution panel boards as defined in Section 26 24 00.

c. Motor circuit protectors.

d. Terminal lugs shall be rated not less than 75 degrees C.

e. Terminal lugs shall be rated for and capable of terminating number and size of cables indicated on drawings.

2. Do not use in switchboards rated over 1200 amps.

3. Provide standard function trip system on circuit breakers rated less than 400 amps unless otherwise indicated.

4. Provide permanent trip unit containing individual thermal and magnetic trip elements in each pole.

5. Thermal trip elements shall be factory preset and sealed. Circuit breakers shall be true rms sensing and thermally responsive to protect circuit conductor(s) in a 40° C ambient temperature.

6. Provide circuit breaker frame sizes above 150 amperes with magnetic trip adjustment located on front of circuit breaker.

7. Provide UL Listed HACR type for two- and three-pole circuit breakers rated up to 250 amperes at 600 VAC.

8. Provide Class A (5 ma) sensitivity breaker where GFCI circuit breakers are indicated.

9. Provide equipment ground fault protection where indicated with following provisions:

a. Modified zero sequence sensing system.

b. Ground fault sensing system:

1) Requiring no external power to trip circuit breaker.

2) Suitable for use on grounded systems and suitable for use on three-phase, three-wire circuits where system neutral is grounded but not carried through system or on three-phase, four-wire systems.

3) Include ground fault memory circuit to sum time increments of intermittent arcing ground faults above pickup point.

4) Shall not affect interrupting rating of companion circuit breaker.

c. Companion circuit breaker equipped with ground-fault shunt trip and capable of group mounting.

d. Field adjustable ground fault pickup current setting and time delay with switch for setting ground fault pickup point and means to seal pickup and delay adjustments.

e. Means of testing ground fault system to meet on-site testing requirements of NEC.

f. Local visual ground fault trip indication.

D. Electronic trip circuit breakers with standard function trip system

1. Provide standard function trip system on circuit breakers rated less than 400 amps unless otherwise indicated.

2. Provide circuit breaker trip system with microprocessor-based true rms sensing design with sensing accuracy through thirteenth (13th) harmonic and sensor ampere ratings as indicated on associated schedules or drawings.

3. Provide integral trip system independent of any external power source and with industrial grade electronic components.

4. Determine ampere rating of circuit breaker by combination of interchangeable rating plug, sensor size and long-time pickup adjustment on circuit breaker. Clearly mark sensor size, rating plug and adjustment positions on face of circuit breaker.

5. Provide circuit breakers UL listed to carry 80 percent of ampere rating continuously.

6. Provide following time/current response adjustments, each with discrete settings independent from other adjustments:

    a. Instantaneous Pickup.

    b. Long time pickup and delay.

    c. Short time pickup.

    d. Short time delay (I^2t IN only).

    e. Ground fault pickup and delay (I^2t OUT only) where indicated.

7. Provide means to seal trip unit adjustments in accordance with NEC.

8. Provide local visual trip indication for overload, short circuit and ground fault trip occurrences as applicable.

9. Provide opening in dead front cover of each panel or switchboard to read breaker trip setting and breaker rating. Use UV rated transparent cover where required.

10. Provide ammeter to individually display all phase currents flowing through circuit breaker including indication of inherent ground fault current flowing in system on circuit breakers with integral ground fault protection. Display current values in true rms with 2 percent accuracy.

11. Provide Long Time Pickup indication to signal when loading approaches or exceeds adjusted ampere rating of circuit breaker.

12. Provide trip system with Long Time memory circuit to sum time increments of intermittent overcurrent conditions above pickup point and means to reset Long Time memory circuit during primary injection testing.

13. Provide circuit breakers equipped with thermal protection in trip unit to protect breaker from catastrophic failure and instantaneous magnetic override set at the withstand rating of the circuit breaker.

14. Provide trip system equipped with externally accessible test port for use with Universal Test Set. Disassembly of circuit breaker shall not be required for testing. 15. LSIG settings shall be fully adjustable.

15. Provide test set capable of verifying operation of trip functions with or without tripping circuit breaker.

E. Electronic trip circuit breakers with full function trip system:

1. Provide full function trip system on circuit breakers rated 400 amps and greater.

2. Provide circuit breaker trip system with microprocessor-based true rms sensing design with sensing accuracy through thirteenth (13th) harmonic and sensor ampere ratings as indicated on associated schedules or drawings.

3. Provide integral trip system independent of any external power source and with industrial grade electronic components.

4. Determine ampere rating of circuit breaker by combination of interchangeable rating plug, sensor size and long-time pickup adjustment on circuit breaker. Clearly mark sensor size, rating plug and adjustment positions on face of circuit breaker.

5. Provide circuit breakers UL listed to carry 80 percent of ampere rating continuously.

6. Provide following time/current response adjustments, each with discrete settings independent from other adjustments:

    a. Instantaneous pickup.
    b. Long time pickup and delay.
    c. Short time pickup.
    d. Short time delay (I^2t IN and I^2t OUT).
    e. Ground fault pickup and delay (I^2t IN and I^2t OUT) where indicated.
    f. Ground fault alarm only where required by NEC.

7. Provide means to seal rating plug and trip unit adjustments in accordance with NEC.

8. Provide ammeter to individually display all phase currents flowing through circuit breaker including indication of inherent ground fault current flowing in system on circuit breakers with integral ground fault protection. Display current values in true rms with 2 percent accuracy.

9. Provide Long Time Pickup indication to signal when loading approaches or exceeds adjusted ampere rating of circuit breaker.

10. Provide trip system with Long Time memory circuit to sum time increments of intermittent overcurrent conditions above pickup point and means to reset Long Time memory circuit during primary injection testing.

11. Provide circuit breakers equipped with thermal protection in trip unit to protect breaker from catastrophic failure and instantaneous magnetic override set at the withstand rating of the circuit breaker.

12. Provide trip system equipped with externally accessible test port for use with Universal Test Set. Disassembly of circuit breaker shall not be required for testing. Provide test set capable of verifying operation of trip functions with or without tripping circuit breaker.

13. Provide communications capabilities for remote monitoring of circuit breaker trip system, to include phase and ground fault currents, pre-trip alarm indication, switch settings, and trip history information. Required communications protocol(s): Modbus

F. Equipment Ground Fault Protection (Electronic Trip Circuit Breakers)

1. Provide circuit breakers with integral equipment ground fault protection where indicated for grounded systems. Provide circuit breaker suitable for use on three-phase, three-wire circuits where system neutral is grounded but not carried through system or on three-phase, four-wire systems.

2. Provide separate neutral current transformer for three-phase four-wire systems as indicated on schedules or drawings.

3. Provide ground fault sensing system with residual sensing, source ground return or modified differential type.

4. Provide trip system with ground fault memory circuit to sum time increments of intermittent ground faults above pickup point.

5. Provide means of testing ground fault system to meet on-site testing requirements of NEC.

6. Provide local visual trip indication for ground fault trip occurrence(s).

7. Electronic trip unit for power circuit breakers shall be Eaton Digitrip 310, 520M, 1150+ or approved equal with the following features and provisions:

2.03 FUSES

A. UL Class L fuses: Dual-element time-delay and current-limiting type fuses; UL Class L listed for 200,000 rms AIC symmetrical; Bussmann "Low-Peak" 600V, 601-6000A, Type KRP-C.

1. Use for main and main feeder devices over 600A, where fuses are indicated.

B. UL Class RK-1 dual-element fuses: Dual-element time-delay and current-limiting rejection type fuses; UL Class RK-1 listed for 200,000 rms AIC symmetrical; Bussmann "Low-Peak" 0-600A, 250V Type LPNRK and 600V Type LPS-RK.

1. Use for main feeder devices 600A and smaller where fuses are indicated.

C. UL Class RK-1 single-element fuses: Fast-acting current-limiting rejection type fuses; UL Class RK-

1 listed for 200,000 rms AIC symmetrical; Bussmann "Limitron" 1/10-600A, 250V Type KTN-RK and 600V Type KTS-RK.

1. Use as indicated.

D. UL Class RK-5 fuses: Dual-element time-delay and current-limiting rejection type fuses; UL Class RK-5 listed for 200,000 rms AIC; Bussman "Fusetron" 1/10-600A, 250V Type FRN-RK and 600V FRS-RK.

1. Use for motor feeder and branch circuit devices where fuses are indicated.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Provide overcurrent protective devices in switchboards, panelboards and motor control centers as indicated in those sections. . Reconfigure assigned circuits as necessary so that circuit breakers associated with multiwire branch circuits are physically adjacent; record changes in panelboard schedules for as-built drawings in the field immediately and notify SLAC engineering design group (cc FCM). Circuit breaker changes impact arc flash study and energization of equipment.

B. Provide individually enclosed overcurrent protective devices:

1. Wall mounted:

a. Finished areas: Attach to studs via unistrut cross members or metal backing bolted or welded to studs where not otherwise shown.

b. Masonry or concrete walls: Attach to wall via unistrut cross members where not otherwise shown.

c. Mounting height shall be from bottom of enclosure not be less than 12 IN above finished floor. Breaker or switch handle shall not be higher than NEC code requirement.

C. Field settings:

1. Perform field adjustments of protective devices as required to place equipment in final operating condition. Settings shall be in accordance with a SLAC-provided power system study.

2. Provide certified calibration report for each protective device.

D. Arc flash labels:

1. Arc flash hazard warning labels will be furnished by SLAC; install on each piece of electrical equipment.

E. Arc flash boundaries:

1. Identify arc flash protection boundaries in front of all electrical switchboards, switchgear, panel boards, motor control centers, UPS distribution panels, automatic transfer switches and individual disconnects and circuit breakers. Provide outline of arc flash protection boundaries with 2 IN wide strip of red/white Seton M6356 OSHA warning tape or approved equivalent.

F. Perform tests and inspections.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.

G. Acceptance Testing Preparation:

1. Test insulation resistance for each enclosed switch and circuit breaker, component, connecting supply, feeder, and control circuit.

2. Test continuity of each circuit.

Provide primary current injection testing of molded case breakers 225A or larger and adjustable trip breakers with SLAC provided settings.

H. Tests and Inspections:

1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.

2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.

3. Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.

4. Place torque seal compound on all field installed lugs.

5. Required test documentation be reviewed before equipment is energized.

I. Enclosed switches will be considered defective if they do not pass tests and inspections.

J. Prepare test and inspection reports, including a certified report that identifies enclosed switches and circuit breakers and that describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

K. Provide inspection and torque seal/check of termination lugs. Cable terminations shall be torqued per manufacturer recommendation.

END OF SECTION 26 28 00

SECTION 26 28 16 – SWITCHES AND CIRCUIT BREAKERS

PART 1 - GENERAL

1.01 SUMMARY

1. Individually enclosed circuit breakers.
2. Fused and unfused disconnect switches.

1.02 RELATED SECTIONS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

B. Section 26 05 70 – Power System Study

C. Section 26 24 00 – Panelboards

D. Section 26 28 00 – Low Voltage Overcurrent Protective Devices.

1.03 REFERENCES

A. NECA - Standard of Installation.

B. NEMA 250 – Enclosures for Electrical Equipment (1000V maximum).

C. NEMA AB 1 – Molded Case Circuit Breakers.

D. NETA-ATS-2017 – International Electrical Testing Association.

E. NFPA 70 – National Electrical Code NEC

F. NEMA KS 1/UL 98 – Enclosed Switches.

G. ANSI/ASME A17-1.

H. NFPA 72.

1.04 DEFINITIONS

A. NC: Normally closed.

B. NO: Normally open.

C. SPDT: Single pole, double throw.

1.05 PERFORMANCE REQUIREMENTS

A. Perform work in accordance with NECA Standard of Installation.

B. Conform to requirements of NFPA 70.

C. Seismic Performance: Enclosed switches and circuit breakers shall withstand the effects of earthquake motions determined according to ASCE/SEI7.

1. The term "withstand" means "the unit will remain in place without separation or any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."

1.06 SUBMITTALS

A. Comply with provisions of Division 1.

B. Product Data: For each type of enclosed switch, circuit breaker, accessory, and component indicated. Include dimensioned elevations, sections, weights, and manufacturers' technical data on features, performance, electrical characteristics, ratings, accessories, and finishes.

1. Enclosure types and details for types other than NEMA 250, Type 1.

2. Current and voltage ratings.

3. Short-circuit current ratings (interrupting and withstand, as appropriate).

a. Refer to specification Section 26 28 00

4. Detail features, characteristics, ratings, and factory settings of individual overcurrent protective devices, accessories, and auxiliary components.

5. Include time-current coordination curves (average melt) for each type and rating of overcurrent protective device; include selectable ranges for each type of overcurrent protective device.

C. Shop Drawings: For enclosed switches and circuit breakers. Include plans, elevations, sections, details, and attachments to other work.

1. Wiring Diagrams: For power, signal, and control wiring.

D. Seismic Qualification Certificates: For enclosed switches and circuit breakers, accessories and components, from manufacturer:

1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.

2. Dimensioned Outline Drawing of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.

3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.

E. Field quality control forms.

1. Test procedures to be used.

2. Testing criteria.

F. Manufacturer's field service Method of Procedure (MOP).

G. Operation and Maintenance Data: For enclosed switches and circuit breakers to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 1 Section "Operation and Maintenance Data," include the following:

1. Manufacturer's written instructions for testing and adjusting enclosed switches and circuit breakers.

2. Time-current coordination curves (average melt) for each type and rating of overcurrent protective device; include selectable ranges for each type of overcurrent protective device.

1.07 QUALITY ASSURANCE

A. Testing Agency Qualifications: Refer to Specification Section “Electrical Testing”.

B. Source Limitations: Obtain enclosed switches and circuit breakers, overcurrent protective devices, components, and accessories, within same product category, from single source from single manufacturer.

C. Product Selection for Restricted Space: Drawings indicate maximum dimensions for enclosed switches and circuit breakers, including clearances between enclosures, and adjacent surfaces and other items. Comply with indicated maximum dimensions.

D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

1.08 PROJECT CONDITIONS

A. Environmental Limitations: Rate equipment for continuous operation under the following conditions unless otherwise indicated:

1. Ambient Temperature: Not less than minus 22 deg F (minus 30 deg C) and not exceeding 104 deg F (40 deg C).

2. Altitude: Not exceeding 6600 feet (2010 m).

1.09 COORDINATION

A. Coordinate layout and installation of switches, circuit breakers, and components with equipment served and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

PART 2 - PRODUCTS

2.01 FUSIBLE AND NON-FUSIBLE SWITCHES

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Eaton Electrical Inc.

2. ABB

3. Siemens

4. SLAC Approved.

B. Fusible, Type HD, Heavy Duty, Single Throw, 600-VAC, 1200 A and Smaller: UL 98 and NEMA KS 1, horsepower rated, with clips or bolt pads to accommodate specified fuses, lockable handle with capability to accept three padlocks, and interlocked with cover in closed position.

C. Non-fusible, Type HD, Heavy Duty, Single Throw, 600-VAC, 1200 A and Smaller: UL 98 and NEMA KS 1, horsepower rated, lockable handle with capability to accept three padlocks, and interlocked with cover in closed position.

2.02 ACCESSORIES:

1. Equipment Ground Kit: Internally mounted and labeled for copper and aluminum ground conductors.

2. Neutral Kit: Internally mounted; insulated, capable of being grounded and bonded; labeled for copper and aluminum neutral conductors.

3. Class R Fuse Kit: Provides rejection of other fuse types when Class R fuses are specified.

4. Auxiliary Contact Kit: One NO/NC (Form "C") auxiliary contact(s), arranged to activate before switch blades open.

5. Lugs: Mechanical type, suitable for number, size, and conductor material.

6. Provide switches with interlock defeaters for maintenance purposes.

7. Provide clear touch safe protective covers over exposed terminals and/or lugs.

2.03 ENCLOSED MOLDED-CASE CIRCUIT BREAKERS

A. Refer to Section 26 28 00 for breaker requirements.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Eaton Electrical Inc.

2. ABB

3. Siemens

4. SLAC Approved.

C. General Requirements: Comply with UL 489, NEMA AB 1, and NEMA AB 3, with interrupting capacity to comply with available fault currents.

D. Thermal-Magnetic Circuit Breakers (125 amps and less): Inverse time-current element for low-level overloads and instantaneous magnetic trip element for short circuits.

1. For circuit breakers larger than 125 amps, refer to specification Section 26 28 00.

2. Voltage rating shall be 600V class.

E. Features and Accessories;

1. Standard frame sizes, trip ratings, and number of poles.

2. Lugs: Mechanical type, suitable for number, size, trip ratings and conductor material.

2.04 ENCLOSURES

A. Enclosed Switches and Circuit Breakers: NEMA AB 1, NEMA KS 1, NEMA 250, and UL 50, to comply with environmental conditions at installed location.

1. Indoor, Dry and Clean Locations: NEMA 250, Type 1.

2. Outdoor or Wet areas: NEMA 3R

3. Corrosive Locations: NEMA 250, Type 4X, stainless steel.

   a. Typical corrosive environment:

      1) Cooling Tower, Chemical Huts, Chemical Storage Areas, Wash down areas, paint booths, and etc.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Examine elements and surfaces to receive enclosed switches and circuit breakers for compliance with installation tolerances and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 INSTALLATION

A. Install enclosed circuit breakers and disconnect switches where indicated, in accordance with manufacturer's instructions.

B. Install enclosed circuit breakers plumb. Provide supports in accordance with local seismic conditions.

C. Install fuses in fusible disconnect switches.

D. Provide bonding of enclosure, metal raceways, and ground bus/lug/stud.

E. Install individual wall-mounted switches and circuit breakers with tops at uniform height unless otherwise indicated.

F. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.

G. Comply with NECA 1.

H. All switches shall be mounted on separate structures. Do not mount on HVAC housings, duct work, pump frames, etc.

3.03 IDENTIFICATION

A. Comply with requirements in Division 26 Section "Identification for Electrical Systems."

1. Identify field-installed conductors, interconnecting wiring, and components; provide warning

signs.

3.04 ENGRAVED LABELS

A. Melamine plastic laminate, white with black core, 1/8-inch thick.

B. UV and weatherproof rated.

C. Engravers standard letter style. Use minimum of 1-inch high capital letters.

D. Drill or punch labels for mechanical fastening. Use self-tapping stainless steel screws. Use proper size screw not to penetrate more than 1/8 exposure.

3.05 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

B. Inspect and test each circuit breaker to NEMA AB 1 or SG 5, as applicable.

C. Inspect and test each circuit breaker as per latest NETA-ATS.

D. Inspect each circuit breaker visually.

E. Perform several mechanical ON-OFF operations on each circuit breaker.

F. Perform tests and inspections.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.

G. Acceptance Testing Preparation:

1. Test insulation resistance for each enclosed switch and circuit breaker, component, connecting supply, feeder, and control circuit.

2. Test continuity of each circuit.

H. Tests and Inspections:

1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.

2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.

3. Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.

4. Place torque seal compound on all field installed lugs

5. Required test documentation be reviewed before equipment is energized.

I. Enclosed switches will be considered defective if they do not pass tests and inspections.

J. Prepare test and inspection reports, including a certified report that identifies enclosed switches and circuit breakers and that describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

3.06 ADJUSTING

A. Adjust moving parts and operable components to function smoothly and lubricate as recommended by manufacturer.

END OF SECTION 26 28 16

SLAC BUILDING INSPECTION OFFICE
Construction Authorization (IFC)
BY: cnadler
DATE: 2/18/2026 PRS#: 26-009
Revised Authorization ☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.

SECTION 26 28 19 – CIRCUIT AND MOTOR DISCONNECTS

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:

1. Toggle type motor rated switches.
2. Motor starters.
3. Safety switches.
4. Combination motor starter with disconnecting means.

1.02 REGULATORY REQUIREMENTS

A. The following documents form part of the Specifications to the extent stated. Bring conflicts between Specifications, Drawings, and the referenced standards to the attention of the Project Manager, in writing, for resolution before taking any related action.

B. Conform to the latest adopted version of the National Electric Code (NEC).

C. Furnish products listed by UL or other testing firm acceptable to AHJ.

1.03 REFERENCES

A. NECA - Standard of Installation.

B. NEMA 250 – Enclosures for Electrical Equipment (1000V maximum).

C. NETA-ATS- -– International Electrical Testing Association.

D. NFPA 70 – National Electrical Code NEC

E. NEMA KS 1/UL 98 – Enclosed Switches.

F. ANSI/ASME A17-1.

G. NFPA 72.

1.04 SUBMITTALS

A. Comply with provisions of Division 1.

B. Product Data: For each type of disconnect switches and manual motor starters, accessory, and component indicated. Include dimensioned elevations, sections, weights, and manufacturers' technical data on features, performance, electrical characteristics, ratings, accessories, and finishes.

1. Enclosure types and details for types other than NEMA 250, Type 1.

2. Current and voltage ratings.

3. Short-circuit current ratings (interrupting and withstand, as appropriate).

4. Auxiliary contacts, start/stop, hand/off/auto (HOA), pilot light, overloads, and other device list with ratings.

5. Detail features, characteristics, ratings, and factory settings of disconnect switches and manual motor starters and auxiliary components.

C. Shop Drawings: For disconnect switches and manual motor starters. Include plans, elevations, sections, details, and attachments to other work.

1. Wiring Diagrams: For power, signal, and control wiring.

D. Seismic Qualification Certificates: For disconnect switches and manual motor starters, accessories and components, from manufacturer:

1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.

2. Dimensioned Outline Drawing of Equipment Unit.

3. Detailed description of equipment mounting hardware.

E. Field quality control forms.

1. Test procedures to be used.

2. Testing criteria.

F. Manufacturer's field service Method of Procedure (MOP).

G. Operation and Maintenance Data: For disconnect switches and manual motor starters to include in emergency, operation, and maintenance manuals. In addition, include the following:

1. Manufacturer's written instructions for testing and adjusting disconnect switches and manual motor starters.

1.05 QUALITY ASSURANCE

A. Testing Agency Qualifications: Refer to Specification Section “Electrical Testing”.

B. Source Limitations: Obtain disconnect switches and manual motor starters, overcurrent protective devices, components, and accessories, within same product category, from single source from single manufacturer.

C. Product Selection for Restricted Space: Drawings indicate maximum dimensions for disconnect switches and manual motor starters, including clearances between enclosures, and adjacent surfaces and other items. Comply with indicated maximum dimensions.

D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

1.06 PROJECT CONDITIONS

A. Environmental Limitations: Rate equipment for continuous operation under the following conditions unless otherwise indicated:

1. Ambient Temperature: Not less than minus 22 deg F (minus 30 deg C) and not exceeding 104 deg F (40 deg C).

2. Altitude: Not exceeding 6600 feet (2010 m).

1.07 COORDINATION

A. Coordinate layout and installation of disconnect switches and manual motor starters, and components with equipment served and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

B. Size the overload per mechanical/motor nameplate data. Coordinate with instrumentation/controls subcontractor for controls.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Toggle Type Disconnect Switches:

1. Eaton

2. Hubbell

3. Leviton,

4. Legrand - Pass & Seymour

B. Manual Motor Starters:

1. Eaton

2. Siemens

3. ABB

C. Safety Switches:

1. Eaton

2. Siemens

3. ABBABB

2.02 MOTORMOTOR RATED TOGGLE SWITCHES

A. Rating: 120 volt, 1 pole, 20 amp, 1 HP maximum.

B. Enclosure: NEMA 1 indoors, NEMA 3R raintight outdoors, NEMA 4X for corrosive environment. Typical corrosive environment are cooling towers, chemical hutches, paint areas, wash down areas, and areas of corrosive chemical storage.

2.03 MANUAL MOTOR STARTERS

A. Characteristics:

1. Quick-make, quick-break.

2. Thermal overload protection.

3. Clearly label device for maximum voltage, current and horsepower.

B. Enclosure: NEMA 1 indoors, NEMA 3R raintight outdoors, NEMA 4X for corrosive environment. Typical corrosive environment are cooling towers, chemical hutches, paint areas, wash down areas, and areas of corrosive chemical storage.

2.04 COMBINATION STARTERS

A. Provide motor circuit protector with instantaneous trip duty. Reference drawings for additional details.

B. Enclosure: NEMA 1 indoors, NEMA 3R raintight outdoors, NEMA 4X for corrosive environment. Typical corrosive environment are cooling towers, chemical hutches, paint areas, wash down areas, and areas of corrosive chemical storage.

C. Clearly mark switches for maximum voltage, current and horsepower.

D. Provide coil voltage coordinated with control requirements.

E. Provide built in controls transformer with fuse protection where required.

F. Provide thermal overload units sized to equipment nameplate rating.

G. Provide one N.C. and one N.O. auxiliary contacts.

H. Provide LED status lights. Field interchangeable label status

1. On status; Red for motor running.

2. Off status: Green for motor not running.

I. Provide prewired hand/off/auto switch and start/stop button.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Furnish and install disconnect switch in sight of each motor location. Switch shall not be located more than 20 feet of motor. Provide locking device on circuit protective device.

B. Provide disconnect switch in site of each motor controller. Motor controller disconnect equipped with lock-out/tag-out padlock provisions shall not require a disconnect switch at the controlled motor location.

C. Recessed fractional horsepower exhaust ceiling or wall fan units: no disconnect switch required at motor if unit is recessed.

D. Switches shall disconnect all phases simultaneously.

E. Coordinate motor protector rating with motor nameplate data.

3.02 IDENTIFICATION

A. Comply with requirements in Division 26 Section "Identification for Electrical Systems."

1. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs.

3.03 ENGRAVED LABELS

A. Melamine plastic laminate, white with black core, 1/8-inch thick.

B. UV and weatherproof rated.

C. Engravers standard letter style. Use minimum of 3/8-inch high capital letters.

D. Drill or punch labels for mechanical fastening. Use self-tapping stainless steel screws. Use proper size screw not to penetrate more than 1/8 exposure.

END OF SECTION 26 28 19

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	PRS#: 26-009
Revised Authorization	☑ No ☐ Yes
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	

SECTION 26 29 13 – ENCLOSED CONTROLLERS

PART 1 - GENERAL

1.01 SUMMARY

A. Section includes the following enclosed controllers rated 600 V and less:

1. Full-voltage manual.

2. Full-voltage magnetic.

1.02 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.03 DEFINITIONS

A. OCPD: Overcurrent protective device.

1.04 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Enclosed controllers shall withstand the effects of earthquake motions determined according to ASCE/SEI7.

1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event".

1.05 ACTION SUBMITTALS

A. Product Data: For each type of enclosed controller. Include manufacturer's technical data on features, performance, electrical characteristics, ratings, and enclosure types and finishes.

B. Shop Drawings: For each enclosed controller. Include dimensioned plans, elevations, sections, details, and required clearances and service spaces around controller enclosures.

1. Show tabulations of the following:

a. Each installed unit's type and details.

b. Factory-installed devices.

c. Nameplate legends.

d. Short-circuit current rating of integrated unit.

e. Listed and labeled for integrated short-circuit current (withstand) rating of OCPDs in combination controllers by an NRTL acceptable to authorities having jurisdiction.

f. Features, characteristics, ratings, and factory settings of individual OCPDs in combination controllers.

2. Wiring Diagrams: For power, signal, and control wiring.

1.06 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified testing agency.

B. Seismic Qualification Certificates: For enclosed controllers, accessories, and components, from manufacturer:

1. Basis of Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.

2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity, locate and describe mounting and anchorage provisions.

3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.

C. Field quality-control reports.

D. Load-Current and Overload-Relay Heater List: Compile after motors have been installed, and arrange to demonstrate that selection of heaters suits actual motor nameplate full-load currents.

E. Load-Current and List of Settings of Adjustable Overload Relays: Compile after motors have been installed, and arrange to demonstrate that switch settings for motor running overload protection suit actual motors to be protected.

1.07 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For enclosed controllers to include in emergency, operation, and maintenance manuals. In addition to items specified in Section 017823 Operation and Maintenance Data: include the following:

1. Routine maintenance requirements for enclosed controllers and installed components.

2. Manufacturer's written instruction for testing and adjusting circuit breaker and MCP trip settings.

3. Manufacturer's written instruction for setting field-adjustable overload relays.

4. Manufacturer's written instruction for testing, adjusting and reprogramming reduced-voltage solid-state controllers.

1.08 QUALITY ASSURANCE

A. Testing Agency Qualifications: Refer to Specification Section 26 08 05.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

C. Comply with NFPA 70.

1.09 DELIVERY, STORAGE, AND HANDLING

A. Store enclosed controllers indoors in clean, dry space with uniform temperature to prevent condensation. Protect enclosed controllers from exposure to dirt, fumes, water, corrosive substances, and physical damage.

1.10 PROJECT CONDITIONS

A. Environmental Limitations: Rate equipment for continuous operation under prevailing environmental conditions at SLAC site in Menlo Park, CA.

1.11 COORDINATION

A. Coordinate layout and installation of enclosed controllers with other construction including conduit, piping, equipment, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

B. Coordinate sizes and locations of concrete bases with actual equipment provided. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified in Division 3.

C. Coordinate installation of roof curbs, equipment supports, and roof penetrations.

PART 2 - PRODUCTS

2.01 FULL-VOLTAGE CONTROLLERS

A. General Requirements for Full-Voltage Controllers: Comply with NEMA ICS 2, general purpose, Class A.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following, unless otherwise indicated:

1. Eaton Electrical Inc.

2. ABB

3. Siemens

4. SLAC Approved

C. Motor-Starting Switches: “Quick-make, quick-break" toggle or push-button action; marked to show whether unit is OFF or ON.

1. Configuration: Non-reversing.

2. Surface mounting.

3. Pilot light.

4. Provide all starting switches with interlock defeaters for maintenance purposes.

D. Fractional Horsepower Manual Controllers: "Quick-make, quick-break" toggle or push-button action; marked to show whether unit is off, on, or tripped.

1. Configuration: Non-reversing.

2. Overload Relays: Inverse-time-current characteristics; NEMA ICS 2, Class 10 tripping characteristics; heaters matched to nameplate full-load current of actual protected motor; external reset push button; bimetallic type.

3. Surface mounting.

4. Pilot lights – per UL 508.

E. Magnetic Controllers: Full voltage, across the line, electrically held.

1. Configuration: Non-reversing.

2. Contactor Coils: Pressure-encapsulated type with coil transient suppressors.

   a. Operating Voltage: Depending on contactor NEMA size and line-voltage rating, manufacturer's standard matching control power or line voltage.

3. Power Contacts: Totally enclosed, double-break, silver-cadmium oxide; assembled to allow inspection and replacement without disturbing line or load wiring.

4. Control Circuits: 120 VAC; obtained from integral CPT, with primary and secondary fuses of sufficient capacity to operate integral devices and remotely located pilot, indicating, and control devices.

   a. CPT Spare Capacity: 100 VA.

5. Solid-State Overload Relay:

   a. Switch or dial selectable for motor running overload protection.
   b. Sensors in each phase.
   c. Class 10/20 selectable tripping characteristic selected to protect motor against voltage and current unbalance and single phasing.
   d. Class II ground-fault protection, with start and run delays to prevent nuisance trip on starting.

6. Combination starters shall be furnished with the following components:

a. Fused disconnect with interlock defeater for maintenance purposes.

b. 120 VAC control transformer, fused.

c. Red and green pilot lights in cover, red light indicating running condition.

d. Holding coil rated 120 VAC.

e. Hand-Off-Auto selector switch in cover for motors requiring automatic control.

f. Stop-Start buttons in cover for motors requiring push button stations.

g. Overload protection.

h. Two spare auxiliary contacts, in addition to the number of auxiliary contacts needed for each application.

i. Disconnect switch interlock where indicated.

7. Minimum size No. 1.

F. Combination Magnetic Controller: Factory-assembled combination of magnetic controller, OCPD, and disconnecting means.

1. Fusible Disconnecting Means:

a. NEMA KS 1, heavy-duty, horsepower-rated, fusible switch with clips or bolt pads to accommodate Class R fuses.

b. Lockable Handle: Accepts three padlocks and interlocks with cover in closed position.

c. Auxiliary Contacts: N.O./N.C., arranged to activate before switch blades open.

2.02 ENCLOSURES

A. Enclosed Controllers: NEMA ICS 6, to comply with environmental conditions at installed location.

1. Dry and Clean Indoor Locations: NEMA 1.

2. Wet or Damp Outdoor Locations: NEMA 4X, stainless steel.

3. Other Wet or Damp Indoor Locations: NEMA 4.

B. ACCESSORIES

C. General Requirements for Control Circuit and Pilot Devices: NEMA ICS 5; factory installed in controller enclosure cover unless otherwise indicated.

1. Push Buttons, Pilot Lights, and Selector Switches: Heavy-duty.

D. Reversible N.C./N.O. auxiliary contact(s).

E. Sun shields installed on fronts, sides, and tops of enclosures installed outdoors and subject to direct and extended sun exposure.

F. Cover gaskets for Type 1 enclosures.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Examine areas and surfaces to receive enclosed controllers, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

B. Examine enclosed controllers before installation. Reject enclosed controllers that are wet, moisture damaged, or mold damaged.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 INSTALLATION

A. Wall-Mounted Controllers: Install enclosed controllers on walls with tops at uniform height unless otherwise indicated, and by bolting units to wall or mounting on lightweight structural-steel channels bolted to wall. For controllers not at walls, provide freestanding racks complying with Division 26 Section "Hangers and Supports for Electrical Systems. "Seismic Bracing: Comply with requirements specified in Section 26 05 48 "Seismic Controls for Electrical Systems."

B. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.

C. Install fuses in each fusible-switch enclosed controller.

D. Install fuses in control circuits if not factory installed.

E. Install heaters in thermal overload relays. Select heaters based on actual nameplate full-load amperes after motors have been installed.

F. Install, connect, and fuse thermal-protector monitoring relays furnished with motor-driven equipment.

G. Comply with NECA 1.

H. Enclosed controllers shall be mounted on separate structures. Do not mount on HVAC housings, ductwork, pump frames, etc.

3.03 IDENTIFICATION

A. Comply with requirements in Division 26 Section 26 05 53.

1. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs.

3.04 ENGRAVED LABELS

A. Melamine plastic laminate, white with black core, 1/8-inch thick.

B. UV and weatherproof rated.

C. Engravers standard letter style. Use minimum of 3/8-inch high capital letters.

D. Drill or punch labels for mechanical fastening. Use self-tapping stainless steel screws. Use proper size screw not to penetrate more than 1/8 exposure.

3.05 CONTROL WIRING INSTALLATION

A. Install wiring between enclosed controllers and remote devices.

B. Bundle, train, and support wiring in enclosures.

C. Connect selector switches and other automatic-control selection devices where applicable.

1. Connect selector switches to bypass only those manual- and automatic-control devices that have no safety functions when switch is in manual-control position.

2. Connect selector switches with enclosed-controller circuit in both manual and automatic positions for safety-type control devices such as low- and high-pressure cutouts, high-temperature cutouts, and motor overload protectors.

3.06 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

B. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust components, assemblies, and equipment installations, including connections.

C. Perform tests and inspections.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.

D. Acceptance Testing Preparation:

1. Test insulation resistance for each enclosed controller, component, connecting supply, feeder, and control circuit.

2. Test continuity of each circuit.

E. Tests and Inspections:

1. Inspect controllers, wiring, components, connections, and equipment installation. Test and

adjust controllers, components, and equipment.

2. Test insulation resistance for each enclosed-controller element, component, connecting motor supply, feeder, and control circuits.

3. Test continuity of each circuit.

4. Verify that voltages at controller locations are within plus or minus 10 percent of motor nameplate rated voltages. If outside this range for any motor, notify Owner before starting the motor(s).

5. Test each motor for proper phase rotation.

6. Perform each electrical test and visual and mechanical inspection stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.

7. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.

8. Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.

F. Enclosed controllers will be considered defective if they do not pass tests and inspections.

G. Prepare test and inspection reports including a certified report that identifies enclosed controllers and that describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

3.07 ADJUSTING

A. Set field-adjustable switches, auxiliary relays, time-delay relays, timers, and overload-relay pickup and trip ranges.

B. Adjust overload-relay heaters or settings if power factor correction capacitors are connected to the load side of the overload relays.

3.08 PROTECTION

A. Temporary Heating: Apply temporary heat to maintain temperature according to manufacturer's written instructions until enclosed controllers are ready to be energized and placed into service.

B. Replace controllers whose interiors have been exposed to water or other liquids prior to Substantial Completion.

3.09 DEMONSTRATION

A. Train Owner's maintenance personnel to adjust, operate, and maintain enclosed controllers.

END OF SECTION 26 29 13

SLAC BUILDING INSPECTION OFFICE	
Construction Authorization (IFC)	
BY: cnadler	
DATE: 2/18/2026	**PRS#:** 26-009
Revised Authorization ☑ **No** ☐ **Yes**	
These plans have been reviewed for compliance with applicable building codes and standards, and ESH program requirements.	
The authorization of these plans shall not be construed to be a permit or an approval for violation of any **applicable codes, standards, or ESH programs**.	
THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.	